GLOBAL HERITAGE / RESEARCH SYNTHESIS / 2026

THE GLOBAL ARCHAEOLOGICAL RECORD

A 1,000-PAGE REFERENCE EDITION

440 curated site and landscape dossiers

39 research-method and thematic chapters

22 global archaeological regions

A worldwide survey from early stone technologies to modern historical archaeology.

Not a complete census of every dig, report or collection ever produced.

Site profiles are concise research orientations: verify details in primary reports.

Research handbooks distinguish observed evidence, inference and proposals.

Prepared 12 November 2026 | Searchable PDF with clickable chapter bookmarks

Scope note: 1,000 pages do not imply exhaustive site-by-site verification.

Regional atlas directory

Sites begin on page 121; each record occupies two facing pages

INDEX | 22 REGIONS AND 440 SITES

Browse by geography below, or use the PDF outline panel for direct links to individual sites and every research chapter. Page references are fixed for this edition; they are not external source citations.

REGION AND STARTING PAGE

01	East African Human Origins	121
02	Southern Africa and Deep Prehistory	161
03	Nile Valley and Sudan	201
04	North Africa and Sahara	241
05	West and Central African Archaeology	281
06	Levant and Arabia	321
07	Mesopotamia and Iranian Plateau	361
08	Anatolia, Cyprus, and Caucasian Fringe	401
09	Aegean and Ancient Greece	441
10	Classical Rome and Mediterranean	481
11	Atlantic and Northern European Prehistory	521
12	European Iron Age, Medieval, and Eastern Europe	561
13	South Asia: Indus and Deep History	601
14	South Asia: Buddhist and Imperial Centers	641
15	East Asia: China, Korea, Japan	681
16	Southeast Asia	721
17	Inner Asia, Steppe, and Siberia	761
18	North America	801
19	Mesoamerica	841
20	South America	881
21	Amazonia, Andes, Caribbean	921
22	Oceania and Pacific	961

HOW TO READ THE TWO-PAGE DOSSIER

Page A records geography, broad period, material focus, interpretive caveats and three excavation questions. Page B defines an evidence-testing protocol, appropriate analytical methods and responsible future research. The proposed methods are research suggestions, not claims that they have already been applied at that site.

Research chapters and source guide

39 chapters, each three pages • cited research portals

INDEX | METHODS, THEMES AND ATTRIBUTION

01	What archaeologists can and cannot know	4
02	History of archaeological practice	7
03	The global timeline: deep time to recent past	10
04	Human evolution and fossil evidence	13
05	Stone technology and technological learning	16
06	Fire, shelter, and early domestic life	19
07	Ancient DNA, proteins, and migration	22
08	Peopling of the Americas	25
09	The first agricultural societies	28
10	Monuments before states	31
11	Ceramics and changing domestic economies	34
12	Urban origins and settlement scale	37
13	States, institutions, and political authority	40
14	Writing, archives, and accounting	43
15	Trade networks and the movement of objects	46
16	Metallurgy, mining, and craft specialization	49
17	Water management and environmental engineering	52
18	The archaeology of household life	55
19	Burials, commemoration, and ancient societies	58
20	Foodways, farming, and climate	61
21	Rock art, imagery, and symbolic behavior	64
22	Gender, age, labor, and inequality	67
23	War, fortifications, and conflict	70
24	Religion, ritual, and sacred landscapes	73
25	Cities after imperial collapse	76
26	Colonial archaeology and connected worlds	79
27	Maritime and underwater archaeology	82
28	Survey, mapping, GIS, and satellite data	85
29	LiDAR, photogrammetry, and 3D recording	88
30	Excavation recording and stratigraphic reasoning	91
31	Radiocarbon dating and chronological modelling	94
32	Other chronometric methods	97
33	Experimental archaeology and use-wear	100
34	Bioarchaeology and health	103
35	Heritage ethics and descendant communities	106
36	Illicit antiquities, restitution, and provenance	109
37	Conservation, climate, and disaster risks	112
38	Research archives, open data, and reproducibility	115
39	How to evaluate a contested claim	118

VERIFIED GENERAL RESEARCH AND PRESERVATION SOURCES

UNESCO World Heritage List — whc.unesco.org/en/list/

Smithsonian Human Origins — humanorigins.si.edu

Historic England archaeological methods — historicengland.org.uk/research/methods/archaeology/

Society for American Archaeology ethics — saa.org

UNESCO underwater heritage — unesco.org/en/underwater-heritage/2001-convention

Additional titled, linked reading paths are supplied on handbook pages. No list can inventory the full global unpublished excavation record.

What archaeologists can and cannot know

SCOPE AND SIGNIFICANCE

What archaeologists can and cannot know is an essential problem in epistemology. The central record involves archaeological context stratigraphy formation processes and inference. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with archaeological context stratigraphy formation processes and inference. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Olduvai Gorge and Pompeii. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that the same deposit may have multiple explanations and missing evidence is not proof of absence. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Olduvai Gorge, Pompeii, Cahokia. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

What archaeologists can and cannot know

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of archaeological context stratigraphy formation processes and inference; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Olduvai Gorge or a comparable setting, formulate the claim and define its geographical/temporal limits. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Pompeii or a comparable setting, trace each claim to an observation and documented context. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Cahokia or a comparable setting, differentiate samples, site-level aggregates and regional inferences. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Olduvai Gorge or a comparable setting, test taphonomic and collection bias as alternative explanations. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Pompeii or a comparable setting, grade confidence according to independent replication. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact	Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval	Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives	Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims	Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for what archaeologists can and cannot know supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of the same deposit may have multiple explanations and missing evidence is not proof of absence. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

What archaeologists can and cannot know

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For what archaeologists can and cannot know, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Olduvai Gorge, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Pompeii, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Cahokia, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether the same deposit may have multiple explanations and missing evidence is not proof of absence and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO World Heritage Centre, World Heritage List. A selective heritage register, not a global excavation census. https://whc.unesco.org/en/list/

2. UNESCO, World Heritage selection criteria. Heritage assessment and significance frameworks. https://whc.unesco.org/en/criteria/

3. Smithsonian Human Origins, stone tools. Early lithic technologies and their interpretation. https://humanorigins.si.edu/evidence/behavior/stone-tools

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in archaeological context stratigraphy formation processes and inference. To move beyond an orientation chapter, consult excavation volumes for Olduvai Gorge, Pompeii, Cahokia, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

History of archaeological practice

SCOPE AND SIGNIFICANCE

History of archaeological practice is an essential problem in historiography. The central record involves antiquarian collections excavation notebooks stratigraphic recording and collaborative archaeology. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with antiquarian collections excavation notebooks stratigraphic recording and collaborative archaeology. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Pompeii and Troy. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that older digs frequently removed artifacts without recording adequate contextual relationships. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Pompeii, Troy, Ur. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

History of archaeological practice

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of antiquarian collections excavation notebooks stratigraphic recording and collaborative archaeology; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Pompeii or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Troy or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Ur or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Pompeii or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Troy or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for history of archaeological practice supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of older digs frequently removed artifacts without recording adequate contextual relationships. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

History of archaeological practice

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For history of archaeological practice, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Pompeii, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Troy, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Ur, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether older digs frequently removed artifacts without recording adequate contextual relationships and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Harmand et al., Lomekwi 3, Nature (2015). Primary report on 3.3-million-year-old lithics.
https://www.nature.com/articles/nature14464

2. Plummer et al., first million years of technology (2025). Peer-reviewed synthetic review.
https://doi.org/10.1146/annurev-anthro-071923-112250

3. Historic England, Archaeological Methods. Excavation and archaeological science.
https://historicengland.org.uk/research/methods/archaeology/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in antiquarian collections excavation notebooks stratigraphic recording and collaborative archaeology. To move beyond an orientation chapter, consult excavation volumes for Pompeii, Troy, Ur, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

The global timeline: deep time to recent past

SCOPE AND SIGNIFICANCE

The global timeline: deep time to recent past is an essential problem in chronology. The central record involves Pliocene Pleistocene Holocene Bronze Age Iron Age and later historical sequences. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with Pliocene Pleistocene Holocene Bronze Age Iron Age and later historical sequences. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Lomekwi 3 and Uruk (Warka). Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that period names are regionally defined rather than universal dates. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Lomekwi 3, Uruk (Warka), Chaco Canyon. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

The global timeline: deep time to recent past

Chapter 3 • Methods and replication

II | FIELD AND LABORATORY DESIGN

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of Pliocene Pleistocene Holocene Bronze Age Iron Age and later historical sequences; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Lomekwi 3 or a comparable setting, define the target event rather than simply the sample age. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Uruk (Warka) or a comparable setting, choose chronometric materials with secure contexts. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Chaco Canyon or a comparable setting, calibrate and propagate uncertainties correctly. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Lomekwi 3 or a comparable setting, compare stratigraphy with independent dates. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Uruk (Warka) or a comparable setting, revisit regional period nomenclature. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact	Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval	Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives	Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims	Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for the global timeline: deep time to recent past supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of period names are regionally defined rather than universal dates. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

The global timeline: deep time to recent past

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For the global timeline: deep time to recent past, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Lomekwi 3, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Uruk (Warka), compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Chaco Canyon, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether period names are regionally defined rather than universal dates and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Historic England, Scientific Dating. Chronometric dating methods.
https://historicengland.org.uk/advice/technical-advice/archaeological-science/scientific-dating/

2. Historic England, Radiocarbon Dating and Chronological Modelling. Radiocarbon best practice and bayesian analysis.
https://historicengland.org.uk/images-books/publications/radiocarbon-dating-chronological-modelling

3. Historic England, Pleistocene scientific dating. Deep time dating and methodological limitations.
https://historicengland.org.uk/images-books/publications/scientific-dating-pleistocene-sites/scientific-dating-methods/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in Pliocene Pleistocene Holocene Bronze Age Iron Age and later historical sequences. To move beyond an orientation chapter, consult excavation volumes for Lomekwi 3, Uruk (Warka), Chaco Canyon, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Human evolution and fossil evidence

SCOPE AND SIGNIFICANCE

Human evolution and fossil evidence is an essential problem in paleoanthropology. The central record involves skeletal morphology sediment associations ancient proteins and genetics. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with skeletal morphology sediment associations ancient proteins and genetics. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Hadar and Rising Star Cave. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that taxonomic classifications and associations can be revised by new finds. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Hadar, Rising Star Cave, Denisova Cave. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Human evolution and fossil evidence

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of skeletal morphology sediment associations ancient proteins and genetics; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Hadar or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Rising Star Cave or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Denisova Cave or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Hadar or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Rising Star Cave or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for human evolution and fossil evidence supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of taxonomic classifications and associations can be revised by new finds. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Human evolution and fossil evidence

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For human evolution and fossil evidence, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Hadar, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Rising Star Cave, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Denisova Cave, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether taxonomic classifications and associations can be revised by new finds and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Society for American Archaeology, Ethics Principles. Professional stewardship and responsibilities. https://www.saa.org/Member/SAAMember/Career-and-Practice/Principles-of-Archaeological-Ethics.aspx

2. ICOMOS, Archaeological Heritage Charter (1990). Preservation and management of sites. https://www.icomos.de/data/pdf/xx-0421-1300-25.pdf

3. UNESCO, Underwater Cultural Heritage Convention. Underwater heritage and anti-looting principles. https://www.unesco.org/en/underwater-heritage/2001-convention

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in skeletal morphology sediment associations ancient proteins and genetics. To move beyond an orientation chapter, consult excavation volumes for Hadar, Rising Star Cave, Denisova Cave, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Stone technology and technological learning

SCOPE AND SIGNIFICANCE

Stone technology and technological learning is an essential problem in lithic archaeology. The central record involves cores flakes refitting raw material sources use wear and manufacturing debris. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with cores flakes refitting raw material sources use wear and manufacturing debris. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Lomekwi 3 and Olduvai Gorge. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that a tool type does not identify a maker or a single task. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Lomekwi 3, Olduvai Gorge, Attirampakkam. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Stone technology and technological learning

Chapter 5 • Methods and replication

II | FIELD AND LABORATORY DESIGN

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of cores flakes refitting raw material sources use wear and manufacturing debris; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Lomekwi 3 or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Olduvai Gorge or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Attirampakkam or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Lomekwi 3 or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Olduvai Gorge or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for stone technology and technological learning supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of a tool type does not identify a maker or a single task. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Stone technology and technological learning

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For stone technology and technological learning, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Lomekwi 3, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Olduvai Gorge, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Attirampakkam, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether a tool type does not identify a maker or a single task and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Göbekli Tepe. Documented early monumental enclosures. https://whc.unesco.org/en/list/1572

2. UNESCO: Çatalhöyük. Neolithic settlement and occupation. https://whc.unesco.org/en/list/1405/

3. UNESCO: Angkor. Khmer urban and hydraulic landscape. https://whc.unesco.org/en/list/668/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in cores flakes refitting raw material sources use wear and manufacturing debris. To move beyond an orientation chapter, consult excavation volumes for Lomekwi 3, Olduvai Gorge, Attirampakkam, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Fire, shelter, and early domestic life

SCOPE AND SIGNIFICANCE

Fire, shelter, and early domestic life is an essential problem in prehistoric archaeology. The central record involves hearths burnt sediments postholes sleeping surfaces and plant residues. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with hearths burnt sediments postholes sleeping surfaces and plant residues. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Wonderwerk Cave and Sibudu Cave. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that natural burning and occupation processes can resemble deliberate fire use. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Wonderwerk Cave, Sibudu Cave, Ohalo II. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Fire, shelter, and early domestic life

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of hearths burnt sediments postholes sleeping surfaces and plant residues; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Wonderwerk Cave or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Sibudu Cave or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Ohalo II or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Wonderwerk Cave or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Sibudu Cave or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for fire, shelter, and early domestic life supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of natural burning and occupation processes can resemble deliberate fire use. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Fire, shelter, and early domestic life

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For fire, shelter, and early domestic life, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Wonderwerk Cave, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Sibudu Cave, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Ohalo II, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether natural burning and occupation processes can resemble deliberate fire use and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Great Zimbabwe. Southern african polity and monument. https://whc.unesco.org/en/list/364/

2. UNESCO: Cahokia. Mississippian settlement and earthworks. https://whc.unesco.org/en/list/198

3. UNESCO: Rapa Nui. Polynesian archaeological landscape. https://whc.unesco.org/en/list/715/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in hearths burnt sediments postholes sleeping surfaces and plant residues. To move beyond an orientation chapter, consult excavation volumes for Wonderwerk Cave, Sibudu Cave, Ohalo II, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Ancient DNA, proteins, and migration

SCOPE AND SIGNIFICANCE

Ancient DNA, proteins, and migration is an essential problem in biomolecular archaeology. The central record involves human bones dental calculus ancient DNA proteomics and isotopes. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with human bones dental calculus ancient DNA proteomics and isotopes. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Denisova Cave and Shum Laka. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that preservation and sampling bias can distort demographic reconstructions. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Denisova Cave, Shum Laka, Paisley Caves. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Ancient DNA, proteins, and migration

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of human bones dental calculus ancient DNA proteomics and isotopes; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Denisova Cave or a comparable setting, document sample provenance and depositional history. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Shum Laka or a comparable setting, check preservation and contamination controls. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Paisley Caves or a comparable setting, compare measurements across independent samples. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Denisova Cave or a comparable setting, test for demographic and survivorship bias. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Shum Laka or a comparable setting, deposit data with community-approved access conditions. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for ancient dna, proteins, and migration supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of preservation and sampling bias can distort demographic reconstructions. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Ancient DNA, proteins, and migration

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For ancient dna, proteins, and migration, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Denisova Cave, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Shum Laka, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Paisley Caves, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether preservation and sampling bias can distort demographic reconstructions and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO World Heritage Centre, World Heritage List. A selective heritage register, not a global excavation census. https://whc.unesco.org/en/list/

2. UNESCO, World Heritage selection criteria. Heritage assessment and significance frameworks. https://whc.unesco.org/en/criteria/

3. Smithsonian Human Origins, stone tools. Early lithic technologies and their interpretation. https://humanorigins.si.edu/evidence/behavior/stone-tools

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in human bones dental calculus ancient DNA proteomics and isotopes. To move beyond an orientation chapter, consult excavation volumes for Denisova Cave, Shum Laka, Paisley Caves, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Peopling of the Americas

SCOPE AND SIGNIFICANCE

Peopling of the Americas is an essential problem in american prehistory. The central record involves stratified occupation sites lithics genetics and paleoenvironmental reconstruction. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with stratified occupation sites lithics genetics and paleoenvironmental reconstruction. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Cooper's Ferry and Paisley Caves. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that older than Clovis claims demand exceptionally secure associations and dating. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Cooper's Ferry, Paisley Caves, Monte Verde. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Peopling of the Americas

Chapter 8 • Methods and replication

II | FIELD AND LABORATORY DESIGN

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of stratified occupation sites lithics genetics and paleoenvironmental reconstruction; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Cooper's Ferry or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Paisley Caves or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Monte Verde or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Cooper's Ferry or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Paisley Caves or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for peopling of the americas supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of older than Clovis claims demand exceptionally secure associations and dating. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Peopling of the Americas

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For peopling of the americas, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Cooper's Ferry, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Paisley Caves, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Monte Verde, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether older than Clovis claims demand exceptionally secure associations and dating and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Harmand et al., Lomekwi 3, Nature (2015). Primary report on 3.3-million-year-old lithics.
https://www.nature.com/articles/nature14464

2. Plummer et al., first million years of technology (2025). Peer-reviewed synthetic review.
https://doi.org/10.1146/annurev-anthro-071923-112250

3. Historic England, Archaeological Methods. Excavation and archaeological science.
https://historicengland.org.uk/research/methods/archaeology/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in stratified occupation sites lithics genetics and paleoenvironmental reconstruction. To move beyond an orientation chapter, consult excavation volumes for Cooper's Ferry, Paisley Caves, Monte Verde, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

The first agricultural societies

SCOPE AND SIGNIFICANCE

The first agricultural societies is an essential problem in neolithic archaeology. The central record involves charred seeds animal bones cultivation traces and sedentary architecture. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with charred seeds animal bones cultivation traces and sedentary architecture. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Abu Hureyra and Mehrgarh. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that food production was not a single innovation occurring at one global date. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Abu Hureyra, Mehrgarh, Kuk Swamp. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

The first agricultural societies

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of charred seeds animal bones cultivation traces and sedentary architecture; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Abu Hureyra or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Mehrgarh or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Kuk Swamp or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Abu Hureyra or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Mehrgarh or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for the first agricultural societies supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of food production was not a single innovation occurring at one global date. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

The first agricultural societies

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For the first agricultural societies, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Abu Hureyra, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Mehrgarh, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Kuk Swamp, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether food production was not a single innovation occurring at one global date and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Historic England, Scientific Dating. Chronometric dating methods.
https://historicengland.org.uk/advice/technical-advice/archaeological-science/scientific-dating/

2. Historic England, Radiocarbon Dating and Chronological Modelling. Radiocarbon best practice and bayesian analysis.
https://historicengland.org.uk/images-books/publications/radiocarbon-dating-chronological-modelling

3. Historic England, Pleistocene scientific dating. Deep time dating and methodological limitations.
https://historicengland.org.uk/images-books/publications/scientific-dating-pleistocene-sites/scientific-dating-methods/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in charred seeds animal bones cultivation traces and sedentary architecture. To move beyond an orientation chapter, consult excavation volumes for Abu Hureyra, Mehrgarh, Kuk Swamp, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Monuments before states

SCOPE AND SIGNIFICANCE

Monuments before states is an essential problem in social archaeology. The central record involves megaliths labor investment communal gathering and ritual features. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with megaliths labor investment communal gathering and ritual features. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Göbekli Tepe and Stonehenge. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that monument size alone cannot diagnose hierarchy or fixed religious beliefs. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Göbekli Tepe, Stonehenge, Poverty Point. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Monuments before states

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of megaliths labor investment communal gathering and ritual features; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Göbekli Tepe or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Stonehenge or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Poverty Point or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Göbekli Tepe or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Stonehenge or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for monuments before states supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of monument size alone cannot diagnose hierarchy or fixed religious beliefs. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Monuments before states

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For monuments before states, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Göbekli Tepe, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Stonehenge, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Poverty Point, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether monument size alone cannot diagnose hierarchy or fixed religious beliefs and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Society for American Archaeology, Ethics Principles. Professional stewardship and responsibilities. https://www.saa.org/Member/SAAMember/Career-and-Practice/Principles-of-Archaeological-Ethics.aspx

2. ICOMOS, Archaeological Heritage Charter (1990). Preservation and management of sites. https://www.icomos.de/data/pdf/xx-0421-1300-25.pdf

3. UNESCO, Underwater Cultural Heritage Convention. Underwater heritage and anti-looting principles. https://www.unesco.org/en/underwater-heritage/2001-convention

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in megaliths labor investment communal gathering and ritual features. To move beyond an orientation chapter, consult excavation volumes for Göbekli Tepe, Stonehenge, Poverty Point, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Ceramics and changing domestic economies

SCOPE AND SIGNIFICANCE

Ceramics and changing domestic economies is an essential problem in material culture. The central record involves pottery fabrics firing methods residue analysis and vessel distributions. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with pottery fabrics firing methods residue analysis and vessel distributions. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Banpo and Jomon Sannai-Maruyama. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that pottery typologies are relative chronologies not universal cultural identities. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Banpo, Jomon Sannai-Maruyama, Nok sites. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Ceramics and changing domestic economies

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of pottery fabrics firing methods residue analysis and vessel distributions; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Banpo or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Jomon Sannai-Maruyama or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Nok sites or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Banpo or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Jomon Sannai-Maruyama or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for ceramics and changing domestic economies supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of pottery typologies are relative chronologies not universal cultural identities. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Ceramics and changing domestic economies

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For ceramics and changing domestic economies, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Banpo, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Jomon Sannai-Maruyama, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Nok sites, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether pottery typologies are relative chronologies not universal cultural identities and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Göbekli Tepe. Documented early monumental enclosures. https://whc.unesco.org/en/list/1572

2. UNESCO: Çatalhöyük. Neolithic settlement and occupation. https://whc.unesco.org/en/list/1405/

3. UNESCO: Angkor. Khmer urban and hydraulic landscape. https://whc.unesco.org/en/list/668/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in pottery fabrics firing methods residue analysis and vessel distributions. To move beyond an orientation chapter, consult excavation volumes for Banpo, Jomon Sannai-Maruyama, Nok sites, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Urban origins and settlement scale

SCOPE AND SIGNIFICANCE

Urban origins and settlement scale is an essential problem in urban archaeology. The central record involves settlement density roads neighborhoods sanitation administration and rural hinterlands. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with settlement density roads neighborhoods sanitation administration and rural hinterlands. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Uruk (Warka) and Mohenjo-daro. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that monumental centers and cities should not be equated without functional evidence. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Uruk (Warka), Mohenjo-daro, Teotihuacan. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Urban origins and settlement scale

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of settlement density roads neighborhoods sanitation administration and rural hinterlands; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Uruk (Warka) or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Mohenjo-daro or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Teotihuacan or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Uruk (Warka) or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Mohenjo-daro or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for urban origins and settlement scale supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of monumental centers and cities should not be equated without functional evidence. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Urban origins and settlement scale

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For urban origins and settlement scale, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Uruk (Warka), seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Mohenjo-daro, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Teotihuacan, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether monumental centers and cities should not be equated without functional evidence and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Great Zimbabwe. Southern african polity and monument. https://whc.unesco.org/en/list/364/

2. UNESCO: Cahokia. Mississippian settlement and earthworks. https://whc.unesco.org/en/list/198

3. UNESCO: Rapa Nui. Polynesian archaeological landscape. https://whc.unesco.org/en/list/715/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in settlement density roads neighborhoods sanitation administration and rural hinterlands. To move beyond an orientation chapter, consult excavation volumes for Uruk (Warka), Mohenjo-daro, Teotihuacan, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

States, institutions, and political authority

SCOPE AND SIGNIFICANCE

States, institutions, and political authority is an essential problem in political archaeology. The central record involves seals palaces tribute archives fortifications and administrative spaces. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with seals palaces tribute archives fortifications and administrative spaces. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Arslantepe and Persepolis. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that elite artifacts alone do not demonstrate a stable centralized bureaucracy. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Arslantepe, Persepolis, Tikal. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

States, institutions, and political authority

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of seals palaces tribute archives fortifications and administrative spaces; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Arslantepe or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Persepolis or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Tikal or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Arslantepe or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Persepolis or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for states, institutions, and political authority supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of elite artifacts alone do not demonstrate a stable centralized bureaucracy. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

States, institutions, and political authority

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For states, institutions, and political authority, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Arslantepe, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Persepolis, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Tikal, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether elite artifacts alone do not demonstrate a stable centralized bureaucracy and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO World Heritage Centre, World Heritage List. A selective heritage register, not a global excavation census. https://whc.unesco.org/en/list/

2. UNESCO, World Heritage selection criteria. Heritage assessment and significance frameworks. https://whc.unesco.org/en/criteria/

3. Smithsonian Human Origins, stone tools. Early lithic technologies and their interpretation. https://humanorigins.si.edu/evidence/behavior/stone-tools

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in seals palaces tribute archives fortifications and administrative spaces. To move beyond an orientation chapter, consult excavation volumes for Arslantepe, Persepolis, Tikal, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Writing, archives, and accounting

SCOPE AND SIGNIFICANCE

Writing, archives, and accounting is an essential problem in textual archaeology. The central record involves tablets inscriptions seals manuscripts tallying devices and archival spaces. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with tablets inscriptions seals manuscripts tallying devices and archival spaces. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Ebla (Tell Mardikh) and Yinxu (Anyang). Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that texts tend to reflect literate institutions more than whole populations. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Ebla (Tell Mardikh), Yinxu (Anyang), Pylos (Palace of Nestor). These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Writing, archives, and accounting

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of tablets inscriptions seals manuscripts tallying devices and archival spaces; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Ebla (Tell Mardikh) or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Yinxu (Anyang) or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Pylos (Palace of Nestor) or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Ebla (Tell Mardikh) or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Yinxu (Anyang) or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for writing, archives, and accounting supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of texts tend to reflect literate institutions more than whole populations. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Writing, archives, and accounting

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For writing, archives, and accounting, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Ebla (Tell Mardikh), seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Yinxu (Anyang), compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Pylos (Palace of Nestor), audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether texts tend to reflect literate institutions more than whole populations and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Harmand et al., Lomekwi 3, Nature (2015). Primary report on 3.3-million-year-old lithics.
https://www.nature.com/articles/nature14464

2. Plummer et al., first million years of technology (2025). Peer-reviewed synthetic review.
https://doi.org/10.1146/annurev-anthro-071923-112250

3. Historic England, Archaeological Methods. Excavation and archaeological science.
https://historicengland.org.uk/research/methods/archaeology/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in tablets inscriptions seals manuscripts tallying devices and archival spaces. To move beyond an orientation chapter, consult excavation volumes for Ebla (Tell Mardikh), Yinxu (Anyang), Pylos (Palace of Nestor), check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Trade networks and the movement of objects

SCOPE AND SIGNIFICANCE

Trade networks and the movement of objects is an essential problem in exchange archaeology. The central record involves material sourcing distribution studies shipwreck cargo and caravan nodes. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with material sourcing distribution studies shipwreck cargo and caravan nodes. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Berenike and Kültepe (Kanesh). Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that an imported object cannot by itself establish sustained contact or travel routes. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Berenike, Kültepe (Kanesh), Merv. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Trade networks and the movement of objects

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of material sourcing distribution studies shipwreck cargo and caravan nodes; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Berenike or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Kültepe (Kanesh) or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Merv or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Berenike or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Kültepe (Kanesh) or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for trade networks and the movement of objects supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of an imported object cannot by itself establish sustained contact or travel routes. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Trade networks and the movement of objects

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For trade networks and the movement of objects, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Berenike, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Kültepe (Kanesh), compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Merv, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether an imported object cannot by itself establish sustained contact or travel routes and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Historic England, Scientific Dating. Chronometric dating methods.
https://historicengland.org.uk/advice/technical-advice/archaeological-science/scientific-dating/

2. Historic England, Radiocarbon Dating and Chronological Modelling. Radiocarbon best practice and bayesian analysis.
https://historicengland.org.uk/images-books/publications/radiocarbon-dating-chronological-modelling

3. Historic England, Pleistocene scientific dating. Deep time dating and methodological limitations.
https://historicengland.org.uk/images-books/publications/scientific-dating-pleistocene-sites/scientific-dating-methods/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in material sourcing distribution studies shipwreck cargo and caravan nodes. To move beyond an orientation chapter, consult excavation volumes for Berenike, Kültepe (Kanesh), Merv, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Metallurgy, mining, and craft specialization

SCOPE AND SIGNIFICANCE

Metallurgy, mining, and craft specialization is an essential problem in archaeometallurgy. The central record involves slag furnaces ores casting debris isotopic signatures and workshop organization. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with slag furnaces ores casting debris isotopic signatures and workshop organization. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Nok sites and Sardis. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that similar metal objects may result from recycling or convergent designs. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Nok sites, Sardis, Sanxingdui. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Metallurgy, mining, and craft specialization

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of slag furnaces ores casting debris isotopic signatures and workshop organization; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Nok sites or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Sardis or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Sanxingdui or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Nok sites or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Sardis or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for metallurgy, mining, and craft specialization supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of similar metal objects may result from recycling or convergent designs. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Metallurgy, mining, and craft specialization

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For metallurgy, mining, and craft specialization, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Nok sites, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Sardis, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Sanxingdui, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether similar metal objects may result from recycling or convergent designs and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Society for American Archaeology, Ethics Principles. Professional stewardship and responsibilities. https://www.saa.org/Member/SAAMember/Career-and-Practice/Principles-of-Archaeological-Ethics.aspx

2. ICOMOS, Archaeological Heritage Charter (1990). Preservation and management of sites. https://www.icomos.de/data/pdf/xx-0421-1300-25.pdf

3. UNESCO, Underwater Cultural Heritage Convention. Underwater heritage and anti-looting principles. https://www.unesco.org/en/underwater-heritage/2001-convention

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in slag furnaces ores casting debris isotopic signatures and workshop organization. To move beyond an orientation chapter, consult excavation volumes for Nok sites, Sardis, Sanxingdui, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Water management and environmental engineering

SCOPE AND SIGNIFICANCE

Water management and environmental engineering is an essential problem in landscape archaeology. The central record involves canals reservoirs terrace fields paleoenvironmental cores and irrigation networks. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with canals reservoirs terrace fields paleoenvironmental cores and irrigation networks. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Dholavira and Angkor Wat. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that climate stress is rarely sufficient as a single cause for political change. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Dholavira, Angkor Wat, Casa Grande Ruins. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Water management and environmental engineering

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of canals reservoirs terrace fields paleoenvironmental cores and irrigation networks; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Dholavira or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Angkor Wat or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Casa Grande Ruins or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Dholavira or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Angkor Wat or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for water management and environmental engineering supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of climate stress is rarely sufficient as a single cause for political change. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Water management and environmental engineering

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For water management and environmental engineering, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Dholavira, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Angkor Wat, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Casa Grande Ruins, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether climate stress is rarely sufficient as a single cause for political change and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Göbekli Tepe. Documented early monumental enclosures. https://whc.unesco.org/en/list/1572

2. UNESCO: Çatalhöyük. Neolithic settlement and occupation. https://whc.unesco.org/en/list/1405/

3. UNESCO: Angkor. Khmer urban and hydraulic landscape. https://whc.unesco.org/en/list/668/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in canals reservoirs terrace fields paleoenvironmental cores and irrigation networks. To move beyond an orientation chapter, consult excavation volumes for Dholavira, Angkor Wat, Casa Grande Ruins, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

The archaeology of household life

SCOPE AND SIGNIFICANCE

The archaeology of household life is an essential problem in household archaeology. The central record involves floor deposits cooking features storage refuse construction sequence and repair. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with floor deposits cooking features storage refuse construction sequence and repair. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Çatalhöyük and Deir el-Medina. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that specialist interpretations may overemphasize palaces and elite residences. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Çatalhöyük, Deir el-Medina, Skara Brae. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

The archaeology of household life

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of floor deposits cooking features storage refuse construction sequence and repair; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Çatalhöyük or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Deir el-Medina or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Skara Brae or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Çatalhöyük or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Deir el-Medina or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for the archaeology of household life supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of specialist interpretations may overemphasize palaces and elite residences. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

The archaeology of household life

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For the archaeology of household life, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Çatalhöyük, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Deir el-Medina, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Skara Brae, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether specialist interpretations may overemphasize palaces and elite residences and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Great Zimbabwe. Southern african polity and monument. https://whc.unesco.org/en/list/364/

2. UNESCO: Cahokia. Mississippian settlement and earthworks. https://whc.unesco.org/en/list/198

3. UNESCO: Rapa Nui. Polynesian archaeological landscape. https://whc.unesco.org/en/list/715/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in floor deposits cooking features storage refuse construction sequence and repair. To move beyond an orientation chapter, consult excavation volumes for Çatalhöyük, Deir el-Medina, Skara Brae, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Burials, commemoration, and ancient societies

SCOPE AND SIGNIFICANCE

Burials, commemoration, and ancient societies is an essential problem in mortuary archaeology. The central record involves grave architecture skeletal remains ornaments funerary treatment and memorial use. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with grave architecture skeletal remains ornaments funerary treatment and memorial use. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Sungir and Varna Necropolis. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that rich burials do not automatically indicate hereditary rank. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Sungir, Varna Necropolis, Sipán. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Burials, commemoration, and ancient societies

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of grave architecture skeletal remains ornaments funerary treatment and memorial use; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Sungir or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Varna Necropolis or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Sipán or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Sungir or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Varna Necropolis or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for burials, commemoration, and ancient societies supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of rich burials do not automatically indicate hereditary rank. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Burials, commemoration, and ancient societies

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For burials, commemoration, and ancient societies, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Sungir, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Varna Necropolis, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Sipán, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether rich burials do not automatically indicate hereditary rank and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO World Heritage Centre, World Heritage List. A selective heritage register, not a global excavation census. https://whc.unesco.org/en/list/

2. UNESCO, World Heritage selection criteria. Heritage assessment and significance frameworks. https://whc.unesco.org/en/criteria/

3. Smithsonian Human Origins, stone tools. Early lithic technologies and their interpretation. https://humanorigins.si.edu/evidence/behavior/stone-tools

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in grave architecture skeletal remains ornaments funerary treatment and memorial use. To move beyond an orientation chapter, consult excavation volumes for Sungir, Varna Necropolis, Sipán, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Foodways, farming, and climate

SCOPE AND SIGNIFICANCE

Foodways, farming, and climate is an essential problem in environmental archaeology. The central record involves pollen phytoliths seeds animal bones isotopes and stable sediments. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with pollen phytoliths seeds animal bones isotopes and stable sediments. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Kuk Swamp and Ohalo II. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that environmental proxies require careful scale and seasonal interpretation. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Kuk Swamp, Ohalo II, Pinnacle Point. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Foodways, farming, and climate

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of pollen phytoliths seeds animal bones isotopes and stable sediments; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Kuk Swamp or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Ohalo II or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Pinnacle Point or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Kuk Swamp or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Ohalo II or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for foodways, farming, and climate supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of environmental proxies require careful scale and seasonal interpretation. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Foodways, farming, and climate

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For foodways, farming, and climate, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Kuk Swamp, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Ohalo II, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Pinnacle Point, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether environmental proxies require careful scale and seasonal interpretation and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Harmand et al., Lomekwi 3, Nature (2015). Primary report on 3.3-million-year-old lithics. https://www.nature.com/articles/nature14464

2. Plummer et al., first million years of technology (2025). Peer-reviewed synthetic review. https://doi.org/10.1146/annurev-anthro-071923-112250

3. Historic England, Archaeological Methods. Excavation and archaeological science. https://historicengland.org.uk/research/methods/archaeology/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in pollen phytoliths seeds animal bones isotopes and stable sediments. To move beyond an orientation chapter, consult excavation volumes for Kuk Swamp, Ohalo II, Pinnacle Point, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Rock art, imagery, and symbolic behavior

SCOPE AND SIGNIFICANCE

Rock art, imagery, and symbolic behavior is an essential problem in art archaeology. The central record involves paint layers iconography superposition pigments and pigment sourcing. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with paint layers iconography superposition pigments and pigment sourcing. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Chauvet Cave and Tassili n'Ajjer. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that modern viewers cannot directly recover artist intention. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Chauvet Cave, Tassili n'Ajjer, Bhimbetka Rock Shelters. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Rock art, imagery, and symbolic behavior

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of paint layers iconography superposition pigments and pigment sourcing; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Chauvet Cave or a comparable setting, map motifs and spatial contexts without imposing an interpretation. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Tassili n'Ajjer or a comparable setting, record surfaces with calibrated photography. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Bhimbetka Rock Shelters or a comparable setting, test relative sequence and direct dating suitability. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Chauvet Cave or a comparable setting, compare local visual conventions and production technology. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Tassili n'Ajjer or a comparable setting, preserve access rules and material conservation needs. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact	Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval	Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives	Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims	Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for rock art, imagery, and symbolic behavior supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of modern viewers cannot directly recover artist intention. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Rock art, imagery, and symbolic behavior

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For rock art, imagery, and symbolic behavior, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Chauvet Cave, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Tassili n'Ajjer, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Bhimbetka Rock Shelters, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether modern viewers cannot directly recover artist intention and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Historic England, Scientific Dating. Chronometric dating methods.
https://historicengland.org.uk/advice/technical-advice/archaeological-science/scientific-dating/

2. Historic England, Radiocarbon Dating and Chronological Modelling. Radiocarbon best practice and bayesian analysis.
https://historicengland.org.uk/images-books/publications/radiocarbon-dating-chronological-modelling

3. Historic England, Pleistocene scientific dating. Deep time dating and methodological limitations.
https://historicengland.org.uk/images-books/publications/scientific-dating-pleistocene-sites/scientific-dating-methods/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in paint layers iconography superposition pigments and pigment sourcing. To move beyond an orientation chapter, consult excavation volumes for Chauvet Cave, Tassili n'Ajjer, Bhimbetka Rock Shelters, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Gender, age, labor, and inequality

SCOPE AND SIGNIFICANCE

Gender, age, labor, and inequality is an essential problem in social archaeology. The central record involves skeletal stress household activity wear distribution diet and spatial access. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with skeletal stress household activity wear distribution diet and spatial access. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Sibudu Cave and Hochdorf Burial. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that contemporary gender roles must not be projected into ancient societies. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Sibudu Cave, Hochdorf Burial, Mawangdui. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Gender, age, labor, and inequality

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of skeletal stress household activity wear distribution diet and spatial access; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Sibudu Cave or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Hochdorf Burial or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Mawangdui or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Sibudu Cave or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Hochdorf Burial or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for gender, age, labor, and inequality supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of contemporary gender roles must not be projected into ancient societies. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Gender, age, labor, and inequality

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For gender, age, labor, and inequality, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Sibudu Cave, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Hochdorf Burial, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Mawangdui, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether contemporary gender roles must not be projected into ancient societies and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Society for American Archaeology, Ethics Principles. Professional stewardship and responsibilities. https://www.saa.org/Member/SAAMember/Career-and-Practice/Principles-of-Archaeological-Ethics.aspx

2. ICOMOS, Archaeological Heritage Charter (1990). Preservation and management of sites. https://www.icomos.de/data/pdf/xx-0421-1300-25.pdf

3. UNESCO, Underwater Cultural Heritage Convention. Underwater heritage and anti-looting principles. https://www.unesco.org/en/underwater-heritage/2001-convention

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in skeletal stress household activity wear distribution diet and spatial access. To move beyond an orientation chapter, consult excavation volumes for Sibudu Cave, Hochdorf Burial, Mawangdui, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

War, fortifications, and conflict

SCOPE AND SIGNIFICANCE

War, fortifications, and conflict is an essential problem in conflict archaeology. The central record involves trauma weapons siege works walls burning layers and mass graves. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with trauma weapons siege works walls burning layers and mass graves. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Lachish and Troy (Hisarlık). Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that destruction deposits can reflect disasters rather than warfare. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Lachish, Troy (Hisarlık), Masada. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

War, fortifications, and conflict

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of trauma weapons siege works walls burning layers and mass graves; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Lachish or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Troy (Hisarlık) or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Masada or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Lachish or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Troy (Hisarlık) or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for war, fortifications, and conflict supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of destruction deposits can reflect disasters rather than warfare. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

War, fortifications, and conflict

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For war, fortifications, and conflict, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Lachish, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Troy (Hisarlık), compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Masada, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether destruction deposits can reflect disasters rather than warfare and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Göbekli Tepe. Documented early monumental enclosures. https://whc.unesco.org/en/list/1572

2. UNESCO: Çatalhöyük. Neolithic settlement and occupation. https://whc.unesco.org/en/list/1405/

3. UNESCO: Angkor. Khmer urban and hydraulic landscape. https://whc.unesco.org/en/list/668/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in trauma weapons siege works walls burning layers and mass graves. To move beyond an orientation chapter, consult excavation volumes for Lachish, Troy (Hisarlık), Masada, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Religion, ritual, and sacred landscapes

SCOPE AND SIGNIFICANCE

Religion, ritual, and sacred landscapes is an essential problem in archaeology of religion. The central record involves offering pits spatial pattern sacred routes repeated ceremonies and iconography. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with offering pits spatial pattern sacred routes repeated ceremonies and iconography. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Chavín de Huántar and Delphi. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that ritual cannot be reconstructed confidently from unusual artifacts alone. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Chavín de Huántar, Delphi, Rapa Nui (Rano Raraku). These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Religion, ritual, and sacred landscapes

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of offering pits spatial pattern sacred routes repeated ceremonies and iconography; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Chavín de Huántar or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Delphi or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Rapa Nui (Rano Raraku) or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Chavín de Huántar or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Delphi or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for religion, ritual, and sacred landscapes supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of ritual cannot be reconstructed confidently from unusual artifacts alone. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Religion, ritual, and sacred landscapes

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For religion, ritual, and sacred landscapes, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Chavín de Huántar, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Delphi, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Rapa Nui (Rano Raraku), audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether ritual cannot be reconstructed confidently from unusual artifacts alone and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Great Zimbabwe. Southern african polity and monument. https://whc.unesco.org/en/list/364/

2. UNESCO: Cahokia. Mississippian settlement and earthworks. https://whc.unesco.org/en/list/198

3. UNESCO: Rapa Nui. Polynesian archaeological landscape. https://whc.unesco.org/en/list/715/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in offering pits spatial pattern sacred routes repeated ceremonies and iconography. To move beyond an orientation chapter, consult excavation volumes for Chavín de Huántar, Delphi, Rapa Nui (Rano Raraku), check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Cities after imperial collapse

SCOPE AND SIGNIFICANCE

Cities after imperial collapse is an essential problem in late antique and medieval archaeology. The central record involves abandonment reuse reoccupation fortification and economic continuity. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with abandonment reuse reoccupation fortification and economic continuity. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Rome Roman Forum and Merv. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that political transitions do not necessarily imply sudden demographic collapse. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Rome Roman Forum, Merv, Jorvik (York). These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Cities after imperial collapse

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of abandonment reuse reoccupation fortification and economic continuity; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Rome Roman Forum or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Merv or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Jorvik (York) or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Rome Roman Forum or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Merv or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for cities after imperial collapse supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of political transitions do not necessarily imply sudden demographic collapse. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Cities after imperial collapse

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For cities after imperial collapse, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Rome Roman Forum, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Merv, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Jorvik (York), audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether political transitions do not necessarily imply sudden demographic collapse and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO World Heritage Centre, World Heritage List. A selective heritage register, not a global excavation census. https://whc.unesco.org/en/list/

2. UNESCO, World Heritage selection criteria. Heritage assessment and significance frameworks. https://whc.unesco.org/en/criteria/

3. Smithsonian Human Origins, stone tools. Early lithic technologies and their interpretation. https://humanorigins.si.edu/evidence/behavior/stone-tools

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in abandonment reuse reoccupation fortification and economic continuity. To move beyond an orientation chapter, consult excavation volumes for Rome Roman Forum, Merv, Jorvik (York), check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Colonial archaeology and connected worlds

SCOPE AND SIGNIFICANCE

Colonial archaeology and connected worlds is an essential problem in historical archaeology. The central record involves ceramics coins trade goods architecture indigenous settlement patterns and archival accounts. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with ceramics coins trade goods architecture indigenous settlement patterns and archival accounts. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as La Isabela and Carthage. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that colonial records overrepresent administrative viewpoints. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: La Isabela, Carthage, Tenochtitlan Templo Mayor. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Colonial archaeology and connected worlds

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of ceramics coins trade goods architecture indigenous settlement patterns and archival accounts; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At La Isabela or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Carthage or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Tenochtitlan Templo Mayor or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At La Isabela or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Carthage or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for colonial archaeology and connected worlds supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of colonial records overrepresent administrative viewpoints. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Colonial archaeology and connected worlds

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For colonial archaeology and connected worlds, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At La Isabela, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Carthage, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Tenochtitlan Templo Mayor, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether colonial records overrepresent administrative viewpoints and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Harmand et al., Lomekwi 3, Nature (2015). Primary report on 3.3-million-year-old lithics.
https://www.nature.com/articles/nature14464

2. Plummer et al., first million years of technology (2025). Peer-reviewed synthetic review.
https://doi.org/10.1146/annurev-anthro-071923-112250

3. Historic England, Archaeological Methods. Excavation and archaeological science.
https://historicengland.org.uk/research/methods/archaeology/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in ceramics coins trade goods architecture indigenous settlement patterns and archival accounts. To move beyond an orientation chapter, consult excavation volumes for La Isabela, Carthage, Tenochtitlan Templo Mayor, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Maritime and underwater archaeology

SCOPE AND SIGNIFICANCE

Maritime and underwater archaeology is an essential problem in maritime archaeology. The central record involves wreck timbers harbor works ceramic cargo and seabed stratigraphy. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with wreck timbers harbor works ceramic cargo and seabed stratigraphy. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Baiae and Portus. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that commercial salvage destroys context needed for historical explanation. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Baiae, Portus, Berenike. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Maritime and underwater archaeology

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of wreck timbers harbor works ceramic cargo and seabed stratigraphy; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Baiae or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Portus or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Berenike or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Baiae or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Portus or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for maritime and underwater archaeology supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of commercial salvage destroys context needed for historical explanation. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Maritime and underwater archaeology

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For maritime and underwater archaeology, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Baiae, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Portus, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Berenike, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether commercial salvage destroys context needed for historical explanation and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Historic England, Scientific Dating. Chronometric dating methods.
https://historicengland.org.uk/advice/technical-advice/archaeological-science/scientific-dating/

2. Historic England, Radiocarbon Dating and Chronological Modelling. Radiocarbon best practice and bayesian analysis.
https://historicengland.org.uk/images-books/publications/radiocarbon-dating-chronological-modelling

3. Historic England, Pleistocene scientific dating. Deep time dating and methodological limitations.
https://historicengland.org.uk/images-books/publications/scientific-dating-pleistocene-sites/scientific-dating-methods/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in wreck timbers harbor works ceramic cargo and seabed stratigraphy. To move beyond an orientation chapter, consult excavation volumes for Baiae, Portus, Berenike, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Survey, mapping, GIS, and satellite data

SCOPE AND SIGNIFICANCE

Survey, mapping, GIS, and satellite data is an essential problem in spatial archaeology. The central record involves surface collections aerial imagery GIS satellite remote sensing and spatial statistics. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with surface collections aerial imagery GIS satellite remote sensing and spatial statistics. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Angkor Thom and Acre Geoglyphs. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that survey visibility varies by land cover erosion and land management. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Angkor Thom, Acre Geoglyphs, Cahokia. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Survey, mapping, GIS, and satellite data

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of surface collections aerial imagery GIS satellite remote sensing and spatial statistics; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Angkor Thom or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Acre Geoglyphs or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Cahokia or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Angkor Thom or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Acre Geoglyphs or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for survey, mapping, gis, and satellite data supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of survey visibility varies by land cover erosion and land management. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Survey, mapping, GIS, and satellite data

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For survey, mapping, gis, and satellite data, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Angkor Thom, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Acre Geoglyphs, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Cahokia, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether survey visibility varies by land cover erosion and land management and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Society for American Archaeology, Ethics Principles. Professional stewardship and responsibilities. https://www.saa.org/Member/SAAMember/Career-and-Practice/Principles-of-Archaeological-Ethics.aspx

2. ICOMOS, Archaeological Heritage Charter (1990). Preservation and management of sites. https://www.icomos.de/data/pdf/xx-0421-1300-25.pdf

3. UNESCO, Underwater Cultural Heritage Convention. Underwater heritage and anti-looting principles. https://www.unesco.org/en/underwater-heritage/2001-convention

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in surface collections aerial imagery GIS satellite remote sensing and spatial statistics. To move beyond an orientation chapter, consult excavation volumes for Angkor Thom, Acre Geoglyphs, Cahokia, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

LiDAR, photogrammetry, and 3D recording

SCOPE AND SIGNIFICANCE

LiDAR, photogrammetry, and 3D recording is an essential problem in digital field methods. The central record involves point clouds orthophotography remote sensing terrain models and digital twins. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with point clouds orthophotography remote sensing terrain models and digital twins. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Caracol and Upano Valley sites. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that a remotely mapped feature remains a hypothesis until suitably verified. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Caracol, Upano Valley sites, Angkor Wat. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

LiDAR, photogrammetry, and 3D recording

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of point clouds orthophotography remote sensing terrain models and digital twins; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Caracol or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Upano Valley sites or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Angkor Wat or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Caracol or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Upano Valley sites or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for lidar, photogrammetry, and 3d recording supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of a remotely mapped feature remains a hypothesis until suitably verified. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

LiDAR, photogrammetry, and 3D recording

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For lidar, photogrammetry, and 3d recording, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Caracol, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Upano Valley sites, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Angkor Wat, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether a remotely mapped feature remains a hypothesis until suitably verified and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Göbekli Tepe. Documented early monumental enclosures. https://whc.unesco.org/en/list/1572

2. UNESCO: Çatalhöyük. Neolithic settlement and occupation. https://whc.unesco.org/en/list/1405/

3. UNESCO: Angkor. Khmer urban and hydraulic landscape. https://whc.unesco.org/en/list/668/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in point clouds orthophotography remote sensing terrain models and digital twins. To move beyond an orientation chapter, consult excavation volumes for Caracol, Upano Valley sites, Angkor Wat, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Excavation recording and stratigraphic reasoning

SCOPE AND SIGNIFICANCE

Excavation recording and stratigraphic reasoning is an essential problem in field methods. The central record involves context sheets Harris matrices drawings section profiles and total station records. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with context sheets Harris matrices drawings section profiles and total station records. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Pompeii and Tell Brak. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that excavation is destructive so documentation is the reproducible scientific archive. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Pompeii, Tell Brak, Harappa. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Excavation recording and stratigraphic reasoning

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of context sheets Harris matrices drawings section profiles and total station records; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Pompeii or a comparable setting, prepare a context recording scheme and sampling grid. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Tell Brak or a comparable setting, record stratigraphic cuts, fills and interfaces. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Harappa or a comparable setting, link finds to georeferenced context IDs. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Pompeii or a comparable setting, retain raw images plans and section records. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Tell Brak or a comparable setting, produce a reproducible archive and deposition plan. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact	Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval	Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives	Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims	Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for excavation recording and stratigraphic reasoning supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of excavation is destructive so documentation is the reproducible scientific archive. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Excavation recording and stratigraphic reasoning

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For excavation recording and stratigraphic reasoning, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Pompeii, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Tell Brak, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Harappa, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether excavation is destructive so documentation is the reproducible scientific archive and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Great Zimbabwe. Southern african polity and monument. https://whc.unesco.org/en/list/364/

2. UNESCO: Cahokia. Mississippian settlement and earthworks. https://whc.unesco.org/en/list/198

3. UNESCO: Rapa Nui. Polynesian archaeological landscape. https://whc.unesco.org/en/list/715/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in context sheets Harris matrices drawings section profiles and total station records. To move beyond an orientation chapter, consult excavation volumes for Pompeii, Tell Brak, Harappa, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Radiocarbon dating and chronological modelling

SCOPE AND SIGNIFICANCE

Radiocarbon dating and chronological modelling is an essential problem in chronometric methods. The central record involves AMS radiocarbon calibrated age distributions reservoir effects and Bayesian sequences. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with AMS radiocarbon calibrated age distributions reservoir effects and Bayesian sequences. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Kebara Cave and Çatalhöyük. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that a radiocarbon measurement dates a sample event not automatically the archaeological event. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Kebara Cave, Çatalhöyük, Chaco Canyon. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Radiocarbon dating and chronological modelling

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of AMS radiocarbon calibrated age distributions reservoir effects and Bayesian sequences; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Kebara Cave or a comparable setting, identify what physical event a measurement dates. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Çatalhöyük or a comparable setting, sample materials using clean protocols and metadata. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Chaco Canyon or a comparable setting, use independent laboratories or replication where appropriate. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Kebara Cave or a comparable setting, model stratigraphic order and probability distributions. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Çatalhöyük or a comparable setting, document reservoir, contamination and inheritance effects. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for radiocarbon dating and chronological modelling supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of a radiocarbon measurement dates a sample event not automatically the archaeological event. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Radiocarbon dating and chronological modelling

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For radiocarbon dating and chronological modelling, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Kebara Cave, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Çatalhöyük, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Chaco Canyon, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether a radiocarbon measurement dates a sample event not automatically the archaeological event and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO World Heritage Centre, World Heritage List. A selective heritage register, not a global excavation census. https://whc.unesco.org/en/list/

2. UNESCO, World Heritage selection criteria. Heritage assessment and significance frameworks. https://whc.unesco.org/en/criteria/

3. Smithsonian Human Origins, stone tools. Early lithic technologies and their interpretation. https://humanorigins.si.edu/evidence/behavior/stone-tools

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in AMS radiocarbon calibrated age distributions reservoir effects and Bayesian sequences. To move beyond an orientation chapter, consult excavation volumes for Kebara Cave, Çatalhöyük, Chaco Canyon, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Other chronometric methods

SCOPE AND SIGNIFICANCE

Other chronometric methods is an essential problem in chronometric methods. The central record involves dendrochronology luminescence uranium series paleomagnetism and tephra. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with dendrochronology luminescence uranium series paleomagnetism and tephra. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Atapuerca and Hadrian's Wall. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that each clock dates a physical process with material and context constraints. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Atapuerca, Hadrian's Wall, Laetoli. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Other chronometric methods

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of dendrochronology luminescence uranium series paleomagnetism and tephra; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Atapuerca or a comparable setting, identify what physical event a measurement dates. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Hadrian's Wall or a comparable setting, sample materials using clean protocols and metadata. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Laetoli or a comparable setting, use independent laboratories or replication where appropriate. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Atapuerca or a comparable setting, model stratigraphic order and probability distributions. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Hadrian's Wall or a comparable setting, document reservoir, contamination and inheritance effects. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for other chronometric methods supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of each clock dates a physical process with material and context constraints. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Other chronometric methods

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For other chronometric methods, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Atapuerca, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Hadrian's Wall, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Laetoli, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether each clock dates a physical process with material and context constraints and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Harmand et al., Lomekwi 3, Nature (2015). Primary report on 3.3-million-year-old lithics.
https://www.nature.com/articles/nature14464

2. Plummer et al., first million years of technology (2025). Peer-reviewed synthetic review.
https://doi.org/10.1146/annurev-anthro-071923-112250

3. Historic England, Archaeological Methods. Excavation and archaeological science.
https://historicengland.org.uk/research/methods/archaeology/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in dendrochronology luminescence uranium series paleomagnetism and tephra. To move beyond an orientation chapter, consult excavation volumes for Atapuerca, Hadrian's Wall, Laetoli, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Experimental archaeology and use-wear

SCOPE AND SIGNIFICANCE

Experimental archaeology and use-wear is an essential problem in experimental archaeology. The central record involves replica tools controlled experiments fracture patterns microwear and residue comparison. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with replica tools controlled experiments fracture patterns microwear and residue comparison. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Olduvai Gorge and Liang Bua. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that replication establishes feasibility not proof of the past activity. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Olduvai Gorge, Liang Bua, Gona. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Experimental archaeology and use-wear

Chapter 33 • Methods and replication

II | FIELD AND LABORATORY DESIGN

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of replica tools controlled experiments fracture patterns microwear and residue comparison; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Olduvai Gorge or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Liang Bua or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Gona or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Olduvai Gorge or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Liang Bua or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for experimental archaeology and use-wear supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of replication establishes feasibility not proof of the past activity. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Experimental archaeology and use-wear

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For experimental archaeology and use-wear, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Olduvai Gorge, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Liang Bua, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Gona, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether replication establishes feasibility not proof of the past activity and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Historic England, Scientific Dating. Chronometric dating methods.
https://historicengland.org.uk/advice/technical-advice/archaeological-science/scientific-dating/

2. Historic England, Radiocarbon Dating and Chronological Modelling. Radiocarbon best practice and bayesian analysis.
https://historicengland.org.uk/images-books/publications/radiocarbon-dating-chronological-modelling

3. Historic England, Pleistocene scientific dating. Deep time dating and methodological limitations.
https://historicengland.org.uk/images-books/publications/scientific-dating-pleistocene-sites/scientific-dating-methods/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in replica tools controlled experiments fracture patterns microwear and residue comparison. To move beyond an orientation chapter, consult excavation volumes for Olduvai Gorge, Liang Bua, Gona, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Bioarchaeology and health

SCOPE AND SIGNIFICANCE

Bioarchaeology and health is an essential problem in bioarchaeology. The central record involves osteology paleopathology dental disease trauma isotope chemistry and demography. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with osteology paleopathology dental disease trauma isotope chemistry and demography. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Sutton Hoo and Taforalt Cave. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that skeletal assemblages are selective samples not complete ancient populations. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Sutton Hoo, Taforalt Cave, Kerma. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Bioarchaeology and health

Chapter 34 • Methods and replication

II | FIELD AND LABORATORY DESIGN

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of osteology paleopathology dental disease trauma isotope chemistry and demography; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Sutton Hoo or a comparable setting, document sample provenance and depositional history. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Taforalt Cave or a comparable setting, check preservation and contamination controls. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Kerma or a comparable setting, compare measurements across independent samples. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Sutton Hoo or a comparable setting, test for demographic and survivorship bias. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Taforalt Cave or a comparable setting, deposit data with community-approved access conditions. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for bioarchaeology and health supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of skeletal assemblages are selective samples not complete ancient populations. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Bioarchaeology and health

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For bioarchaeology and health, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Sutton Hoo, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Taforalt Cave, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Kerma, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether skeletal assemblages are selective samples not complete ancient populations and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Society for American Archaeology, Ethics Principles. Professional stewardship and responsibilities. https://www.saa.org/Member/SAAMember/Career-and-Practice/Principles-of-Archaeological-Ethics.aspx

2. ICOMOS, Archaeological Heritage Charter (1990). Preservation and management of sites. https://www.icomos.de/data/pdf/xx-0421-1300-25.pdf

3. UNESCO, Underwater Cultural Heritage Convention. Underwater heritage and anti-looting principles. https://www.unesco.org/en/underwater-heritage/2001-convention

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in osteology paleopathology dental disease trauma isotope chemistry and demography. To move beyond an orientation chapter, consult excavation volumes for Sutton Hoo, Taforalt Cave, Kerma, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Heritage ethics and descendant communities

SCOPE AND SIGNIFICANCE

Heritage ethics and descendant communities is an essential problem in heritage stewardship. The central record involves consultation stewardship human-remains care repatriation and research governance. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with consultation stewardship human-remains care repatriation and research governance. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Caguana Ceremonial Park and Shum Laka. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that research access does not supersede legal rights or community obligations. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Caguana Ceremonial Park, Shum Laka, Rapa Nui (Rano Raraku). These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Heritage ethics and descendant communities

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of consultation stewardship human-remains care repatriation and research governance; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Caguana Ceremonial Park or a comparable setting, identify legitimate custodians communities and legal protections. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Shum Laka or a comparable setting, consult before destructive sampling or display. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Rapa Nui (Rano Raraku) or a comparable setting, agree accessible curation and data access conditions. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Caguana Ceremonial Park or a comparable setting, minimize physical intervention and photograph change. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Shum Laka or a comparable setting, publish decisions and preserve rights of review. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for heritage ethics and descendant communities supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of research access does not supersede legal rights or community obligations. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Heritage ethics and descendant communities

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For heritage ethics and descendant communities, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Caguana Ceremonial Park, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Shum Laka, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Rapa Nui (Rano Raraku), audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether research access does not supersede legal rights or community obligations and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Göbekli Tepe. Documented early monumental enclosures. https://whc.unesco.org/en/list/1572

2. UNESCO: Çatalhöyük. Neolithic settlement and occupation. https://whc.unesco.org/en/list/1405/

3. UNESCO: Angkor. Khmer urban and hydraulic landscape. https://whc.unesco.org/en/list/668/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in consultation stewardship human-remains care repatriation and research governance. To move beyond an orientation chapter, consult excavation volumes for Caguana Ceremonial Park, Shum Laka, Rapa Nui (Rano Raraku), check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Illicit antiquities, restitution, and provenance

SCOPE AND SIGNIFICANCE

Illicit antiquities, restitution, and provenance is an essential problem in heritage protection. The central record involves collection histories export permits inventories object chains of custody and damage reports. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with collection histories export permits inventories object chains of custody and damage reports. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Nimrud and Carthage. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that unprovenanced finds lose relationships that matter more than aesthetic value. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Nimrud, Carthage, Benin City Earthworks. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Illicit antiquities, restitution, and provenance

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of collection histories export permits inventories object chains of custody and damage reports; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Nimrud or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Carthage or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Benin City Earthworks or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Nimrud or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Carthage or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for illicit antiquities, restitution, and provenance supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of unprovenanced finds lose relationships that matter more than aesthetic value. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Illicit antiquities, restitution, and provenance

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For illicit antiquities, restitution, and provenance, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Nimrud, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Carthage, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Benin City Earthworks, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether unprovenanced finds lose relationships that matter more than aesthetic value and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO: Great Zimbabwe. Southern african polity and monument. https://whc.unesco.org/en/list/364/

2. UNESCO: Cahokia. Mississippian settlement and earthworks. https://whc.unesco.org/en/list/198

3. UNESCO: Rapa Nui. Polynesian archaeological landscape. https://whc.unesco.org/en/list/715/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in collection histories export permits inventories object chains of custody and damage reports. To move beyond an orientation chapter, consult excavation volumes for Nimrud, Carthage, Benin City Earthworks, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Conservation, climate, and disaster risks

SCOPE AND SIGNIFICANCE

Conservation, climate, and disaster risks is an essential problem in conservation. The central record involves salt corrosion temperature fluctuation flooding wildfire erosion and conflict damage. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with salt corrosion temperature fluctuation flooding wildfire erosion and conflict damage. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Pompeii and Timbuktu. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that display and tourism can accelerate deterioration if unmanaged. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Pompeii, Timbuktu, Rapa Nui (Rano Raraku). These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Conservation, climate, and disaster risks

Chapter 37 • Methods and replication

II | FIELD AND LABORATORY DESIGN

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of salt corrosion temperature fluctuation flooding wildfire erosion and conflict damage; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Pompeii or a comparable setting, baseline measure deterioration and site microclimate. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Timbuktu or a comparable setting, identify threats through repeat photographic and sensor surveys. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Rapa Nui (Rano Raraku) or a comparable setting, prioritize reversible low-impact interventions. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Pompeii or a comparable setting, assess visitor pressure and emergency access. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Timbuktu or a comparable setting, monitor the effect of conservation over time. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact	Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval	Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives	Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims	Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for conservation, climate, and disaster risks supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of display and tourism can accelerate deterioration if unmanaged. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Conservation, climate, and disaster risks

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For conservation, climate, and disaster risks, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Pompeii, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Timbuktu, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Rapa Nui (Rano Raraku), audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether display and tourism can accelerate deterioration if unmanaged and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. UNESCO World Heritage Centre, World Heritage List. A selective heritage register, not a global excavation census. https://whc.unesco.org/en/list/

2. UNESCO, World Heritage selection criteria. Heritage assessment and significance frameworks. https://whc.unesco.org/en/criteria/

3. Smithsonian Human Origins, stone tools. Early lithic technologies and their interpretation. https://humanorigins.si.edu/evidence/behavior/stone-tools

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in salt corrosion temperature fluctuation flooding wildfire erosion and conflict damage. To move beyond an orientation chapter, consult excavation volumes for Pompeii, Timbuktu, Rapa Nui (Rano Raraku), check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Research archives, open data, and reproducibility

SCOPE AND SIGNIFICANCE

Research archives, open data, and reproducibility is an essential problem in data stewardship. The central record involves catalogues excavation databases image archives lab data provenance and durable identifiers. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with catalogues excavation databases image archives lab data provenance and durable identifiers. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Harappa and Çatalhöyük. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that orphaned spreadsheets and undocumented transformations impede replication. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Harappa, Çatalhöyük, Chaco Canyon. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

Research archives, open data, and reproducibility

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of catalogues excavation databases image archives lab data provenance and durable identifiers; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Harappa or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Çatalhöyük or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Chaco Canyon or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Harappa or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Çatalhöyük or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for research archives, open data, and reproducibility supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of orphaned spreadsheets and undocumented transformations impede replication. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

Research archives, open data, and reproducibility

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For research archives, open data, and reproducibility, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Harappa, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Çatalhöyük, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Chaco Canyon, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether orphaned spreadsheets and undocumented transformations impede replication and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Harmand et al., Lomekwi 3, Nature (2015). Primary report on 3.3-million-year-old lithics.
https://www.nature.com/articles/nature14464

2. Plummer et al., first million years of technology (2025). Peer-reviewed synthetic review.
https://doi.org/10.1146/annurev-anthro-071923-112250

3. Historic England, Archaeological Methods. Excavation and archaeological science.
https://historicengland.org.uk/research/methods/archaeology/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in catalogues excavation databases image archives lab data provenance and durable identifiers. To move beyond an orientation chapter, consult excavation volumes for Harappa, Çatalhöyük, Chaco Canyon, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

How to evaluate a contested claim

SCOPE AND SIGNIFICANCE

How to evaluate a contested claim is an essential problem in critical reasoning. The central record involves independent dating secure stratigraphy replication transparent uncertainty and publication scrutiny. Archaeologists study the frequency, context, date, spatial organization and preservation of these materials in order to reconstruct processes that are no longer directly observable. Each conclusion should identify what was measured, the temporal scale of the inference, and which alternative readings remain compatible with the evidence.

INTERPRETIVE ARCHITECTURE

OBSERVATION

Begin with independent dating secure stratigraphy replication transparent uncertainty and publication scrutiny. Record the physical object or feature and its find context separately from the interpretation attributed to it. Sampling strategies should make negative results meaningful.

SEQUENCE

Ask whether patterns belong to one event, a long-duration occupation, or several later uses. Relative dating and scientific chronometry must be reconciled rather than treated as competing authorities.

COMPARISON

Set the local dataset beside examples such as Gunung Padang and Rising Star Cave. Similar forms can reflect shared ancestry, exchange, parallel solutions or unrelated reuse; choose tests that can tell these apart.

LIMITATION

A central warning in this field is that extraordinary dates or sensational claims should not bypass ordinary archaeological standards. Even a strong hypothesis should identify the conditions under which it would be revised or rejected.

CROSS-REGIONAL CASES

Working comparisons: Gunung Padang, Rising Star Cave, Lothal. These examples represent different research settings; they do not form a single evolutionary ladder. Compare environmental context, chronology, sample size, excavation history, curatorial status and who has authority to interpret or manage the archaeological materials.

WHAT WOULD CHANGE THE CONCLUSION?

A newly dated closed context, a laboratory reanalysis, or a complete record of an overlooked occupation phase could revise a familiar account. The strongest review specifies in advance what evidence would support or contradict a proposed explanation, and what could remain unresolved after further excavation.

How to evaluate a contested claim

A REPRODUCIBLE INVESTIGATION

The aim is not simply to accumulate more examples of independent dating secure stratigraphy replication transparent uncertainty and publication scrutiny; it is to resolve an explicit historical question with measurable observations. A study plan needs the sampling universe, inclusion and exclusion rules, recording conventions, chronological strategy, analytical chain, conservation plan and reporting destination. Select non-destructive alternatives first where these can answer the question adequately.

PROTOCOL 01

At Gunung Padang or a comparable setting, define the research question at a testable geographic and temporal scale. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 02

At Rising Star Cave or a comparable setting, compile excavation data with consistent context identifiers. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 03

At Lothal or a comparable setting, apply one independent method to check the central inference. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 04

At Gunung Padang or a comparable setting, compare rival explanations and quantifiable uncertainties. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

PROTOCOL 05

At Rising Star Cave or a comparable setting, archive the evidence and publish a transparent synthesis. Log context IDs, measurements, assumptions, controls and potential contradictory evidence. Repeat the analysis with independent data where feasible.

CRITICAL EVIDENCE MATRIX

Observation	Measured material or spatial fact Site/context ID; instrument; recovery conditions
Dating	Secure order or calibrated age interval Direct and indirect dates; uncertainty; sample event
Inference	Historical explanation with alternatives Predictions; comparison sample; rival hypotheses
Archive	Reproducible provenance of claims Raw files; codes; permits; collection location

REPORTING QUALITY

A credible final report for how to evaluate a contested claim supplies context plans, artifact catalogues, analytical results, uncertainty statements and a reasoned discussion of extraordinary dates or sensational claims should not bypass ordinary archaeological standards. If evidence is not accessible for scrutiny, qualify the conclusion instead of using false precision.

How to evaluate a contested claim

COMPETING EXPLANATIONS

A sound account should not assume that the most vivid story is the best-supported one. For how to evaluate a contested claim, begin with at least three families of models: continuity and local development; interaction, borrowing or population movement; and apparent change created by preservation, sampling or site-formation processes. These are methodological possibilities, not an assertion that particular sites actually fit all three models.

ILLUSTRATIVE CRITICAL TESTS

- At Gunung Padang, seek a well-stratified sequence that can discriminate gradual change from episodic remodeling.
- At Rising Star Cave, compare independent chronology and regional material networks rather than dating solely from visual resemblance.
- At Lothal, audit recovery history for missing contexts, conservation bias and later interventions.
- For all cases, test whether extraordinary dates or sensational claims should not bypass ordinary archaeological standards and identify the minimum evidence needed to make a narrower claim.

STEWARDSHIP AND INTELLECTUAL RESPONSIBILITY

Sites and collections are finite resources. The desirability of an explanation cannot justify unpermitted excavation, destructive sampling without a defensible question, traffic in unprovenanced artifacts, or exclusion of communities with legitimate interests in a place. Curatorial infrastructure, accessible reporting and respectful consultation are part of archaeological knowledge production, not afterthoughts.

READING PATHWAY

1. Historic England, Scientific Dating. Chronometric dating methods.
https://historicengland.org.uk/advice/technical-advice/archaeological-science/scientific-dating/

2. Historic England, Radiocarbon Dating and Chronological Modelling. Radiocarbon best practice and bayesian analysis.
https://historicengland.org.uk/images-books/publications/radiocarbon-dating-chronological-modelling

3. Historic England, Pleistocene scientific dating. Deep time dating and methodological limitations.
https://historicengland.org.uk/images-books/publications/scientific-dating-pleistocene-sites/scientific-dating-methods/

SYNTHESIS AND UNRESOLVED QUESTIONS

The objective is a bounded, revisable explanation grounded in independent dating secure stratigraphy replication transparent uncertainty and publication scrutiny. To move beyond an orientation chapter, consult excavation volumes for Gunung Padang, Rising Star Cave, Lothal, check subsequent peer-reviewed corrections, obtain original dating tables and review any claims made without secure context. A detailed site-level bibliography must be assembled separately for research publication.

EVIDENCE LABELS USED IN THIS VOLUME

Observed = directly documented object or context; Inferred = reasoned explanation consistent with observations; Proposed = question or working model awaiting adequate tests; Contested = published interpretations have not converged. These labels refer to the type of claim, not to the reputation of an excavation team.

Lomekwi 3

East African Human Origins • Evidence dossier 1 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kenya	Pliocene	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Lomekwi 3 is a site or landscape associated with pliocene research in Kenya. The defining record highlighted in this atlas is large stone artifacts and early knapping. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large stone artifacts and early knapping
Chronological frame	Pliocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large stone artifacts and early knapping linked to the cited phase of Pliocene?
- What natural and human processes altered deposits at Lomekwi 3 after original use?
- Which nearby sites, especially Laetoli, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lomekwi 3 with Laetoli, Bodo D'ar and Nariokotome. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Sonia Harmand and colleagues reported in 2015 a distinctive assemblage dated to around 3.3 million years ago, substantially earlier than classic Oldowan industries.

RESEARCH POINTER: https://www.nature.com/articles/nature14464

Lomekwi 3

Kenya • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Lomekwi 3 begins with a bounded claim. For the material described here as large stone artifacts and early knapping, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lomekwi 3 using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bodo D'ar and Nariokotome using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nyayanga

East African Human Origins • Evidence dossier 2 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kenya	Early Stone Age	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Nyayanga is a site or landscape associated with early stone age research in Kenya. The defining record highlighted in this atlas is Oldowan implements and butchered animal remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Oldowan implements and butchered animal remains
Chronological frame	Early Stone Age; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Oldowan implements and butchered animal remains linked to the cited phase of Early Stone Age?
- What natural and human processes altered deposits at Nyayanga after original use?
- Which nearby sites, especially Hadar, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nyayanga with Hadar, Omo Kibish and Kariandusi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

A 2023 study led by Tom Plummer and colleagues reported Oldowan tools near cut-marked animal bones and Paranthropus teeth; association does not identify the maker.

RESEARCH POINTER: https://humanorigins.si.edu/research/whats-hot-human-origins/oldowan-tools-3-million-years-ago

Nyayanga

Kenya • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nyayanga begins with a bounded claim. For the material described here as Oldowan implements and butchered animal remains, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nyayanga using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Omo Kibish and Kariandusi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Olduvai Gorge

East African Human Origins • Evidence dossier 3 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tanzania	**Pleistocene**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Olduvai Gorge is a site or landscape associated with pleistocene research in Tanzania. The defining record highlighted in this atlas is Oldowan and Acheulean stone industries. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Oldowan and Acheulean stone industries
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Oldowan and Acheulean stone industries linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Olduvai Gorge after original use?
- Which nearby sites, especially Dikika, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Olduvai Gorge with Dikika, Herto and Isimila. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Louis and Mary Leakey and later researchers made the gorge pivotal for linking stone technology with hominin remains and a long Pleistocene sedimentary sequence.

Olduvai Gorge

Tanzania • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Olduvai Gorge begins with a bounded claim. For the material described here as Oldowan and Acheulean stone industries, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Olduvai Gorge using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Herto and Isimila using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Laetoli

East African Human Origins • Evidence dossier 4 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tanzania	**Pliocene**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Laetoli is a site or landscape associated with pliocene research in Tanzania. The defining record highlighted in this atlas is fossil hominin footprints in volcanic ash. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fossil hominin footprints in volcanic ash
Chronological frame	Pliocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fossil hominin footprints in volcanic ash linked to the cited phase of Pliocene?
- What natural and human processes altered deposits at Laetoli after original use?
- Which nearby sites, especially Ledi-Geraru, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Laetoli with Ledi-Geraru, Konso and Mumba Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Mary Leakey's team documented fossil hominin footprint trails, widely used to investigate early bipedal locomotion.

Laetoli

Tanzania • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Laetoli begins with a bounded claim. For the material described here as fossil hominin footprints in volcanic ash, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Laetoli using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Konso and Mumba Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hadar

East African Human Origins • Evidence dossier 5 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Pliocene**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Hadar is a site or landscape associated with pliocene research in Ethiopia. The defining record highlighted in this atlas is Australopithecus afarensis fossil record. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Australopithecus afarensis fossil record
Chronological frame	Pliocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Australopithecus afarensis fossil record linked to the cited phase of Pliocene?
- What natural and human processes altered deposits at Hadar after original use?
- Which nearby sites, especially Gona, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hadar with Gona, Olorgesailie and Kondoa Rock Art. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Donald Johanson and colleagues' 1974 discovery of the Australopithecus afarensis skeleton popularly called Lucy shaped debate about bipedalism.

Hadar

Ethiopia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Hadar begins with a bounded claim. For the material described here as Australopithecus afarensis fossil record, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hadar using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Olorgesailie and Kondoa Rock Art using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dikika

East African Human Origins • Evidence dossier 6 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Pliocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Dikika is a site or landscape associated with pliocene research in Ethiopia. The defining record highlighted in this atlas is juvenile australopith remains and early behavior debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	juvenile australopith remains and early behavior debates
Chronological frame	Pliocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are juvenile australopith remains and early behavior debates linked to the cited phase of Pliocene?
- What natural and human processes altered deposits at Dikika after original use?
- Which nearby sites, especially Melka Kunture, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dikika with Melka Kunture, Koobi Fora and Lomekwi 3. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dikika

Ethiopia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dikika begins with a bounded claim. For the material described here as juvenile australopith remains and early behavior debates, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dikika using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Koobi Fora and Lomekwi 3 using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ledi-Geraru

East African Human Origins • Evidence dossier 7 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Pliocene-Pleistocene**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Ledi-Geraru is a site or landscape associated with pliocene-pleistocene research in Ethiopia. The defining record highlighted in this atlas is early Homo fossils and Oldowan artifacts. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Homo fossils and Oldowan artifacts
Chronological frame	Pliocene-Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Homo fossils and Oldowan artifacts linked to the cited phase of Pliocene-Pleistocene?
- What natural and human processes altered deposits at Ledi-Geraru after original use?
- Which nearby sites, especially Bodo D'ar, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ledi-Geraru with Bodo D'ar, Nariokotome and Nyayanga. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ledi-Geraru

Ethiopia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ledi-Geraru begins with a bounded claim. For the material described here as early Homo fossils and Oldowan artifacts, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ledi-Geraru using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nariokotome and Nyayanga using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Gona

East African Human Origins • Evidence dossier 8 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Early Stone Age**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Gona is a site or landscape associated with early stone age research in Ethiopia. The defining record highlighted in this atlas is early flaked stone assemblages. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early flaked stone assemblages
Chronological frame	Early Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early flaked stone assemblages linked to the cited phase of Early Stone Age?
- What natural and human processes altered deposits at Gona after original use?
- Which nearby sites, especially Omo Kibish, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Gona with Omo Kibish, Kariandusi and Olduvai Gorge. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Early Stone Age finds became a key comparative sequence for understanding the antiquity and spread of Oldowan stoneworking.

Gona

Ethiopia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Gona begins with a bounded claim. For the material described here as early flaked stone assemblages, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Gona using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kariandusi and Olduvai Gorge using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Melka Kunture

East African Human Origins • Evidence dossier 9 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Pleistocene**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Melka Kunture is a site or landscape associated with pleistocene research in Ethiopia. The defining record highlighted in this atlas is multi-period lithic workshops and occupation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	multi-period lithic workshops and occupation
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are multi-period lithic workshops and occupation linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Melka Kunture after original use?
- Which nearby sites, especially Herto, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Melka Kunture with Herto, Isimila and Laetoli. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Melka Kunture

Ethiopia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Melka Kunture begins with a bounded claim. For the material described here as multi-period lithic workshops and occupation, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Melka Kunture using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Isimila and Laetoli using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bodo D'ar

East African Human Origins • Evidence dossier 10 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Middle Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Bodo D'ar is a site or landscape associated with middle pleistocene research in Ethiopia. The defining record highlighted in this atlas is archaic human cranial remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	archaic human cranial remains
Chronological frame	Middle Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are archaic human cranial remains linked to the cited phase of Middle Pleistocene?
- What natural and human processes altered deposits at Bodo D'ar after original use?
- Which nearby sites, especially Konso, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bodo D'ar with Konso, Mumba Cave and Hadar. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Bodo D'ar

Ethiopia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Bodo D'ar begins with a bounded claim. For the material described here as archaic human cranial remains, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bodo D'ar using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mumba Cave and Hadar using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Omo Kibish

East African Human Origins • Evidence dossier 11 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Middle Stone Age**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Omo Kibish is a site or landscape associated with middle stone age research in Ethiopia. The defining record highlighted in this atlas is early Homo sapiens remains and stratigraphy. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Homo sapiens remains and stratigraphy
Chronological frame	Middle Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Homo sapiens remains and stratigraphy linked to the cited phase of Middle Stone Age?
- What natural and human processes altered deposits at Omo Kibish after original use?
- Which nearby sites, especially Olorgesailie, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Omo Kibish with Olorgesailie, Kondoa Rock Art and Dikika. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Omo Kibish

Ethiopia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Omo Kibish begins with a bounded claim. For the material described here as early Homo sapiens remains and stratigraphy, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Omo Kibish using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kondoa Rock Art and Dikika using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Herto

East African Human Origins • Evidence dossier 12 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Middle Pleistocene**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Herto is a site or landscape associated with middle pleistocene research in Ethiopia. The defining record highlighted in this atlas is early Homo sapiens fossils. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Homo sapiens fossils
Chronological frame	Middle Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Homo sapiens fossils linked to the cited phase of Middle Pleistocene?
- What natural and human processes altered deposits at Herto after original use?
- Which nearby sites, especially Koobi Fora, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Herto with Koobi Fora, Lomekwi 3 and Ledi-Geraru. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Herto

Ethiopia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Herto begins with a bounded claim. For the material described here as early Homo sapiens fossils, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Herto using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lomekwi 3 and Ledi-Geraru using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Konso

East African Human Origins • Evidence dossier 13 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ethiopia	**Early Pleistocene**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Konso is a site or landscape associated with early pleistocene research in Ethiopia. The defining record highlighted in this atlas is Acheulean handaxe sequences. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Acheulean handaxe sequences
Chronological frame	Early Pleistocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Acheulean handaxe sequences linked to the cited phase of Early Pleistocene?
- What natural and human processes altered deposits at Konso after original use?
- Which nearby sites, especially Nariokotome, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Konso with Nariokotome, Nyayanga and Gona. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Konso

Ethiopia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Konso begins with a bounded claim. For the material described here as Acheulean handaxe sequences, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Konso using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nyayanga and Gona using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Olorgesailie

East African Human Origins • Evidence dossier 14 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kenya	Pleistocene	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Olorgesailie is a site or landscape associated with pleistocene research in Kenya. The defining record highlighted in this atlas is Acheulean landscapes and Middle Stone Age transition. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Acheulean landscapes and Middle Stone Age transition
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Acheulean landscapes and Middle Stone Age transition linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Olorgesailie after original use?
- Which nearby sites, especially Kariandusi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Olorgesailie with Kariandusi, Olduvai Gorge and Melka Kunture. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Olorgesailie

Kenya • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Olorgesailie begins with a bounded claim. For the material described here as Acheulean landscapes and Middle Stone Age transition, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Olorgesailie using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Olduvai Gorge and Melka Kunture using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Koobi Fora

East African Human Origins • Evidence dossier 15 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kenya	Pleistocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Koobi Fora is a site or landscape associated with pleistocene research in Kenya. The defining record highlighted in this atlas is hominin fossils and stone assemblages. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	hominin fossils and stone assemblages
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are hominin fossils and stone assemblages linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Koobi Fora after original use?
- Which nearby sites, especially Isimila, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Koobi Fora with Isimila, Laetoli and Bodo D'ar. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Koobi Fora

Kenya • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Koobi Fora begins with a bounded claim. For the material described here as hominin fossils and stone assemblages, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Koobi Fora using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Laetoli and Bodo D'ar using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nariokotome

East African Human Origins • Evidence dossier 16 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kenya	Early Pleistocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Nariokotome is a site or landscape associated with early pleistocene research in Kenya. The defining record highlighted in this atlas is Homo erectus skeleton and environment. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Homo erectus skeleton and environment
Chronological frame	Early Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Homo erectus skeleton and environment linked to the cited phase of Early Pleistocene?
- What natural and human processes altered deposits at Nariokotome after original use?
- Which nearby sites, especially Mumba Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nariokotome with Mumba Cave, Hadar and Omo Kibish. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nariokotome

Kenya • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nariokotome begins with a bounded claim. For the material described here as Homo erectus skeleton and environment, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nariokotome using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hadar and Omo Kibish using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kariandusi

East African Human Origins • Evidence dossier 17 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kenya	**Acheulean**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Kariandusi is a site or landscape associated with acheulean research in Kenya. The defining record highlighted in this atlas is dense handaxe concentrations. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	dense handaxe concentrations
Chronological frame	Acheulean; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are dense handaxe concentrations linked to the cited phase of Acheulean?
- What natural and human processes altered deposits at Kariandusi after original use?
- Which nearby sites, especially Kondoa Rock Art, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kariandusi with Kondoa Rock Art, Dikika and Herto. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kariandusi

Kenya • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Kariandusi begins with a bounded claim. For the material described here as dense handaxe concentrations, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kariandusi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dikika and Herto using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Isimila

East African Human Origins • Evidence dossier 18 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tanzania	**Acheulean**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Isimila is a site or landscape associated with acheulean research in Tanzania. The defining record highlighted in this atlas is stone tools in eroded sediment exposures. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	stone tools in eroded sediment exposures
Chronological frame	Acheulean; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are stone tools in eroded sediment exposures linked to the cited phase of Acheulean?
- What natural and human processes altered deposits at Isimila after original use?
- Which nearby sites, especially Lomekwi 3, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Isimila with Lomekwi 3, Ledi-Geraru and Konso. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Isimila

Tanzania • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Isimila begins with a bounded claim. For the material described here as stone tools in eroded sediment exposures, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Isimila using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ledi-Geraru and Konso using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mumba Cave

East African Human Origins • Evidence dossier 19 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tanzania	**Middle-Later Stone Age**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Mumba Cave is a site or landscape associated with middle-later stone age research in Tanzania. The defining record highlighted in this atlas is long stratified cave occupation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long stratified cave occupation
Chronological frame	Middle-Later Stone Age; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long stratified cave occupation linked to the cited phase of Middle-Later Stone Age?
- What natural and human processes altered deposits at Mumba Cave after original use?
- Which nearby sites, especially Nyayanga, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mumba Cave with Nyayanga, Gona and Olorgesailie. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mumba Cave

Tanzania • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mumba Cave begins with a bounded claim. For the material described here as long stratified cave occupation, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mumba Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Gona and Olorgesailie using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kondoa Rock Art

East African Human Origins • Evidence dossier 20 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tanzania	**Holocene**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Kondoa Rock Art is a site or landscape associated with holocene research in Tanzania. The defining record highlighted in this atlas is rock paintings and cultural landscapes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: east african origins is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock paintings and cultural landscapes
Chronological frame	Holocene; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock paintings and cultural landscapes linked to the cited phase of Holocene?
- What natural and human processes altered deposits at Kondoa Rock Art after original use?
- Which nearby sites, especially Olduvai Gorge, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kondoa Rock Art with Olduvai Gorge, Melka Kunture and Koobi Fora. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kondoa Rock Art

Tanzania • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kondoa Rock Art begins with a bounded claim. For the material described here as rock paintings and cultural landscapes, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kondoa Rock Art using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Melka Kunture and Koobi Fora using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Blombos Cave

Southern Africa and Deep Prehistory • Evidence dossier 21 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Middle Stone Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Blombos Cave is a site or landscape associated with middle stone age research in South Africa. The defining record highlighted in this atlas is engraved ochre and shell beads. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	engraved ochre and shell beads
Chronological frame	Middle Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are engraved ochre and shell beads linked to the cited phase of Middle Stone Age?
- What natural and human processes altered deposits at Blombos Cave after original use?
- Which nearby sites, especially Sibudu Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Blombos Cave with Sibudu Cave, Drimolen and Apollo 11 Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Research led by Christopher Henshilwood and collaborators placed engraved ochre and shell ornaments at the center of discussions of early symbolic behavior.

Blombos Cave

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Blombos Cave begins with a bounded claim. For the material described here as engraved ochre and shell beads, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Blombos Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Drimolen and Apollo 11 Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pinnacle Point

Southern Africa and Deep Prehistory • Evidence dossier 22 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Middle Stone Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Pinnacle Point is a site or landscape associated with middle stone age research in South Africa. The defining record highlighted in this atlas is coastal adaptation and shellfish use. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	coastal adaptation and shellfish use
Chronological frame	Middle Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are coastal adaptation and shellfish use linked to the cited phase of Middle Stone Age?
- What natural and human processes altered deposits at Pinnacle Point after original use?
- Which nearby sites, especially Border Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pinnacle Point with Border Cave, Rising Star Cave and Twyfelfontein. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Pinnacle Point

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Pinnacle Point begins with a bounded claim. For the material described here as coastal adaptation and shellfish use, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pinnacle Point using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Rising Star Cave and Twyfelfontein using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Klasies River Caves

Southern Africa and Deep Prehistory • Evidence dossier 23 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Middle Stone Age	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Klasies River Caves is a site or landscape associated with middle stone age research in South Africa. The defining record highlighted in this atlas is human fossils and occupation layers. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	human fossils and occupation layers
Chronological frame	Middle Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are human fossils and occupation layers linked to the cited phase of Middle Stone Age?
- What natural and human processes altered deposits at Klasies River Caves after original use?
- Which nearby sites, especially Diepkloof Rock Shelter, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Klasies River Caves with Diepkloof Rock Shelter, Makapansgat and Tsodilo Hills. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Klasies River Caves

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Klasies River Caves begins with a bounded claim. For the material described here as human fossils and occupation layers, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Klasies River Caves using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Makapansgat and Tsodilo Hills using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sibudu Cave

Southern Africa and Deep Prehistory • Evidence dossier 24 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Middle Stone Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Sibudu Cave is a site or landscape associated with middle stone age research in South Africa. The defining record highlighted in this atlas is adhesives bedding and complex technologies. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	adhesives bedding and complex technologies
Chronological frame	Middle Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are adhesives bedding and complex technologies linked to the cited phase of Middle Stone Age?
- What natural and human processes altered deposits at Sibudu Cave after original use?
- Which nearby sites, especially Wonderwerk Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sibudu Cave with Wonderwerk Cave, Taung and Kalambo Falls. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sibudu Cave

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sibudu Cave begins with a bounded claim. For the material described here as adhesives bedding and complex technologies, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sibudu Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Taung and Kalambo Falls using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Border Cave

Southern Africa and Deep Prehistory • Evidence dossier 25 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Middle-Later Stone Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Border Cave is a site or landscape associated with middle-later stone age research in South Africa. The defining record highlighted in this atlas is early modern human lifeways. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early modern human lifeways
Chronological frame	Middle-Later Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early modern human lifeways linked to the cited phase of Middle-Later Stone Age?
- What natural and human processes altered deposits at Border Cave after original use?
- Which nearby sites, especially Sterkfontein Caves, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Border Cave with Sterkfontein Caves, Great Zimbabwe and Ishango. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Border Cave

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Border Cave begins with a bounded claim. For the material described here as early modern human lifeways, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Border Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Great Zimbabwe and Ishango using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Diepkloof Rock Shelter

Southern Africa and Deep Prehistory • Evidence dossier 26 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Middle Stone Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Diepkloof Rock Shelter is a site or landscape associated with middle stone age research in South Africa. The defining record highlighted in this atlas is engraved ostrich eggshell containers. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	engraved ostrich eggshell containers
Chronological frame	Middle Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are engraved ostrich eggshell containers linked to the cited phase of Middle Stone Age?
- What natural and human processes altered deposits at Diepkloof Rock Shelter after original use?
- Which nearby sites, especially Swartkrans, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Diepkloof Rock Shelter with Swartkrans, Boomplaas Cave and Blombos Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Diepkloof Rock Shelter

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Diepkloof Rock Shelter begins with a bounded claim. For the material described here as engraved ostrich eggshell containers, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Diepkloof Rock Shelter using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Boomplaas Cave and Blombos Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Wonderwerk Cave

Southern Africa and Deep Prehistory • Evidence dossier 27 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Early Stone Age onward	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Wonderwerk Cave is a site or landscape associated with early stone age onward research in South Africa. The defining record highlighted in this atlas is deep occupation deposits and fire debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	deep occupation deposits and fire debates
Chronological frame	Early Stone Age onward; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are deep occupation deposits and fire debates linked to the cited phase of Early Stone Age onward?
- What natural and human processes altered deposits at Wonderwerk Cave after original use?
- Which nearby sites, especially Drimolen, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Wonderwerk Cave with Drimolen, Apollo 11 Cave and Pinnacle Point. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Wonderwerk Cave

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Wonderwerk Cave begins with a bounded claim. For the material described here as deep occupation deposits and fire debates, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Wonderwerk Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Apollo 11 Cave and Pinnacle Point using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sterkfontein Caves

Southern Africa and Deep Prehistory • Evidence dossier 28 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Pliocene-Pleistocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Sterkfontein Caves is a site or landscape associated with pliocene-pleistocene research in South Africa. The defining record highlighted in this atlas is australopith fossils in karst deposits. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	australopith fossils in karst deposits
Chronological frame	Pliocene-Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are australopith fossils in karst deposits linked to the cited phase of Pliocene-Pleistocene?
- What natural and human processes altered deposits at Sterkfontein Caves after original use?
- Which nearby sites, especially Rising Star Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sterkfontein Caves with Rising Star Cave, Twyfelfontein and Klasies River Caves. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Robert Broom's investigations helped establish South Africa's fossil-rich karst caves as crucial evidence for australopith evolution.

Sterkfontein Caves

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sterkfontein Caves begins with a bounded claim. For the material described here as australopith fossils in karst deposits, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sterkfontein Caves using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Twyfelfontein and Klasies River Caves using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Swartkrans

Southern Africa and Deep Prehistory • Evidence dossier 29 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Pleistocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Swartkrans is a site or landscape associated with pleistocene research in South Africa. The defining record highlighted in this atlas is hominin remains and bone tools. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	hominin remains and bone tools
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are hominin remains and bone tools linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Swartkrans after original use?
- Which nearby sites, especially Makapansgat, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Swartkrans with Makapansgat, Tsodilo Hills and Sibudu Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Swartkrans

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Swartkrans begins with a bounded claim. For the material described here as hominin remains and bone tools, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Swartkrans using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tsodilo Hills and Sibudu Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Drimolen

Southern Africa and Deep Prehistory • Evidence dossier 30 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Early Pleistocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Drimolen is a site or landscape associated with early pleistocene research in South Africa. The defining record highlighted in this atlas is early Homo and Paranthropus fossils. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Homo and Paranthropus fossils
Chronological frame	Early Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Homo and Paranthropus fossils linked to the cited phase of Early Pleistocene?
- What natural and human processes altered deposits at Drimolen after original use?
- Which nearby sites, especially Taung, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Drimolen with Taung, Kalambo Falls and Border Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Drimolen

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Drimolen begins with a bounded claim. For the material described here as early Homo and Paranthropus fossils, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Drimolen using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kalambo Falls and Border Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Rising Star Cave

Southern Africa and Deep Prehistory • Evidence dossier 31 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	**Middle Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Rising Star Cave is a site or landscape associated with middle pleistocene research in South Africa. The defining record highlighted in this atlas is Homo naledi remains and depositional debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Homo naledi remains and depositional debates
Chronological frame	Middle Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Homo naledi remains and depositional debates linked to the cited phase of Middle Pleistocene?
- What natural and human processes altered deposits at Rising Star Cave after original use?
- Which nearby sites, especially Great Zimbabwe, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Rising Star Cave with Great Zimbabwe, Ishango and Diepkloof Rock Shelter. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Lee Berger and collaborators described Homo naledi from an extraordinary concentration of human-relative remains, raising active debates about dating and deposition.

Rising Star Cave

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Rising Star Cave begins with a bounded claim. For the material described here as Homo naledi remains and depositional debates, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Rising Star Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ishango and Diepkloof Rock Shelter using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Makapansgat

Southern Africa and Deep Prehistory • Evidence dossier 32 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Pliocene-Pleistocene	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Makapansgat is a site or landscape associated with pliocene-pleistocene research in South Africa. The defining record highlighted in this atlas is australopith cave assemblages. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	australopith cave assemblages
Chronological frame	Pliocene-Pleistocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are australopith cave assemblages linked to the cited phase of Pliocene-Pleistocene?
- What natural and human processes altered deposits at Makapansgat after original use?
- Which nearby sites, especially Boomplaas Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Makapansgat with Boomplaas Cave, Blombos Cave and Wonderwerk Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Makapansgat

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Makapansgat begins with a bounded claim. For the material described here as australopith cave assemblages, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Makapansgat using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Blombos Cave and Wonderwerk Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Taung

Southern Africa and Deep Prehistory • Evidence dossier 33 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Pliocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Taung is a site or landscape associated with pliocene research in South Africa. The defining record highlighted in this atlas is Taung child fossil discovery context. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Taung child fossil discovery context
Chronological frame	Pliocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Taung child fossil discovery context linked to the cited phase of Pliocene?
- What natural and human processes altered deposits at Taung after original use?
- Which nearby sites, especially Apollo 11 Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Taung with Apollo 11 Cave, Pinnacle Point and Sterkfontein Caves. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Raymond Dart described the Taung Child in 1925; the find transformed debates over the African origins of humans.

Taung

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Taung begins with a bounded claim. For the material described here as Taung child fossil discovery context, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Taung using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pinnacle Point and Sterkfontein Caves using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Great Zimbabwe

Southern Africa and Deep Prehistory • Evidence dossier 34 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Zimbabwe	**medieval**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Great Zimbabwe is a site or landscape associated with medieval research in Zimbabwe. The defining record highlighted in this atlas is dry-stone enclosures and southern African urban trade. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	dry-stone enclosures and southern African urban trade
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are dry-stone enclosures and southern African urban trade linked to the cited phase of medieval?
- What natural and human processes altered deposits at Great Zimbabwe after original use?
- Which nearby sites, especially Twyfelfontein, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Great Zimbabwe with Twyfelfontein, Klasies River Caves and Swartkrans. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Early colonial interpretations frequently denied African authorship. Excavations and scholarship, including work by Gertrude Caton-Thompson, established the monument in the history of southern African societies.

RESEARCH POINTER: https://whc.unesco.org/en/list/364/

Great Zimbabwe

Zimbabwe • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Great Zimbabwe begins with a bounded claim. For the material described here as dry-stone enclosures and southern African urban trade, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Great Zimbabwe using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Klasies River Caves and Swartkrans using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Boomplaas Cave

Southern Africa and Deep Prehistory • Evidence dossier 35 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Africa	Middle-Later Stone Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Boomplaas Cave is a site or landscape associated with middle-later stone age research in South Africa. The defining record highlighted in this atlas is long environmental and occupation sequence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long environmental and occupation sequence
Chronological frame	Middle-Later Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long environmental and occupation sequence linked to the cited phase of Middle-Later Stone Age?
- What natural and human processes altered deposits at Boomplaas Cave after original use?
- Which nearby sites, especially Tsodilo Hills, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Boomplaas Cave with Tsodilo Hills, Sibudu Cave and Drimolen. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Boomplaas Cave

South Africa • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Boomplaas Cave begins with a bounded claim. For the material described here as long environmental and occupation sequence, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Boomplaas Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sibudu Cave and Drimolen using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Apollo 11 Cave

Southern Africa and Deep Prehistory • Evidence dossier 36 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Namibia	**Later Stone Age**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Apollo 11 Cave is a site or landscape associated with later stone age research in Namibia. The defining record highlighted in this atlas is painted stone plaques. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	painted stone plaques
Chronological frame	Later Stone Age; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are painted stone plaques linked to the cited phase of Later Stone Age?
- What natural and human processes altered deposits at Apollo 11 Cave after original use?
- Which nearby sites, especially Kalambo Falls, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Apollo 11 Cave with Kalambo Falls, Border Cave and Rising Star Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Apollo 11 Cave

Namibia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Apollo 11 Cave begins with a bounded claim. For the material described here as painted stone plaques, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Apollo 11 Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Border Cave and Rising Star Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Twyfelfontein

Southern Africa and Deep Prehistory • Evidence dossier 37 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Namibia	**Holocene**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Twyfelfontein is a site or landscape associated with holocene research in Namibia. The defining record highlighted in this atlas is rock engravings and landscape use. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock engravings and landscape use
Chronological frame	Holocene; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock engravings and landscape use linked to the cited phase of Holocene?
- What natural and human processes altered deposits at Twyfelfontein after original use?
- Which nearby sites, especially Ishango, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Twyfelfontein with Ishango, Diepkloof Rock Shelter and Makapansgat. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Twyfelfontein

Namibia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Twyfelfontein begins with a bounded claim. For the material described here as rock engravings and landscape use, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Twyfelfontein using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Diepkloof Rock Shelter and Makapansgat using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tsodilo Hills

Southern Africa and Deep Prehistory • Evidence dossier 38 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Botswana	**Holocene**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Tsodilo Hills is a site or landscape associated with holocene research in Botswana. The defining record highlighted in this atlas is rock art and long-term sacred landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock art and long-term sacred landscape
Chronological frame	Holocene; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock art and long-term sacred landscape linked to the cited phase of Holocene?
- What natural and human processes altered deposits at Tsodilo Hills after original use?
- Which nearby sites, especially Blombos Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tsodilo Hills with Blombos Cave, Wonderwerk Cave and Taung. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tsodilo Hills

Botswana • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tsodilo Hills begins with a bounded claim. For the material described here as rock art and long-term sacred landscape, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tsodilo Hills using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Wonderwerk Cave and Taung using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kalambo Falls

Southern Africa and Deep Prehistory • Evidence dossier 39 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Zambia	**Pleistocene**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Kalambo Falls is a site or landscape associated with pleistocene research in Zambia. The defining record highlighted in this atlas is woodworking and stone tools in waterlogged contexts. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	woodworking and stone tools in waterlogged contexts
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are woodworking and stone tools in waterlogged contexts linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Kalambo Falls after original use?
- Which nearby sites, especially Pinnacle Point, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kalambo Falls with Pinnacle Point, Sterkfontein Caves and Great Zimbabwe. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kalambo Falls

Zambia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kalambo Falls begins with a bounded claim. For the material described here as woodworking and stone tools in waterlogged contexts, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kalambo Falls using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sterkfontein Caves and Great Zimbabwe using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ishango

Southern Africa and Deep Prehistory • Evidence dossier 40 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
DR Congo	Late Pleistocene	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Ishango is a site or landscape associated with late pleistocene research in DR Congo. The defining record highlighted in this atlas is barbed bone points and notched bone. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: southern regions is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	barbed bone points and notched bone
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are barbed bone points and notched bone linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Ishango after original use?
- Which nearby sites, especially Klasies River Caves, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ishango with Klasies River Caves, Swartkrans and Boomplaas Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ishango

DR Congo • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ishango begins with a bounded claim. For the material described here as barbed bone points and notched bone, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ishango using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Swartkrans and Boomplaas Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Giza Plateau

Nile Valley and Sudan • Evidence dossier 41 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Old Kingdom	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Giza Plateau is a site or landscape associated with old kingdom research in Egypt. The defining record highlighted in this atlas is pyramids tombs and workers' settlements. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	pyramids tombs and workers' settlements
Chronological frame	Old Kingdom; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are pyramids tombs and workers' settlements linked to the cited phase of Old Kingdom?
- What natural and human processes altered deposits at Giza Plateau after original use?
- Which nearby sites, especially Dahshur, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Giza Plateau with Dahshur, Karnak and Kerma. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Excavations beyond the pyramids, notably at settlement and labor-support areas, have broadened inquiry from royal monuments toward workforces and supply systems.

Giza Plateau

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Giza Plateau begins with a bounded claim. For the material described here as pyramids tombs and workers' settlements, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Giza Plateau using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Karnak and Kerma using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Saqqara

Nile Valley and Sudan • Evidence dossier 42 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	**Early Dynastic onward**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Saqqara is a site or landscape associated with early dynastic onward research in Egypt. The defining record highlighted in this atlas is Step Pyramid and royal necropolises. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Step Pyramid and royal necropolises
Chronological frame	Early Dynastic onward; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Step Pyramid and royal necropolises linked to the cited phase of Early Dynastic onward?
- What natural and human processes altered deposits at Saqqara after original use?
- Which nearby sites, especially Abydos, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Saqqara with Abydos, Luxor Temple and Jebel Barkal. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Saqqara

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Saqqara begins with a bounded claim. For the material described here as Step Pyramid and royal necropolises, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Saqqara using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Luxor Temple and Jebel Barkal using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Abusir

Nile Valley and Sudan • Evidence dossier 43 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Old Kingdom	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Abusir is a site or landscape associated with old kingdom research in Egypt. The defining record highlighted in this atlas is pyramid complexes and administrative papyri. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	pyramid complexes and administrative papyri
Chronological frame	Old Kingdom; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are pyramid complexes and administrative papyri linked to the cited phase of Old Kingdom?
- What natural and human processes altered deposits at Abusir after original use?
- Which nearby sites, especially Hierakonpolis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Abusir with Hierakonpolis, Amarna and Meroe. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Abusir

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Abusir begins with a bounded claim. For the material described here as pyramid complexes and administrative papyri, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Abusir using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Amarna and Meroe using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dahshur

Nile Valley and Sudan • Evidence dossier 44 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Old Kingdom	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Dahshur is a site or landscape associated with old kingdom research in Egypt. The defining record highlighted in this atlas is early pyramid engineering. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early pyramid engineering
Chronological frame	Old Kingdom; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early pyramid engineering linked to the cited phase of Old Kingdom?
- What natural and human processes altered deposits at Dahshur after original use?
- Which nearby sites, especially Naqada, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dahshur with Naqada, Elephantine and Nuri. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dahshur

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dahshur begins with a bounded claim. For the material described here as early pyramid engineering, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dahshur using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Elephantine and Nuri using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Abydos

Nile Valley and Sudan • Evidence dossier 45 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	**Predynastic onward**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Abydos is a site or landscape associated with predynastic onward research in Egypt. The defining record highlighted in this atlas is royal tombs and cult archaeology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal tombs and cult archaeology
Chronological frame	Predynastic onward; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal tombs and cult archaeology linked to the cited phase of Predynastic onward?
- What natural and human processes altered deposits at Abydos after original use?
- Which nearby sites, especially Deir el-Medina, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Abydos with Deir el-Medina, Berenike and Qasr Ibrim. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Early dynastic royal burials and later cult structures make the site a reference point for state formation and mortuary change along the Nile.

Abydos

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Abydos begins with a bounded claim. For the material described here as royal tombs and cult archaeology, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Abydos using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Berenike and Qasr Ibrim using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hierakonpolis

Nile Valley and Sudan • Evidence dossier 46 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Predynastic	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Hierakonpolis is a site or landscape associated with predynastic research in Egypt. The defining record highlighted in this atlas is early urbanism and elite burials. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early urbanism and elite burials
Chronological frame	Predynastic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early urbanism and elite burials linked to the cited phase of Predynastic?
- What natural and human processes altered deposits at Hierakonpolis after original use?
- Which nearby sites, especially Valley of the Kings, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hierakonpolis with Valley of the Kings, Oxyrhynchus and Giza Plateau. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hierakonpolis

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Hierakonpolis begins with a bounded claim. For the material described here as early urbanism and elite burials, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hierakonpolis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Oxyrhynchus and Giza Plateau using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Naqada

Nile Valley and Sudan • Evidence dossier 47 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Predynastic	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Naqada is a site or landscape associated with predynastic research in Egypt. The defining record highlighted in this atlas is cemeteries and cultural sequence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	cemeteries and cultural sequence
Chronological frame	Predynastic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are cemeteries and cultural sequence linked to the cited phase of Predynastic?
- What natural and human processes altered deposits at Naqada after original use?
- Which nearby sites, especially Karnak, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Naqada with Karnak, Kerma and Saqqara. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

William Flinders Petrie's material-sequence work at Egyptian cemeteries contributed to archaeological seriation and Predynastic chronology.

Naqada

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Naqada begins with a bounded claim. For the material described here as cemeteries and cultural sequence, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Naqada using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kerma and Saqqara using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Deir el-Medina

Nile Valley and Sudan • Evidence dossier 48 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	New Kingdom	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Deir el-Medina is a site or landscape associated with new kingdom research in Egypt. The defining record highlighted in this atlas is workers' village and written records. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	workers' village and written records
Chronological frame	New Kingdom; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are workers' village and written records linked to the cited phase of New Kingdom?
- What natural and human processes altered deposits at Deir el-Medina after original use?
- Which nearby sites, especially Luxor Temple, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Deir el-Medina with Luxor Temple, Jebel Barkal and Abusir. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Deir el-Medina

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Deir el-Medina begins with a bounded claim. For the material described here as workers' village and written records, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Deir el-Medina using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Jebel Barkal and Abusir using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Valley of the Kings

Nile Valley and Sudan • Evidence dossier 49 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	New Kingdom	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Valley of the Kings is a site or landscape associated with new kingdom research in Egypt. The defining record highlighted in this atlas is royal tombs and burial equipment. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal tombs and burial equipment
Chronological frame	New Kingdom; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal tombs and burial equipment linked to the cited phase of New Kingdom?
- What natural and human processes altered deposits at Valley of the Kings after original use?
- Which nearby sites, especially Amarna, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Valley of the Kings with Amarna, Meroe and Dahshur. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Howard Carter's 1922 excavation of Tutankhamun's tomb shows both the information potential and the conservation complexities of a furnished burial.

Valley of the Kings

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Valley of the Kings begins with a bounded claim. For the material described here as royal tombs and burial equipment, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Valley of the Kings using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Meroe and Dahshur using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Karnak

Nile Valley and Sudan • Evidence dossier 50 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Middle Kingdom onward	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Karnak is a site or landscape associated with middle kingdom onward research in Egypt. The defining record highlighted in this atlas is temple complex and inscriptions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple complex and inscriptions
Chronological frame	Middle Kingdom onward; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple complex and inscriptions linked to the cited phase of Middle Kingdom onward?
- What natural and human processes altered deposits at Karnak after original use?
- Which nearby sites, especially Elephantine, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Karnak with Elephantine, Nuri and Abydos. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Karnak

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Karnak begins with a bounded claim. For the material described here as temple complex and inscriptions, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Karnak using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nuri and Abydos using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Luxor Temple

Nile Valley and Sudan • Evidence dossier 51 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	New Kingdom onward	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Luxor Temple is a site or landscape associated with new kingdom onward research in Egypt. The defining record highlighted in this atlas is monumental cult architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental cult architecture
Chronological frame	New Kingdom onward; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental cult architecture linked to the cited phase of New Kingdom onward?
- What natural and human processes altered deposits at Luxor Temple after original use?
- Which nearby sites, especially Berenike, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Luxor Temple with Berenike, Qasr Ibrim and Hierakonpolis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Luxor Temple

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Luxor Temple begins with a bounded claim. For the material described here as monumental cult architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Luxor Temple using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Qasr Ibrim and Hierakonpolis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Amarna

Nile Valley and Sudan • Evidence dossier 52 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	**New Kingdom**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Amarna is a site or landscape associated with new kingdom research in Egypt. The defining record highlighted in this atlas is planned royal city and domestic archaeology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	planned royal city and domestic archaeology
Chronological frame	New Kingdom; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are planned royal city and domestic archaeology linked to the cited phase of New Kingdom?
- What natural and human processes altered deposits at Amarna after original use?
- Which nearby sites, especially Oxyrhynchus, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Amarna with Oxyrhynchus, Giza Plateau and Naqada. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Modern research, associated with Barry Kemp and others, integrates houses, royal architecture, cemeteries and craft spaces to study an unusually short-lived royal capital.

Amarna

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Amarna begins with a bounded claim. For the material described here as planned royal city and domestic archaeology, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Amarna using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Giza Plateau and Naqada using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Elephantine

Nile Valley and Sudan • Evidence dossier 53 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	**Dynastic onward**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Elephantine is a site or landscape associated with dynastic onward research in Egypt. The defining record highlighted in this atlas is frontier settlement and multilingual documents. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	frontier settlement and multilingual documents
Chronological frame	Dynastic onward; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are frontier settlement and multilingual documents linked to the cited phase of Dynastic onward?
- What natural and human processes altered deposits at Elephantine after original use?
- Which nearby sites, especially Kerma, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Elephantine with Kerma, Saqqara and Deir el-Medina. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Elephantine

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Elephantine begins with a bounded claim. For the material described here as frontier settlement and multilingual documents, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Elephantine using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Saqqara and Deir el-Medina using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Berenike

Nile Valley and Sudan • Evidence dossier 54 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Hellenistic-Roman	Network

01 WHAT SURVIVES AND WHY IT MATTERS

Berenike is a site or landscape associated with hellenistic-roman research in Egypt. The defining record highlighted in this atlas is Red Sea trade port. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Red Sea trade port
Chronological frame	Hellenistic-Roman; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Red Sea trade port linked to the cited phase of Hellenistic-Roman?
- What natural and human processes altered deposits at Berenike after original use?
- Which nearby sites, especially Jebel Barkal, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Berenike with Jebel Barkal, Abusir and Valley of the Kings. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Berenike

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Berenike begins with a bounded claim. For the material described here as Red Sea trade port, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Berenike using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Abusir and Valley of the Kings using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Oxyrhynchus

Nile Valley and Sudan • Evidence dossier 55 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Greco-Roman	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Oxyrhynchus is a site or landscape associated with greco-roman research in Egypt. The defining record highlighted in this atlas is papyri and urban refuse deposits. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	papyri and urban refuse deposits
Chronological frame	Greco-Roman; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are papyri and urban refuse deposits linked to the cited phase of Greco-Roman?
- What natural and human processes altered deposits at Oxyrhynchus after original use?
- Which nearby sites, especially Meroe, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Oxyrhynchus with Meroe, Dahshur and Karnak. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Oxyrhynchus

Egypt • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Oxyrhynchus begins with a bounded claim. For the material described here as papyri and urban refuse deposits, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Oxyrhynchus using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dahshur and Karnak using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kerma

Nile Valley and Sudan • Evidence dossier 56 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sudan	Bronze Age	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Kerma is a site or landscape associated with bronze age research in Sudan. The defining record highlighted in this atlas is Nubian urban center and funerary monuments. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Nubian urban center and funerary monuments
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Nubian urban center and funerary monuments linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Kerma after original use?
- Which nearby sites, especially Nuri, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kerma with Nuri, Abydos and Luxor Temple. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Long research histories, including work by Charles Bonnet, document an influential Nubian center with distinctive religious and funerary architecture.

Kerma

Sudan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kerma begins with a bounded claim. For the material described here as Nubian urban center and funerary monuments, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kerma using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Abydos and Luxor Temple using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Jebel Barkal

Nile Valley and Sudan • Evidence dossier 57 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sudan	Napatan-Meroitic	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Jebel Barkal is a site or landscape associated with napatan-meroitic research in Sudan. The defining record highlighted in this atlas is sacred landscape and royal temples. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	sacred landscape and royal temples
Chronological frame	Napatan-Meroitic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are sacred landscape and royal temples linked to the cited phase of Napatan-Meroitic?
- What natural and human processes altered deposits at Jebel Barkal after original use?
- Which nearby sites, especially Qasr Ibrim, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Jebel Barkal with Qasr Ibrim, Hierakonpolis and Amarna. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Jebel Barkal

Sudan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Jebel Barkal begins with a bounded claim. For the material described here as sacred landscape and royal temples, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Jebel Barkal using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hierakonpolis and Amarna using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Meroe

Nile Valley and Sudan • Evidence dossier 58 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sudan	**Meroitic**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Meroe is a site or landscape associated with meroitic research in Sudan. The defining record highlighted in this atlas is pyramids urbanism and ironworking. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	pyramids urbanism and ironworking
Chronological frame	Meroitic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are pyramids urbanism and ironworking linked to the cited phase of Meroitic?
- What natural and human processes altered deposits at Meroe after original use?
- Which nearby sites, especially Giza Plateau, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Meroe with Giza Plateau, Naqada and Elephantine. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Meroe

Sudan • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Meroe begins with a bounded claim. For the material described here as pyramids urbanism and ironworking, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Meroe using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Naqada and Elephantine using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nuri

Nile Valley and Sudan • Evidence dossier 59 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sudan	**Napatan**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Nuri is a site or landscape associated with napatan research in Sudan. The defining record highlighted in this atlas is royal cemetery and pyramid tombs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal cemetery and pyramid tombs
Chronological frame	Napatan; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal cemetery and pyramid tombs linked to the cited phase of Napatan?
- What natural and human processes altered deposits at Nuri after original use?
- Which nearby sites, especially Saqqara, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nuri with Saqqara, Deir el-Medina and Berenike. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nuri

Sudan • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nuri begins with a bounded claim. For the material described here as royal cemetery and pyramid tombs, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nuri using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Deir el-Medina and Berenike using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Qasr Ibrim

Nile Valley and Sudan • Evidence dossier 60 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	Nubian-Roman-medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Qasr Ibrim is a site or landscape associated with nubian-roman-medieval research in Egypt. The defining record highlighted in this atlas is fortress and exceptional documentary preservation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: nile and nubia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortress and exceptional documentary preservation
Chronological frame	Nubian-Roman-medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortress and exceptional documentary preservation linked to the cited phase of Nubian-Roman-medieval?
- What natural and human processes altered deposits at Qasr Ibrim after original use?
- Which nearby sites, especially Abusir, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Qasr Ibrim with Abusir, Valley of the Kings and Oxyrhynchus. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Qasr Ibrim

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Qasr Ibrim begins with a bounded claim. For the material described here as fortress and exceptional documentary preservation, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Qasr Ibrim using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Valley of the Kings and Oxyrhynchus using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tassili n'Ajjer

North Africa and Sahara • Evidence dossier 61 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Algeria	**Holocene**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Tassili n'Ajjer is a site or landscape associated with holocene research in Algeria. The defining record highlighted in this atlas is Saharan rock art and changing ecosystems. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Saharan rock art and changing ecosystems
Chronological frame	Holocene; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Saharan rock art and changing ecosystems linked to the cited phase of Holocene?
- What natural and human processes altered deposits at Tassili n'Ajjer after original use?
- Which nearby sites, especially Wadi Sura, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tassili n'Ajjer with Wadi Sura, Lixus and Leptis Magna. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tassili n'Ajjer

Algeria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tassili n'Ajjer begins with a bounded claim. For the material described here as Saharan rock art and changing ecosystems, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tassili n'Ajjer using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lixus and Leptis Magna using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tadrart Acacus

North Africa and Sahara • Evidence dossier 62 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Libya	**Holocene**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Tadrart Acacus is a site or landscape associated with holocene research in Libya. The defining record highlighted in this atlas is rock shelters and painted landscapes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock shelters and painted landscapes
Chronological frame	Holocene; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock shelters and painted landscapes linked to the cited phase of Holocene?
- What natural and human processes altered deposits at Tadrart Acacus after original use?
- Which nearby sites, especially Taforalt Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tadrart Acacus with Taforalt Cave, Carthage and Cyrene. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tadrart Acacus

Libya • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tadrart Acacus begins with a bounded claim. For the material described here as rock shelters and painted landscapes, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tadrart Acacus using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Carthage and Cyrene using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nabta Playa

North Africa and Sahara • Evidence dossier 63 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Nabta Playa is a site or landscape associated with neolithic research in Egypt. The defining record highlighted in this atlas is pastoral ceremonial stones and seasonal settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	pastoral ceremonial stones and seasonal settlement
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are pastoral ceremonial stones and seasonal settlement linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Nabta Playa after original use?
- Which nearby sites, especially Jebel Irhoud, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nabta Playa with Jebel Irhoud, Dougga and Garamantian Germa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nabta Playa

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Nabta Playa begins with a bounded claim. For the material described here as pastoral ceremonial stones and seasonal settlement, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nabta Playa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dougga and Garamantian Germa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Wadi Sura

North Africa and Sahara • Evidence dossier 64 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Egypt	**Holocene**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Wadi Sura is a site or landscape associated with holocene research in Egypt. The defining record highlighted in this atlas is painted rock shelters in Gilf Kebir. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	painted rock shelters in Gilf Kebir
Chronological frame	Holocene; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are painted rock shelters in Gilf Kebir linked to the cited phase of Holocene?
- What natural and human processes altered deposits at Wadi Sura after original use?
- Which nearby sites, especially Aterian sites of Temara, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Wadi Sura with Aterian sites of Temara, Kerkuane and Ghadames Oasis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Wadi Sura

Egypt • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Wadi Sura begins with a bounded claim. For the material described here as painted rock shelters in Gilf Kebir, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Wadi Sura using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kerkuane and Ghadames Oasis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Taforalt Cave

North Africa and Sahara • Evidence dossier 65 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Morocco	**Late Pleistocene**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Taforalt Cave is a site or landscape associated with late pleistocene research in Morocco. The defining record highlighted in this atlas is Iberomaurusian burials and subsistence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Iberomaurusian burials and subsistence
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Iberomaurusian burials and subsistence linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Taforalt Cave after original use?
- Which nearby sites, especially Sidi Abderrahman, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Taforalt Cave with Sidi Abderrahman, Bulla Regia and M'Zab Valley. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Taforalt Cave

Morocco • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Taforalt Cave begins with a bounded claim. For the material described here as Iberomaurusian burials and subsistence, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Taforalt Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bulla Regia and M'Zab Valley using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Jebel Irhoud

North Africa and Sahara • Evidence dossier 66 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Morocco	Middle Stone Age	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Jebel Irhoud is a site or landscape associated with middle stone age research in Morocco. The defining record highlighted in this atlas is early Homo sapiens fossils. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Homo sapiens fossils
Chronological frame	Middle Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Homo sapiens fossils linked to the cited phase of Middle Stone Age?
- What natural and human processes altered deposits at Jebel Irhoud after original use?
- Which nearby sites, especially Volubilis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Jebel Irhoud with Volubilis, Sabratha and Tassili n'Ajjer. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

New dating and analysis of fossil material shifted debates about the diversity and geographic origins of early Homo sapiens.

Jebel Irhoud

Morocco • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Jebel Irhoud begins with a bounded claim. For the material described here as early Homo sapiens fossils, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Jebel Irhoud using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sabratha and Tassili n'Ajjer using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Aterian sites of Temara

North Africa and Sahara • Evidence dossier 67 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Morocco	Middle Stone Age	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Aterian sites of Temara is a site or landscape associated with middle stone age research in Morocco. The defining record highlighted in this atlas is tanged stone tools and coastal use. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	tanged stone tools and coastal use
Chronological frame	Middle Stone Age; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are tanged stone tools and coastal use linked to the cited phase of Middle Stone Age?
- What natural and human processes altered deposits at Aterian sites of Temara after original use?
- Which nearby sites, especially Lixus, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Aterian sites of Temara with Lixus, Leptis Magna and Tadrart Acacus. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Aterian sites of Temara

Morocco • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Aterian sites of Temara begins with a bounded claim. For the material described here as tanged stone tools and coastal use, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Aterian sites of Temara using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Leptis Magna and Tadrart Acacus using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sidi Abderrahman

North Africa and Sahara • Evidence dossier 68 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Morocco	**Pleistocene**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Sidi Abderrahman is a site or landscape associated with pleistocene research in Morocco. The defining record highlighted in this atlas is Acheulean and fossil-bearing deposits. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Acheulean and fossil-bearing deposits
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Acheulean and fossil-bearing deposits linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Sidi Abderrahman after original use?
- Which nearby sites, especially Carthage, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sidi Abderrahman with Carthage, Cyrene and Nabta Playa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sidi Abderrahman

Morocco • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sidi Abderrahman begins with a bounded claim. For the material described here as Acheulean and fossil-bearing deposits, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sidi Abderrahman using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cyrene and Nabta Playa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Volubilis

North Africa and Sahara • Evidence dossier 69 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Morocco	**Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Volubilis is a site or landscape associated with roman research in Morocco. The defining record highlighted in this atlas is provincial urban planning and mosaics. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	provincial urban planning and mosaics
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are provincial urban planning and mosaics linked to the cited phase of Roman?
- What natural and human processes altered deposits at Volubilis after original use?
- Which nearby sites, especially Dougga, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Volubilis with Dougga, Garamantian Germa and Wadi Sura. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Volubilis

Morocco • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Volubilis begins with a bounded claim. For the material described here as provincial urban planning and mosaics, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Volubilis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Garamantian Germa and Wadi Sura using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lixus

North Africa and Sahara • Evidence dossier 70 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Morocco	**Phoenician-Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Lixus is a site or landscape associated with phoenician-roman research in Morocco. The defining record highlighted in this atlas is coastal settlement and fish-processing installations. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	coastal settlement and fish-processing installations
Chronological frame	Phoenician-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are coastal settlement and fish-processing installations linked to the cited phase of Phoenician-Roman?
- What natural and human processes altered deposits at Lixus after original use?
- Which nearby sites, especially Kerkuane, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lixus with Kerkuane, Ghadames Oasis and Taforalt Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lixus

Morocco • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Lixus begins with a bounded claim. For the material described here as coastal settlement and fish-processing installations, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lixus using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ghadames Oasis and Taforalt Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Carthage

North Africa and Sahara • Evidence dossier 71 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tunisia	**Punic-Roman**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Carthage is a site or landscape associated with punic-roman research in Tunisia. The defining record highlighted in this atlas is port installations and urban stratigraphy. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	port installations and urban stratigraphy
Chronological frame	Punic-Roman; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are port installations and urban stratigraphy linked to the cited phase of Punic-Roman?
- What natural and human processes altered deposits at Carthage after original use?
- Which nearby sites, especially Bulla Regia, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Carthage with Bulla Regia, M'Zab Valley and Jebel Irhoud. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Punic and Roman phases together illustrate how conquest, reuse and excavation history complicate the reconstruction of an ancient urban landscape.

Carthage

Tunisia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Carthage begins with a bounded claim. For the material described here as port installations and urban stratigraphy, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Carthage using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with M'Zab Valley and Jebel Irhoud using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dougga

North Africa and Sahara • Evidence dossier 72 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tunisia	**Numidian-Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Dougga is a site or landscape associated with numidian-roman research in Tunisia. The defining record highlighted in this atlas is monuments and provincial city plan. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monuments and provincial city plan
Chronological frame	Numidian-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monuments and provincial city plan linked to the cited phase of Numidian-Roman?
- What natural and human processes altered deposits at Dougga after original use?
- Which nearby sites, especially Sabratha, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dougga with Sabratha, Tassili n'Ajjer and Aterian sites of Temara. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dougga

Tunisia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dougga begins with a bounded claim. For the material described here as monuments and provincial city plan, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dougga using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tassili n'Ajjer and Aterian sites of Temara using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kerkuane

North Africa and Sahara • Evidence dossier 73 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tunisia	**Punic**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Kerkuane is a site or landscape associated with punic research in Tunisia. The defining record highlighted in this atlas is preserved Punic town plan. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	preserved Punic town plan
Chronological frame	Punic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are preserved Punic town plan linked to the cited phase of Punic?
- What natural and human processes altered deposits at Kerkuane after original use?
- Which nearby sites, especially Leptis Magna, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kerkuane with Leptis Magna, Tadrart Acacus and Sidi Abderrahman. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kerkuane

Tunisia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kerkuane begins with a bounded claim. For the material described here as preserved Punic town plan, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kerkuane using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tadrart Acacus and Sidi Abderrahman using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bulla Regia

North Africa and Sahara • Evidence dossier 74 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tunisia	**Roman**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Bulla Regia is a site or landscape associated with roman research in Tunisia. The defining record highlighted in this atlas is subterranean villas and mosaics. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	subterranean villas and mosaics
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are subterranean villas and mosaics linked to the cited phase of Roman?
- What natural and human processes altered deposits at Bulla Regia after original use?
- Which nearby sites, especially Cyrene, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bulla Regia with Cyrene, Nabta Playa and Volubilis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Bulla Regia

Tunisia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Bulla Regia begins with a bounded claim. For the material described here as subterranean villas and mosaics, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bulla Regia using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nabta Playa and Volubilis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Sabratha

North Africa and Sahara • Evidence dossier 75 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Libya	Roman	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Sabratha is a site or landscape associated with roman research in Libya. The defining record highlighted in this atlas is theatre temples and North African urbanism. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	theatre temples and North African urbanism
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are theatre temples and North African urbanism linked to the cited phase of Roman?
- What natural and human processes altered deposits at Sabratha after original use?
- Which nearby sites, especially Garamantian Germa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sabratha with Garamantian Germa, Wadi Sura and Lixus. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sabratha

Libya • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sabratha begins with a bounded claim. For the material described here as theatre temples and North African urbanism, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sabratha using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Wadi Sura and Lixus using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Leptis Magna

North Africa and Sahara • Evidence dossier 76 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Libya	**Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Leptis Magna is a site or landscape associated with roman research in Libya. The defining record highlighted in this atlas is monumental imperial city. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental imperial city
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental imperial city linked to the cited phase of Roman?
- What natural and human processes altered deposits at Leptis Magna after original use?
- Which nearby sites, especially Ghadames Oasis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Leptis Magna with Ghadames Oasis, Taforalt Cave and Carthage. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Leptis Magna

Libya • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Leptis Magna begins with a bounded claim. For the material described here as monumental imperial city, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Leptis Magna using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Taforalt Cave and Carthage using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cyrene

North Africa and Sahara • Evidence dossier 77 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Libya	Greek-Roman	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Cyrene is a site or landscape associated with greek-roman research in Libya. The defining record highlighted in this atlas is colonial city and sanctuaries. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	colonial city and sanctuaries
Chronological frame	Greek-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are colonial city and sanctuaries linked to the cited phase of Greek-Roman?
- What natural and human processes altered deposits at Cyrene after original use?
- Which nearby sites, especially M'Zab Valley, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cyrene with M'Zab Valley, Jebel Irhoud and Dougga. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Cyrene

Libya • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Cyrene begins with a bounded claim. For the material described here as colonial city and sanctuaries, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cyrene using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Jebel Irhoud and Dougga using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Garamantian Germa

North Africa and Sahara • Evidence dossier 78 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Libya	**Iron Age-antique**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Garamantian Germa is a site or landscape associated with iron age-antique research in Libya. The defining record highlighted in this atlas is oasis settlement and irrigation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	oasis settlement and irrigation
Chronological frame	Iron Age-antique; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are oasis settlement and irrigation linked to the cited phase of Iron Age-antique?
- What natural and human processes altered deposits at Garamantian Germa after original use?
- Which nearby sites, especially Tassili n'Ajjer, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Garamantian Germa with Tassili n'Ajjer, Aterian sites of Temara and Kerkuane. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Garamantian Germa

Libya • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Garamantian Germa begins with a bounded claim. For the material described here as oasis settlement and irrigation, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Garamantian Germa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Aterian sites of Temara and Kerkuane using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ghadames Oasis

North Africa and Sahara • Evidence dossier 79 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Libya	**antique-medieval**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Ghadames Oasis is a site or landscape associated with antique-medieval research in Libya. The defining record highlighted in this atlas is Saharan architecture and trade routes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Saharan architecture and trade routes
Chronological frame	antique-medieval; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Saharan architecture and trade routes linked to the cited phase of antique-medieval?
- What natural and human processes altered deposits at Ghadames Oasis after original use?
- Which nearby sites, especially Tadrart Acacus, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ghadames Oasis with Tadrart Acacus, Sidi Abderrahman and Bulla Regia. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ghadames Oasis

Libya • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ghadames Oasis begins with a bounded claim. For the material described here as Saharan architecture and trade routes, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ghadames Oasis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sidi Abderrahman and Bulla Regia using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

M'Zab Valley

North Africa and Sahara • Evidence dossier 80 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Algeria	**medieval**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

M'Zab Valley is a site or landscape associated with medieval research in Algeria. The defining record highlighted in this atlas is oasis settlements and urban ecology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: northern and saharan is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	oasis settlements and urban ecology
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are oasis settlements and urban ecology linked to the cited phase of medieval?
- What natural and human processes altered deposits at M'Zab Valley after original use?
- Which nearby sites, especially Nabta Playa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare M'Zab Valley with Nabta Playa, Volubilis and Sabratha. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

M'Zab Valley

Algeria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of M'Zab Valley begins with a bounded claim. For the material described here as oasis settlements and urban ecology, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for M'Zab Valley using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Volubilis and Sabratha using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dhar Tichitt

West and Central African Archaeology • Evidence dossier 81 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mauritania	**Neolithic-Iron Age**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Dhar Tichitt is a site or landscape associated with neolithic-iron age research in Mauritania. The defining record highlighted in this atlas is dry-stone settlements and pastoral systems. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	dry-stone settlements and pastoral systems
Chronological frame	Neolithic-Iron Age; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are dry-stone settlements and pastoral systems linked to the cited phase of Neolithic-Iron Age?
- What natural and human processes altered deposits at Dhar Tichitt after original use?
- Which nearby sites, especially Timbuktu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dhar Tichitt with Timbuktu, Ife and Kintampo sites. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dhar Tichitt

Mauritania • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dhar Tichitt begins with a bounded claim. For the material described here as dry-stone settlements and pastoral systems, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dhar Tichitt using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ife and Kintampo sites using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ounjougou

West and Central African Archaeology • Evidence dossier 82 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mali	Pleistocene-Holocene	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Ounjougou is a site or landscape associated with pleistocene-holocene research in Mali. The defining record highlighted in this atlas is long pottery and landscape sequence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long pottery and landscape sequence
Chronological frame	Pleistocene-Holocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long pottery and landscape sequence linked to the cited phase of Pleistocene-Holocene?
- What natural and human processes altered deposits at Ounjougou after original use?
- Which nearby sites, especially Gao-Saney, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ounjougou with Gao-Saney, Benin City Earthworks and Begho. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ounjougou

Mali • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ounjougou begins with a bounded claim. For the material described here as long pottery and landscape sequence, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ounjougou using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Benin City Earthworks and Begho using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Djenné-Djenno

West and Central African Archaeology • Evidence dossier 83 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mali	Iron Age-medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Djenné-Djenno is a site or landscape associated with iron age-medieval research in Mali. The defining record highlighted in this atlas is early West African urban settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early West African urban settlement
Chronological frame	Iron Age-medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early West African urban settlement linked to the cited phase of Iron Age-medieval?
- What natural and human processes altered deposits at Djenné-Djenno after original use?
- Which nearby sites, especially Tadmekka (Essouk), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Djenné-Djenno with Tadmekka (Essouk), Sungbo's Eredo and Bono Manso. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Djenné-Djenno

Mali • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Djenné-Djenno begins with a bounded claim. For the material described here as early West African urban settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Djenné-Djenno using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sungbo's Eredo and Bono Manso using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Timbuktu

West and Central African Archaeology • Evidence dossier 84 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mali	medieval-modern	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Timbuktu is a site or landscape associated with medieval-modern research in Mali. The defining record highlighted in this atlas is trade city and manuscript architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	trade city and manuscript architecture
Chronological frame	medieval-modern; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are trade city and manuscript architecture linked to the cited phase of medieval-modern?
- What natural and human processes altered deposits at Timbuktu after original use?
- Which nearby sites, especially Kumbi Saleh, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Timbuktu with Kumbi Saleh, Sukur Cultural Landscape and Kongo Dia Ntotila (Mbanza Kongo). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Timbuktu

Mali • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Timbuktu begins with a bounded claim. For the material described here as trade city and manuscript architecture, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Timbuktu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sukur Cultural Landscape and Kongo Dia Ntotila (Mbanza Kongo) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Gao-Saney

West and Central African Archaeology • Evidence dossier 85 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mali	medieval	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Gao-Saney is a site or landscape associated with medieval research in Mali. The defining record highlighted in this atlas is Sahelian cemeteries and exchange networks. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Sahelian cemeteries and exchange networks
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Sahelian cemeteries and exchange networks linked to the cited phase of medieval?
- What natural and human processes altered deposits at Gao-Saney after original use?
- Which nearby sites, especially Nok sites, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Gao-Saney with Nok sites, Dufuna and Shum Laka. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Gao-Saney

Mali • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Gao-Saney begins with a bounded claim. For the material described here as Sahelian cemeteries and exchange networks, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Gao-Saney using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dufuna and Shum Laka using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tadmekka (Essouk)

West and Central African Archaeology • Evidence dossier 86 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mali	medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Tadmekka (Essouk) is a site or landscape associated with medieval research in Mali. The defining record highlighted in this atlas is trans-Saharan commerce and coin manufacture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	trans-Saharan commerce and coin manufacture
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are trans-Saharan commerce and coin manufacture linked to the cited phase of medieval?
- What natural and human processes altered deposits at Tadmekka (Essouk) after original use?
- Which nearby sites, especially Igbo-Ukwu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tadmekka (Essouk) with Igbo-Ukwu, Birnin Lafiya and Dhar Tichitt. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tadmekka (Essouk)

Mali • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tadmekka (Essouk) begins with a bounded claim. For the material described here as trans-Saharan commerce and coin manufacture, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tadmekka (Essouk) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Birnin Lafiya and Dhar Tichitt using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kumbi Saleh

West and Central African Archaeology • Evidence dossier 87 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mauritania	**medieval**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Kumbi Saleh is a site or landscape associated with medieval research in Mauritania. The defining record highlighted in this atlas is settlement associated with Ghana Empire debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	settlement associated with Ghana Empire debates
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are settlement associated with Ghana Empire debates linked to the cited phase of medieval?
- What natural and human processes altered deposits at Kumbi Saleh after original use?
- Which nearby sites, especially Ife, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kumbi Saleh with Ife, Kintampo sites and Ounjougou. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kumbi Saleh

Mauritania • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kumbi Saleh begins with a bounded claim. For the material described here as settlement associated with Ghana Empire debates, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kumbi Saleh using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kintampo sites and Ounjougou using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nok sites

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nigeria	Iron Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Nok sites is a site or landscape associated with iron age research in Nigeria. The defining record highlighted in this atlas is terracotta sculpture and ironworking. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	terracotta sculpture and ironworking
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are terracotta sculpture and ironworking linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Nok sites after original use?
- Which nearby sites, especially Benin City Earthworks, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nok sites with Benin City Earthworks, Begho and Djenné-Djenno. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nok sites

Nigeria • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nok sites begins with a bounded claim. For the material described here as terracotta sculpture and ironworking, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nok sites using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Begho and Djenné-Djenno using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Igbo-Ukwu

West and Central African Archaeology • Evidence dossier 89 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nigeria	early medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Igbo-Ukwu is a site or landscape associated with early medieval research in Nigeria. The defining record highlighted in this atlas is elaborate copper-alloy objects and ritual deposits. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	elaborate copper-alloy objects and ritual deposits
Chronological frame	early medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are elaborate copper-alloy objects and ritual deposits linked to the cited phase of early medieval?
- What natural and human processes altered deposits at Igbo-Ukwu after original use?
- Which nearby sites, especially Sungbo's Eredo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Igbo-Ukwu with Sungbo's Eredo, Bono Manso and Timbuktu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Igbo-Ukwu

Nigeria • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Igbo-Ukwu begins with a bounded claim. For the material described here as elaborate copper-alloy objects and ritual deposits, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Igbo-Ukwu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bono Manso and Timbuktu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ife

West and Central African Archaeology • Evidence dossier 90 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nigeria	medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Ife is a site or landscape associated with medieval research in Nigeria. The defining record highlighted in this atlas is naturalistic sculptural traditions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	naturalistic sculptural traditions
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are naturalistic sculptural traditions linked to the cited phase of medieval?
- What natural and human processes altered deposits at Ife after original use?
- Which nearby sites, especially Sukur Cultural Landscape, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ife with Sukur Cultural Landscape, Kongo Dia Ntotila (Mbanza Kongo) and Gao-Saney. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ife

Nigeria • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ife begins with a bounded claim. For the material described here as naturalistic sculptural traditions, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ife using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kongo Dia Ntotila (Mbanza Kongo) and Gao-Saney using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Benin City Earthworks

West and Central African Archaeology • Evidence dossier 91 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nigeria	medieval-modern	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Benin City Earthworks is a site or landscape associated with medieval-modern research in Nigeria. The defining record highlighted in this atlas is earthworks and urban history. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	earthworks and urban history
Chronological frame	medieval-modern; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are earthworks and urban history linked to the cited phase of medieval-modern?
- What natural and human processes altered deposits at Benin City Earthworks after original use?
- Which nearby sites, especially Dufuna, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Benin City Earthworks with Dufuna, Shum Laka and Tadmekka (Essouk). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Benin City Earthworks

Nigeria • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Benin City Earthworks begins with a bounded claim. For the material described here as earthworks and urban history, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Benin City Earthworks using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Shum Laka and Tadmekka (Essouk) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sungbo's Eredo

West and Central African Archaeology • Evidence dossier 92 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nigeria	medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Sungbo's Eredo is a site or landscape associated with medieval research in Nigeria. The defining record highlighted in this atlas is vast engineered embankment systems. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	vast engineered embankment systems
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are vast engineered embankment systems linked to the cited phase of medieval?
- What natural and human processes altered deposits at Sungbo's Eredo after original use?
- Which nearby sites, especially Birnin Lafiya, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sungbo's Eredo with Birnin Lafiya, Dhar Tichitt and Kumbi Saleh. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sungbo's Eredo

Nigeria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sungbo's Eredo begins with a bounded claim. For the material described here as vast engineered embankment systems, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sungbo's Eredo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dhar Tichitt and Kumbi Saleh using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sukur Cultural Landscape

West and Central African Archaeology • Evidence dossier 93 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nigeria	historic	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Sukur Cultural Landscape is a site or landscape associated with historic research in Nigeria. The defining record highlighted in this atlas is terraced landscape iron production. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	terraced landscape iron production
Chronological frame	historic; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are terraced landscape iron production linked to the cited phase of historic?
- What natural and human processes altered deposits at Sukur Cultural Landscape after original use?
- Which nearby sites, especially Kintampo sites, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sukur Cultural Landscape with Kintampo sites, Ounjougou and Nok sites. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sukur Cultural Landscape

Nigeria • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Sukur Cultural Landscape begins with a bounded claim. For the material described here as terraced landscape iron production, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sukur Cultural Landscape using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ounjougou and Nok sites using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dufuna

West and Central African Archaeology • Evidence dossier 94 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nigeria	Holocene	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Dufuna is a site or landscape associated with holocene research in Nigeria. The defining record highlighted in this atlas is ancient watercraft discovery. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ancient watercraft discovery
Chronological frame	Holocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ancient watercraft discovery linked to the cited phase of Holocene?
- What natural and human processes altered deposits at Dufuna after original use?
- Which nearby sites, especially Begho, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dufuna with Begho, Djenné-Djenno and Igbo-Ukwu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dufuna

Nigeria • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Dufuna begins with a bounded claim. For the material described here as ancient watercraft discovery, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dufuna using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Djenné-Djenno and Igbo-Ukwu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Birnin Lafiya

West and Central African Archaeology • Evidence dossier 95 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Benin	**Iron Age-medieval**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Birnin Lafiya is a site or landscape associated with iron age-medieval research in Benin. The defining record highlighted in this atlas is settlement mounds and regional chronology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	settlement mounds and regional chronology
Chronological frame	Iron Age-medieval; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are settlement mounds and regional chronology linked to the cited phase of Iron Age-medieval?
- What natural and human processes altered deposits at Birnin Lafiya after original use?
- Which nearby sites, especially Bono Manso, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Birnin Lafiya with Bono Manso, Timbuktu and Ife. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Birnin Lafiya

Benin • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Birnin Lafiya begins with a bounded claim. For the material described here as settlement mounds and regional chronology, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Birnin Lafiya using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Timbuktu and Ife using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kintampo sites

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ghana	Late Stone Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Kintampo sites is a site or landscape associated with late stone age research in Ghana. The defining record highlighted in this atlas is early food production and pottery. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early food production and pottery
Chronological frame	Late Stone Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early food production and pottery linked to the cited phase of Late Stone Age?
- What natural and human processes altered deposits at Kintampo sites after original use?
- Which nearby sites, especially Kongo Dia Ntotila (Mbanza Kongo), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kintampo sites with Kongo Dia Ntotila (Mbanza Kongo), Gao-Saney and Benin City Earthworks. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kintampo sites

Ghana • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Kintampo sites begins with a bounded claim. For the material described here as early food production and pottery, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kintampo sites using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Gao-Saney and Benin City Earthworks using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Begho

West and Central African Archaeology • Evidence dossier 97 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ghana	medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Begho is a site or landscape associated with medieval research in Ghana. The defining record highlighted in this atlas is market settlement and craft industries. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	market settlement and craft industries
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are market settlement and craft industries linked to the cited phase of medieval?
- What natural and human processes altered deposits at Begho after original use?
- Which nearby sites, especially Shum Laka, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Begho with Shum Laka, Tadmekka (Essouk) and Sungbo's Eredo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Begho

Ghana • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Begho begins with a bounded claim. For the material described here as market settlement and craft industries, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Begho using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tadmekka (Essouk) and Sungbo's Eredo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bono Manso

West and Central African Archaeology • Evidence dossier 98 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ghana	medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Bono Manso is a site or landscape associated with medieval research in Ghana. The defining record highlighted in this atlas is forest-savanna political center. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	forest-savanna political center
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are forest-savanna political center linked to the cited phase of medieval?
- What natural and human processes altered deposits at Bono Manso after original use?
- Which nearby sites, especially Dhar Tichitt, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bono Manso with Dhar Tichitt, Kumbi Saleh and Sukur Cultural Landscape. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Bono Manso

Ghana • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Bono Manso begins with a bounded claim. For the material described here as forest-savanna political center, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bono Manso using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kumbi Saleh and Sukur Cultural Landscape using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kongo Dia Ntotila (Mbanza Kongo)

West and Central African Archaeology • Evidence dossier 99 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Angola	**medieval-modern**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Kongo Dia Ntotila (Mbanza Kongo) is a site or landscape associated with medieval-modern research in Angola. The defining record highlighted in this atlas is royal town and colonial-era transformations. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal town and colonial-era transformations
Chronological frame	medieval-modern; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal town and colonial-era transformations linked to the cited phase of medieval-modern?
- What natural and human processes altered deposits at Kongo Dia Ntotila (Mbanza Kongo) after original use?
- Which nearby sites, especially Ounjougou, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kongo Dia Ntotila (Mbanza Kongo) with Ounjougou, Nok sites and Dufuna. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kongo Dia Ntotila (Mbanza Kongo)

Angola • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Kongo Dia Ntotila (Mbanza Kongo) begins with a bounded claim. For the material described here as royal town and colonial-era transformations, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kongo Dia Ntotila (Mbanza Kongo) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nok sites and Dufuna using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Shum Laka

West and Central African Archaeology • Evidence dossier 100 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cameroon	Late Pleistocene-Holocene	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Shum Laka is a site or landscape associated with late pleistocene-holocene research in Cameroon. The defining record highlighted in this atlas is rock shelter burials and ancient DNA. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in africa: west and central is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock shelter burials and ancient DNA
Chronological frame	Late Pleistocene-Holocene; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock shelter burials and ancient DNA linked to the cited phase of Late Pleistocene-Holocene?
- What natural and human processes altered deposits at Shum Laka after original use?
- Which nearby sites, especially Djenné-Djenno, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Shum Laka with Djenné-Djenno, Igbo-Ukwu and Birnin Lafiya. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Shum Laka

Cameroon • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Shum Laka begins with a bounded claim. For the material described here as rock shelter burials and ancient DNA, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Shum Laka using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Igbo-Ukwu and Birnin Lafiya using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Jericho (Tell es-Sultan)

Levant and Arabia • Evidence dossier 101 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Palestine	**Pre-Pottery Neolithic on...**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Jericho (Tell es-Sultan) is a site or landscape associated with pre-pottery neolithic onward research in Palestine. The defining record highlighted in this atlas is early walled settlement and tower. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early walled settlement and tower
Chronological frame	Pre-Pottery Neolithic onward; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early walled settlement and tower linked to the cited phase of Pre-Pottery Neolithic onward?
- What natural and human processes altered deposits at Jericho (Tell es-Sultan) after original use?
- Which nearby sites, especially Mureybet, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Jericho (Tell es-Sultan) with Mureybet, Qafzeh Cave and Masada. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Early excavations by Kathleen Kenyon helped establish the significance of stratigraphic methods for interpreting prehistoric settlement sequences.

Jericho (Tell es-Sultan)

Palestine • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Jericho (Tell es-Sultan) begins with a bounded claim. For the material described here as early walled settlement and tower, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Jericho (Tell es-Sultan) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Qafzeh Cave and Masada using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ain Ghazal

Levant and Arabia • Evidence dossier 102 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Jordan	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Ain Ghazal is a site or landscape associated with neolithic research in Jordan. The defining record highlighted in this atlas is plaster statues and village aggregation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	plaster statues and village aggregation
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are plaster statues and village aggregation linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Ain Ghazal after original use?
- Which nearby sites, especially Tell Qaramel, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ain Ghazal with Tell Qaramel, Ohalo II and Petra. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Large plaster human figures preserve evidence of distinctive Neolithic craft and social practices; functions are inferred rather than directly observed.

Ain Ghazal

Jordan • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ain Ghazal begins with a bounded claim. For the material described here as plaster statues and village aggregation, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ain Ghazal using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ohalo II and Petra using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Abu Hureyra

Levant and Arabia • Evidence dossier 103 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Syria	Epipaleolithic-Neolithic	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Abu Hureyra is a site or landscape associated with epipaleolithic-neolithic research in Syria. The defining record highlighted in this atlas is foraging to farming transition. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	foraging to farming transition
Chronological frame	Epipaleolithic-Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are foraging to farming transition linked to the cited phase of Epipaleolithic-Neolithic?
- What natural and human processes altered deposits at Abu Hureyra after original use?
- Which nearby sites, especially Jarmo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Abu Hureyra with Jarmo, Nahal Hemar Cave and Hegra (Madain Salih). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Abu Hureyra

Syria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Abu Hureyra begins with a bounded claim. For the material described here as foraging to farming transition, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Abu Hureyra using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nahal Hemar Cave and Hegra (Madain Salih) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mureybet

Levant and Arabia • Evidence dossier 104 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Syria	Epipaleolithic-Neolithic	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Mureybet is a site or landscape associated with epipaleolithic-neolithic research in Syria. The defining record highlighted in this atlas is early village architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early village architecture
Chronological frame	Epipaleolithic-Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early village architecture linked to the cited phase of Epipaleolithic-Neolithic?
- What natural and human processes altered deposits at Mureybet after original use?
- Which nearby sites, especially Kebara Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mureybet with Kebara Cave, Megiddo and AlUla / Dadan. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mureybet

Syria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mureybet begins with a bounded claim. For the material described here as early village architecture, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mureybet using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Megiddo and AlUla / Dadan using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tell Qaramel

Levant and Arabia • Evidence dossier 105 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Syria	Pre-Pottery Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Tell Qaramel is a site or landscape associated with pre-pottery neolithic research in Syria. The defining record highlighted in this atlas is early stone towers and settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early stone towers and settlement
Chronological frame	Pre-Pottery Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early stone towers and settlement linked to the cited phase of Pre-Pottery Neolithic?
- What natural and human processes altered deposits at Tell Qaramel after original use?
- Which nearby sites, especially Tabun Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tell Qaramel with Tabun Cave, Hazor and Qaryat al-Faw. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tell Qaramel

Syria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tell Qaramel begins with a bounded claim. For the material described here as early stone towers and settlement, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tell Qaramel using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hazor and Qaryat al-Faw using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Jarmo

Levant and Arabia • Evidence dossier 106 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Jarmo is a site or landscape associated with neolithic research in Iraq. The defining record highlighted in this atlas is early farming village. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early farming village
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early farming village linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Jarmo after original use?
- Which nearby sites, especially Skhul Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Jarmo with Skhul Cave, Lachish and Jericho (Tell es-Sultan). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Jarmo

Iraq • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Jarmo begins with a bounded claim. For the material described here as early farming village, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Jarmo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lachish and Jericho (Tell es-Sultan) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kebara Cave

Levant and Arabia • Evidence dossier 107 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	Middle Paleolithic	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Kebara Cave is a site or landscape associated with middle paleolithic research in Israel. The defining record highlighted in this atlas is Neanderthal remains and cave sequence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Neanderthal remains and cave sequence
Chronological frame	Middle Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Neanderthal remains and cave sequence linked to the cited phase of Middle Paleolithic?
- What natural and human processes altered deposits at Kebara Cave after original use?
- Which nearby sites, especially Qafzeh Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kebara Cave with Qafzeh Cave, Masada and Ain Ghazal. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kebara Cave

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kebara Cave begins with a bounded claim. For the material described here as Neanderthal remains and cave sequence, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kebara Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Masada and Ain Ghazal using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tabun Cave

Levant and Arabia • Evidence dossier 108 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Tabun Cave is a site or landscape associated with paleolithic research in Israel. The defining record highlighted in this atlas is long Levantine occupation stratigraphy. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long Levantine occupation stratigraphy
Chronological frame	Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long Levantine occupation stratigraphy linked to the cited phase of Paleolithic?
- What natural and human processes altered deposits at Tabun Cave after original use?
- Which nearby sites, especially Ohalo II, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tabun Cave with Ohalo II, Petra and Abu Hureyra. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tabun Cave

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tabun Cave begins with a bounded claim. For the material described here as long Levantine occupation stratigraphy, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tabun Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Petra and Abu Hureyra using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Skhul Cave

Levant and Arabia • Evidence dossier 109 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Middle Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Skhul Cave is a site or landscape associated with middle paleolithic research in Israel. The defining record highlighted in this atlas is early modern human burials. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early modern human burials
Chronological frame	Middle Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early modern human burials linked to the cited phase of Middle Paleolithic?
- What natural and human processes altered deposits at Skhul Cave after original use?
- Which nearby sites, especially Nahal Hemar Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Skhul Cave with Nahal Hemar Cave, Hegra (Madain Salih) and Mureybet. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Skhul Cave

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Skhul Cave begins with a bounded claim. For the material described here as early modern human burials, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Skhul Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hegra (Madain Salih) and Mureybet using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Qafzeh Cave

Levant and Arabia • Evidence dossier 110 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Middle Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Qafzeh Cave is a site or landscape associated with middle paleolithic research in Israel. The defining record highlighted in this atlas is early Homo sapiens remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Homo sapiens remains
Chronological frame	Middle Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Homo sapiens remains linked to the cited phase of Middle Paleolithic?
- What natural and human processes altered deposits at Qafzeh Cave after original use?
- Which nearby sites, especially Megiddo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Qafzeh Cave with Megiddo, AlUla / Dadan and Tell Qaramel. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Qafzeh Cave

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Qafzeh Cave begins with a bounded claim. For the material described here as early Homo sapiens remains, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Qafzeh Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with AlUla / Dadan and Tell Qaramel using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ohalo II

Levant and Arabia • Evidence dossier 111 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Epipaleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Ohalo II is a site or landscape associated with epipaleolithic research in Israel. The defining record highlighted in this atlas is waterlogged lakeshore camp and plant remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	waterlogged lakeshore camp and plant remains
Chronological frame	Epipaleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are waterlogged lakeshore camp and plant remains linked to the cited phase of Epipaleolithic?
- What natural and human processes altered deposits at Ohalo II after original use?
- Which nearby sites, especially Hazor, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ohalo II with Hazor, Qaryat al-Faw and Jarmo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ohalo II

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ohalo II begins with a bounded claim. For the material described here as waterlogged lakeshore camp and plant remains, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ohalo II using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Qaryat al-Faw and Jarmo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nahal Hemar Cave

Levant and Arabia • Evidence dossier 112 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Pre-Pottery Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Nahal Hemar Cave is a site or landscape associated with pre-pottery neolithic research in Israel. The defining record highlighted in this atlas is ritual deposits and organic artifacts. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ritual deposits and organic artifacts
Chronological frame	Pre-Pottery Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ritual deposits and organic artifacts linked to the cited phase of Pre-Pottery Neolithic?
- What natural and human processes altered deposits at Nahal Hemar Cave after original use?
- Which nearby sites, especially Lachish, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nahal Hemar Cave with Lachish, Jericho (Tell es-Sultan) and Kebara Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nahal Hemar Cave

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Nahal Hemar Cave begins with a bounded claim. For the material described here as ritual deposits and organic artifacts, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nahal Hemar Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Jericho (Tell es-Sultan) and Kebara Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Megiddo

Levant and Arabia • Evidence dossier 113 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Bronze-Iron Age**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Megiddo is a site or landscape associated with bronze-iron age research in Israel. The defining record highlighted in this atlas is tell stratigraphy and fortifications. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	tell stratigraphy and fortifications
Chronological frame	Bronze-Iron Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are tell stratigraphy and fortifications linked to the cited phase of Bronze-Iron Age?
- What natural and human processes altered deposits at Megiddo after original use?
- Which nearby sites, especially Masada, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Megiddo with Masada, Ain Ghazal and Tabun Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Megiddo

Israel • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Megiddo begins with a bounded claim. For the material described here as tell stratigraphy and fortifications, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Megiddo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ain Ghazal and Tabun Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hazor

Levant and Arabia • Evidence dossier 114 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Bronze-Iron Age**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Hazor is a site or landscape associated with bronze-iron age research in Israel. The defining record highlighted in this atlas is urban center and destruction layers. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	urban center and destruction layers
Chronological frame	Bronze-Iron Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are urban center and destruction layers linked to the cited phase of Bronze-Iron Age?
- What natural and human processes altered deposits at Hazor after original use?
- Which nearby sites, especially Petra, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hazor with Petra, Abu Hureyra and Skhul Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hazor

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hazor begins with a bounded claim. For the material described here as urban center and destruction layers, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hazor using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Abu Hureyra and Skhul Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lachish

Levant and Arabia • Evidence dossier 115 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Bronze-Iron Age**	**Text**

01 WHAT SURVIVES AND WHY IT MATTERS

Lachish is a site or landscape associated with bronze-iron age research in Israel. The defining record highlighted in this atlas is siege works and inscriptions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	siege works and inscriptions
Chronological frame	Bronze-Iron Age; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are siege works and inscriptions linked to the cited phase of Bronze-Iron Age?
- What natural and human processes altered deposits at Lachish after original use?
- Which nearby sites, especially Hegra (Madain Salih), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lachish with Hegra (Madain Salih), Mureybet and Qafzeh Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lachish

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Lachish begins with a bounded claim. For the material described here as siege works and inscriptions, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lachish using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mureybet and Qafzeh Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Masada

Levant and Arabia • Evidence dossier 116 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Israel	**Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Masada is a site or landscape associated with roman research in Israel. The defining record highlighted in this atlas is desert fortress and siege archaeology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	desert fortress and siege archaeology
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are desert fortress and siege archaeology linked to the cited phase of Roman?
- What natural and human processes altered deposits at Masada after original use?
- Which nearby sites, especially AlUla / Dadan, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Masada with AlUla / Dadan, Tell Qaramel and Ohalo II. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Masada

Israel • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Masada begins with a bounded claim. For the material described here as desert fortress and siege archaeology, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Masada using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tell Qaramel and Ohalo II using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Petra

Levant and Arabia • Evidence dossier 117 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Jordan	**Nabataean-Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Petra is a site or landscape associated with nabataean-roman research in Jordan. The defining record highlighted in this atlas is rock-cut architecture and hydraulic engineering. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock-cut architecture and hydraulic engineering
Chronological frame	Nabataean-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock-cut architecture and hydraulic engineering linked to the cited phase of Nabataean-Roman?
- What natural and human processes altered deposits at Petra after original use?
- Which nearby sites, especially Qaryat al-Faw, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Petra with Qaryat al-Faw, Jarmo and Nahal Hemar Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The carved monuments form only part of a Nabataean landscape also containing canals, cisterns, domestic zones and agricultural installations.

Petra

Jordan • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Petra begins with a bounded claim. For the material described here as rock-cut architecture and hydraulic engineering, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Petra using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Jarmo and Nahal Hemar Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hegra (Madain Salih)

Levant and Arabia • Evidence dossier 118 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Saudi Arabia	**Nabataean**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Hegra (Madain Salih) is a site or landscape associated with nabataean research in Saudi Arabia. The defining record highlighted in this atlas is rock-cut tomb facades. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock-cut tomb facades
Chronological frame	Nabataean; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock-cut tomb facades linked to the cited phase of Nabataean?
- What natural and human processes altered deposits at Hegra (Madain Salih) after original use?
- Which nearby sites, especially Jericho (Tell es-Sultan), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hegra (Madain Salih) with Jericho (Tell es-Sultan), Kebara Cave and Megiddo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hegra (Madain Salih)

Saudi Arabia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Hegra (Madain Salih) begins with a bounded claim. For the material described here as rock-cut tomb facades, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hegra (Madain Salih) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kebara Cave and Megiddo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

AlUla / Dadan

Levant and Arabia • Evidence dossier 119 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Saudi Arabia	**Iron Age-Nabataean**	**Text**

01 WHAT SURVIVES AND WHY IT MATTERS

AlUla / Dadan is a site or landscape associated with iron age-nabataean research in Saudi Arabia. The defining record highlighted in this atlas is oasis polities and inscriptions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	oasis polities and inscriptions
Chronological frame	Iron Age-Nabataean; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are oasis polities and inscriptions linked to the cited phase of Iron Age-Nabataean?
- What natural and human processes altered deposits at AlUla / Dadan after original use?
- Which nearby sites, especially Ain Ghazal, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare AlUla / Dadan with Ain Ghazal, Tabun Cave and Hazor. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

AlUla / Dadan

Saudi Arabia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of AlUla / Dadan begins with a bounded claim. For the material described here as oasis polities and inscriptions, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for AlUla / Dadan using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tabun Cave and Hazor using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Qaryat al-Faw

Levant and Arabia • Evidence dossier 120 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Saudi Arabia	**pre-Islamic**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Qaryat al-Faw is a site or landscape associated with pre-islamic research in Saudi Arabia. The defining record highlighted in this atlas is caravan town and South Arabian connections. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: levant and arabian peninsula is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	caravan town and South Arabian connections
Chronological frame	pre-Islamic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are caravan town and South Arabian connections linked to the cited phase of pre-Islamic?
- What natural and human processes altered deposits at Qaryat al-Faw after original use?
- Which nearby sites, especially Abu Hureyra, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Qaryat al-Faw with Abu Hureyra, Skhul Cave and Lachish. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Qaryat al-Faw

Saudi Arabia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Qaryat al-Faw begins with a bounded claim. For the material described here as caravan town and South Arabian connections, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Qaryat al-Faw using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Skhul Cave and Lachish using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Uruk (Warka)

Mesopotamia and Iranian Plateau • Evidence dossier 121 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Late Chalcolithic	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Uruk (Warka) is a site or landscape associated with late chalcolithic research in Iraq. The defining record highlighted in this atlas is early urbanism and writing systems. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early urbanism and writing systems
Chronological frame	Late Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early urbanism and writing systems linked to the cited phase of Late Chalcolithic?
- What natural and human processes altered deposits at Uruk (Warka) after original use?
- Which nearby sites, especially Nippur, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Uruk (Warka) with Nippur, Khorsabad (Dur-Sharrukin) and Chogha Mish. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Late fourth-millennium administrative devices, monumental precincts and early writing are key evidence for institutional change, without proving a single path to cities.

Uruk (Warka)

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Uruk (Warka) begins with a bounded claim. For the material described here as early urbanism and writing systems, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Uruk (Warka) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Khorsabad (Dur-Sharrukin) and Chogha Mish using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ur

Mesopotamia and Iranian Plateau • Evidence dossier 122 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Early Dynastic onward	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Ur is a site or landscape associated with early dynastic onward research in Iraq. The defining record highlighted in this atlas is royal cemetery and city institutions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal cemetery and city institutions
Chronological frame	Early Dynastic onward; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal cemetery and city institutions linked to the cited phase of Early Dynastic onward?
- What natural and human processes altered deposits at Ur after original use?
- Which nearby sites, especially Kish, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ur with Kish, Babylon and Susa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Leonard Woolley's work at the Royal Cemetery made early Mesopotamian elite burial customs and their material assemblages famous.

Ur

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ur begins with a bounded claim. For the material described here as royal cemetery and city institutions, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ur using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Babylon and Susa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Eridu

Mesopotamia and Iranian Plateau • Evidence dossier 123 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	**Ubaid onward**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Eridu is a site or landscape associated with ubaid onward research in Iraq. The defining record highlighted in this atlas is temple sequences and early settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple sequences and early settlement
Chronological frame	Ubaid onward; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple sequences and early settlement linked to the cited phase of Ubaid onward?
- What natural and human processes altered deposits at Eridu after original use?
- Which nearby sites, especially Lagash (Tell al-Hiba), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Eridu with Lagash (Tell al-Hiba), Ashur and Persepolis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Eridu

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Eridu begins with a bounded claim. For the material described here as temple sequences and early settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Eridu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ashur and Persepolis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Nippur

Mesopotamia and Iranian Plateau • Evidence dossier 124 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Sumerian onward	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Nippur is a site or landscape associated with sumerian onward research in Iraq. The defining record highlighted in this atlas is scribal tablets and sacred precinct. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	scribal tablets and sacred precinct
Chronological frame	Sumerian onward; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are scribal tablets and sacred precinct linked to the cited phase of Sumerian onward?
- What natural and human processes altered deposits at Nippur after original use?
- Which nearby sites, especially Girsu (Tello), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nippur with Girsu (Tello), Mari (Tell Hariri) and Pasargadae. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nippur

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nippur begins with a bounded claim. For the material described here as scribal tablets and sacred precinct, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nippur using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mari (Tell Hariri) and Pasargadae using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Kish

Mesopotamia and Iranian Plateau • Evidence dossier 125 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Early Dynastic	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Kish is a site or landscape associated with early dynastic research in Iraq. The defining record highlighted in this atlas is city-states and burials. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	city-states and burials
Chronological frame	Early Dynastic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are city-states and burials linked to the cited phase of Early Dynastic?
- What natural and human processes altered deposits at Kish after original use?
- Which nearby sites, especially Nineveh, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kish with Nineveh, Ebla (Tell Mardikh) and Shahr-e Sukhteh. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kish

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Kish begins with a bounded claim. For the material described here as city-states and burials, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kish using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ebla (Tell Mardikh) and Shahr-e Sukhteh using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Lagash (Tell al-Hiba)

Mesopotamia and Iranian Plateau • Evidence dossier 126 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Early Dynastic	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Lagash (Tell al-Hiba) is a site or landscape associated with early dynastic research in Iraq. The defining record highlighted in this atlas is political urbanism and administration. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	political urbanism and administration
Chronological frame	Early Dynastic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are political urbanism and administration linked to the cited phase of Early Dynastic?
- What natural and human processes altered deposits at Lagash (Tell al-Hiba) after original use?
- Which nearby sites, especially Nimrud, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lagash (Tell al-Hiba) with Nimrud, Tell Brak and Uruk (Warka). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lagash (Tell al-Hiba)

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Lagash (Tell al-Hiba) begins with a bounded claim. For the material described here as political urbanism and administration, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lagash (Tell al-Hiba) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tell Brak and Uruk (Warka) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Girsu (Tello)

Mesopotamia and Iranian Plateau • Evidence dossier 127 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Sumerian	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Girsu (Tello) is a site or landscape associated with sumerian research in Iraq. The defining record highlighted in this atlas is temples inscriptions and state governance. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temples inscriptions and state governance
Chronological frame	Sumerian; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temples inscriptions and state governance linked to the cited phase of Sumerian?
- What natural and human processes altered deposits at Girsu (Tello) after original use?
- Which nearby sites, especially Khorsabad (Dur-Sharrukin), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Girsu (Tello) with Khorsabad (Dur-Sharrukin), Chogha Mish and Ur. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Girsu (Tello)

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Girsu (Tello) begins with a bounded claim. For the material described here as temples inscriptions and state governance, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Girsu (Tello) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chogha Mish and Ur using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nineveh

Mesopotamia and Iranian Plateau • Evidence dossier 128 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Assyrian	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Nineveh is a site or landscape associated with assyrian research in Iraq. The defining record highlighted in this atlas is imperial capital and library tablets. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	imperial capital and library tablets
Chronological frame	Assyrian; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are imperial capital and library tablets linked to the cited phase of Assyrian?
- What natural and human processes altered deposits at Nineveh after original use?
- Which nearby sites, especially Babylon, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nineveh with Babylon, Susa and Eridu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The palace libraries and reliefs have been studied alongside the history of large-scale nineteenth-century excavations and the dispersal of collections.

Nineveh

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nineveh begins with a bounded claim. For the material described here as imperial capital and library tablets, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nineveh using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Susa and Eridu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nimrud

Mesopotamia and Iranian Plateau • Evidence dossier 129 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Neo-Assyrian	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Nimrud is a site or landscape associated with neo-assyrian research in Iraq. The defining record highlighted in this atlas is palaces reliefs and ivories. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palaces reliefs and ivories
Chronological frame	Neo-Assyrian; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palaces reliefs and ivories linked to the cited phase of Neo-Assyrian?
- What natural and human processes altered deposits at Nimrud after original use?
- Which nearby sites, especially Ashur, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nimrud with Ashur, Persepolis and Nippur. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nimrud

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nimrud begins with a bounded claim. For the material described here as palaces reliefs and ivories, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nimrud using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Persepolis and Nippur using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Khorsabad (Dur-Sharrukin)

Mesopotamia and Iranian Plateau • Evidence dossier 130 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	**Neo-Assyrian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Khorsabad (Dur-Sharrukin) is a site or landscape associated with neo-assyrian research in Iraq. The defining record highlighted in this atlas is planned royal capital. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	planned royal capital
Chronological frame	Neo-Assyrian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are planned royal capital linked to the cited phase of Neo-Assyrian?
- What natural and human processes altered deposits at Khorsabad (Dur-Sharrukin) after original use?
- Which nearby sites, especially Mari (Tell Hariri), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Khorsabad (Dur-Sharrukin) with Mari (Tell Hariri), Pasargadae and Kish. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Khorsabad (Dur-Sharrukin)

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Khorsabad (Dur-Sharrukin) begins with a bounded claim. For the material described here as planned royal capital, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Khorsabad (Dur-Sharrukin) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pasargadae and Kish using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Babylon

Mesopotamia and Iranian Plateau • Evidence dossier 131 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	Neo-Babylonian	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Babylon is a site or landscape associated with neo-babylonian research in Iraq. The defining record highlighted in this atlas is ceremonial gateways and urban fabric. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ceremonial gateways and urban fabric
Chronological frame	Neo-Babylonian; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ceremonial gateways and urban fabric linked to the cited phase of Neo-Babylonian?
- What natural and human processes altered deposits at Babylon after original use?
- Which nearby sites, especially Ebla (Tell Mardikh), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Babylon with Ebla (Tell Mardikh), Shahr-e Sukhteh and Lagash (Tell al-Hiba). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Babylon

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Babylon begins with a bounded claim. For the material described here as ceremonial gateways and urban fabric, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Babylon using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Shahr-e Sukhteh and Lagash (Tell al-Hiba) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ashur

Mesopotamia and Iranian Plateau • Evidence dossier 132 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iraq	**Assyrian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Ashur is a site or landscape associated with assyrian research in Iraq. The defining record highlighted in this atlas is long-lived imperial center. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long-lived imperial center
Chronological frame	Assyrian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long-lived imperial center linked to the cited phase of Assyrian?
- What natural and human processes altered deposits at Ashur after original use?
- Which nearby sites, especially Tell Brak, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ashur with Tell Brak, Uruk (Warka) and Girsu (Tello). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ashur

Iraq • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ashur begins with a bounded claim. For the material described here as long-lived imperial center, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ashur using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Uruk (Warka) and Girsu (Tello) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mari (Tell Hariri)

Mesopotamia and Iranian Plateau • Evidence dossier 133 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Syria	Bronze Age	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Mari (Tell Hariri) is a site or landscape associated with bronze age research in Syria. The defining record highlighted in this atlas is palace archive and urban diplomacy. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palace archive and urban diplomacy
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palace archive and urban diplomacy linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Mari (Tell Hariri) after original use?
- Which nearby sites, especially Chogha Mish, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mari (Tell Hariri) with Chogha Mish, Ur and Nineveh. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mari (Tell Hariri)

Syria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mari (Tell Hariri) begins with a bounded claim. For the material described here as palace archive and urban diplomacy, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mari (Tell Hariri) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ur and Nineveh using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ebla (Tell Mardikh)

Mesopotamia and Iranian Plateau • Evidence dossier 134 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Syria	Bronze Age	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Ebla (Tell Mardikh) is a site or landscape associated with bronze age research in Syria. The defining record highlighted in this atlas is palace archives and city networks. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palace archives and city networks
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palace archives and city networks linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Ebla (Tell Mardikh) after original use?
- Which nearby sites, especially Susa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ebla (Tell Mardikh) with Susa, Eridu and Nimrud. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ebla (Tell Mardikh)

Syria • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ebla (Tell Mardikh) begins with a bounded claim. For the material described here as palace archives and city networks, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ebla (Tell Mardikh) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Eridu and Nimrud using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Tell Brak

Mesopotamia and Iranian Plateau • Evidence dossier 135 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Syria	Chalcolithic-Bronze	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Tell Brak is a site or landscape associated with chalcolithic-bronze research in Syria. The defining record highlighted in this atlas is early urban settlement and eye idols. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early urban settlement and eye idols
Chronological frame	Chalcolithic-Bronze; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early urban settlement and eye idols linked to the cited phase of Chalcolithic-Bronze?
- What natural and human processes altered deposits at Tell Brak after original use?
- Which nearby sites, especially Persepolis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tell Brak with Persepolis, Nippur and Khorsabad (Dur-Sharrukin). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tell Brak

Syria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tell Brak begins with a bounded claim. For the material described here as early urban settlement and eye idols, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tell Brak using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nippur and Khorsabad (Dur-Sharrukin) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chogha Mish

Mesopotamia and Iranian Plateau • Evidence dossier 136 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iran	Chalcolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Chogha Mish is a site or landscape associated with chalcolithic research in Iran. The defining record highlighted in this atlas is administrative and ceramic sequences. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	administrative and ceramic sequences
Chronological frame	Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are administrative and ceramic sequences linked to the cited phase of Chalcolithic?
- What natural and human processes altered deposits at Chogha Mish after original use?
- Which nearby sites, especially Pasargadae, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chogha Mish with Pasargadae, Kish and Babylon. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Chogha Mish

Iran • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Chogha Mish begins with a bounded claim. For the material described here as administrative and ceramic sequences, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chogha Mish using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kish and Babylon using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Susa

Mesopotamia and Iranian Plateau • Evidence dossier 137 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iran	Chalcolithic-Achaemenid	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Susa is a site or landscape associated with chalcolithic-achaemenid research in Iran. The defining record highlighted in this atlas is urban chronology and administrative material. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	urban chronology and administrative material
Chronological frame	Chalcolithic-Achaemenid; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are urban chronology and administrative material linked to the cited phase of Chalcolithic-Achaemenid?
- What natural and human processes altered deposits at Susa after original use?
- Which nearby sites, especially Shahr-e Sukhteh, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Susa with Shahr-e Sukhteh, Lagash (Tell al-Hiba) and Ashur. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Susa

Iran • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Susa begins with a bounded claim. For the material described here as urban chronology and administrative material, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Susa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lagash (Tell al-Hiba) and Ashur using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Persepolis

Mesopotamia and Iranian Plateau • Evidence dossier 138 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iran	Achaemenid	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Persepolis is a site or landscape associated with achaemenid research in Iran. The defining record highlighted in this atlas is royal terraces reliefs and inscriptions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal terraces reliefs and inscriptions
Chronological frame	Achaemenid; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal terraces reliefs and inscriptions linked to the cited phase of Achaemenid?
- What natural and human processes altered deposits at Persepolis after original use?
- Which nearby sites, especially Uruk (Warka), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Persepolis with Uruk (Warka), Girsu (Tello) and Mari (Tell Hariri). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Persepolis' architecture, reliefs and administrative tablets must be read together to distinguish royal display from institutional practice.

Persepolis

Iran • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Persepolis begins with a bounded claim. For the material described here as royal terraces reliefs and inscriptions, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Persepolis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Girsu (Tello) and Mari (Tell Hariri) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pasargadae

Mesopotamia and Iranian Plateau • Evidence dossier 139 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iran	**Achaemenid**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Pasargadae is a site or landscape associated with achaemenid research in Iran. The defining record highlighted in this atlas is early Persian royal garden and tomb. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Persian royal garden and tomb
Chronological frame	Achaemenid; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Persian royal garden and tomb linked to the cited phase of Achaemenid?
- What natural and human processes altered deposits at Pasargadae after original use?
- Which nearby sites, especially Ur, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pasargadae with Ur, Nineveh and Ebla (Tell Mardikh). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Pasargadae

Iran • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Pasargadae begins with a bounded claim. For the material described here as early Persian royal garden and tomb, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pasargadae using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nineveh and Ebla (Tell Mardikh) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Shahr-e Sukhteh

Mesopotamia and Iranian Plateau • Evidence dossier 140 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Iran	Bronze Age	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Shahr-e Sukhteh is a site or landscape associated with bronze age research in Iran. The defining record highlighted in this atlas is crafts burial grounds and trade. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: mesopotamia and iran is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	crafts burial grounds and trade
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are crafts burial grounds and trade linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Shahr-e Sukhteh after original use?
- Which nearby sites, especially Eridu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Shahr-e Sukhteh with Eridu, Nimrud and Tell Brak. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Shahr-e Sukhteh

Iran • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Shahr-e Sukhteh begins with a bounded claim. For the material described here as crafts burial grounds and trade, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Shahr-e Sukhteh using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nimrud and Tell Brak using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Göbekli Tepe

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 141 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Pre-Pottery Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Göbekli Tepe is a site or landscape associated with pre-pottery neolithic research in Türkiye. The defining record highlighted in this atlas is monumental T-pillared enclosures. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental T-pillared enclosures
Chronological frame	Pre-Pottery Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental T-pillared enclosures linked to the cited phase of Pre-Pottery Neolithic?
- What natural and human processes altered deposits at Göbekli Tepe after original use?
- Which nearby sites, especially Aşıklı Höyük, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Göbekli Tepe with Aşıklı Höyük, Alaca Höyük and Pergamon. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Klaus Schmidt's excavations made the site's T-shaped carved pillars central to the study of communal building before or during the emergence of farming.

RESEARCH POINTER: https://whc.unesco.org/en/list/1572

Göbekli Tepe

Türkiye • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Göbekli Tepe begins with a bounded claim. For the material described here as monumental T-pillared enclosures, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Göbekli Tepe using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Alaca Höyük and Pergamon using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Karahan Tepe

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 142 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Pre-Pottery Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Karahan Tepe is a site or landscape associated with pre-pottery neolithic research in Türkiye. The defining record highlighted in this atlas is monumental carved stone structures. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental carved stone structures
Chronological frame	Pre-Pottery Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental carved stone structures linked to the cited phase of Pre-Pottery Neolithic?
- What natural and human processes altered deposits at Karahan Tepe after original use?
- Which nearby sites, especially Boncuklu Höyük, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Karahan Tepe with Boncuklu Höyük, Troy (Hisarlık) and Kültepe (Kanesh). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Karahan Tepe

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Karahan Tepe begins with a bounded claim. For the material described here as monumental carved stone structures, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Karahan Tepe using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Troy (Hisarlık) and Kültepe (Kanesh) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Çatalhöyük

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 143 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Çatalhöyük is a site or landscape associated with neolithic research in Türkiye. The defining record highlighted in this atlas is dense roof-accessed settlement and wall art. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	dense roof-accessed settlement and wall art
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are dense roof-accessed settlement and wall art linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Çatalhöyük after original use?
- Which nearby sites, especially Çayönü, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Çatalhöyük with Çayönü, Gordion and Mersin-Yumuktepe. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

James Mellaart's discoveries and Ian Hodder's later collaborative project reshaped approaches to tightly packed Neolithic households and interpretive archaeology.

RESEARCH POINTER: https://whc.unesco.org/en/list/1405/

Çatalhöyük

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Çatalhöyük begins with a bounded claim. For the material described here as dense roof-accessed settlement and wall art, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Çatalhöyük using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Gordion and Mersin-Yumuktepe using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Aşıklı Höyük

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 144 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Pre-Pottery Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Aşıklı Höyük is a site or landscape associated with pre-pottery neolithic research in Türkiye. The defining record highlighted in this atlas is sedentary village and animal management. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	sedentary village and animal management
Chronological frame	Pre-Pottery Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are sedentary village and animal management linked to the cited phase of Pre-Pottery Neolithic?
- What natural and human processes altered deposits at Aşıklı Höyük after original use?
- Which nearby sites, especially Nevalı Çori, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Aşıklı Höyük with Nevalı Çori, Sardis and Khirokitia. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Aşıklı Höyük

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Aşıklı Höyük begins with a bounded claim. For the material described here as sedentary village and animal management, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Aşıklı Höyük using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sardis and Khirokitia using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Boncuklu Höyük

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 145 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Boncuklu Höyük is a site or landscape associated with neolithic research in Türkiye. The defining record highlighted in this atlas is early village lifeways. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early village lifeways
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early village lifeways linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Boncuklu Höyük after original use?
- Which nearby sites, especially Arslantepe, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Boncuklu Höyük with Arslantepe, Ephesus and Enkomi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Boncuklu Höyük

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Boncuklu Höyük begins with a bounded claim. For the material described here as early village lifeways, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Boncuklu Höyük using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ephesus and Enkomi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Çayönü

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 146 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Pre-Pottery Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Çayönü is a site or landscape associated with pre-pottery neolithic research in Türkiye. The defining record highlighted in this atlas is communal buildings and domestication research. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	communal buildings and domestication research
Chronological frame	Pre-Pottery Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are communal buildings and domestication research linked to the cited phase of Pre-Pottery Neolithic?
- What natural and human processes altered deposits at Çayönü after original use?
- Which nearby sites, especially Hattusa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Çayönü with Hattusa, Aphrodisias and Göbekli Tepe. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Çayönü

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Çayönü begins with a bounded claim. For the material described here as communal buildings and domestication research, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Çayönü using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Aphrodisias and Göbekli Tepe using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nevalı Çori

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 147 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Pre-Pottery Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Nevalı Çori is a site or landscape associated with pre-pottery neolithic research in Türkiye. The defining record highlighted in this atlas is sculpture and early village ritual. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	sculpture and early village ritual
Chronological frame	Pre-Pottery Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are sculpture and early village ritual linked to the cited phase of Pre-Pottery Neolithic?
- What natural and human processes altered deposits at Nevalı Çori after original use?
- Which nearby sites, especially Alaca Höyük, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nevalı Çori with Alaca Höyük, Pergamon and Karahan Tepe. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nevalı Çori

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Nevalı Çori begins with a bounded claim. For the material described here as sculpture and early village ritual, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nevalı Çori using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pergamon and Karahan Tepe using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Arslantepe

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 148 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Chalcolithic-Early Bronze**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Arslantepe is a site or landscape associated with chalcolithic-early bronze research in Türkiye. The defining record highlighted in this atlas is administration and elite architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	administration and elite architecture
Chronological frame	Chalcolithic-Early Bronze; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are administration and elite architecture linked to the cited phase of Chalcolithic-Early Bronze?
- What natural and human processes altered deposits at Arslantepe after original use?
- Which nearby sites, especially Troy (Hisarlık), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Arslantepe with Troy (Hisarlık), Kültepe (Kanesh) and Çatalhöyük. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Arslantepe

Türkiye • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Arslantepe begins with a bounded claim. For the material described here as administration and elite architecture, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Arslantepe using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kültepe (Kanesh) and Çatalhöyük using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hattusa

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 149 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Hittite**	**Text**

01 WHAT SURVIVES AND WHY IT MATTERS

Hattusa is a site or landscape associated with hittite research in Türkiye. The defining record highlighted in this atlas is fortified capital and cuneiform archives. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortified capital and cuneiform archives
Chronological frame	Hittite; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortified capital and cuneiform archives linked to the cited phase of Hittite?
- What natural and human processes altered deposits at Hattusa after original use?
- Which nearby sites, especially Gordion, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hattusa with Gordion, Mersin-Yumuktepe and Aşıklı Höyük. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The Hittite capital's cuneiform archives provided major evidence for a Bronze Age state and its diplomacy.

Hattusa

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hattusa begins with a bounded claim. For the material described here as fortified capital and cuneiform archives, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hattusa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mersin-Yumuktepe and Aşıklı Höyük using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Alaca Höyük

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 150 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Early Bronze-Hittite**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Alaca Höyük is a site or landscape associated with early bronze-hittite research in Türkiye. The defining record highlighted in this atlas is rich burials and monumental architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rich burials and monumental architecture
Chronological frame	Early Bronze-Hittite; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rich burials and monumental architecture linked to the cited phase of Early Bronze-Hittite?
- What natural and human processes altered deposits at Alaca Höyük after original use?
- Which nearby sites, especially Sardis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Alaca Höyük with Sardis, Khirokitia and Boncuklu Höyük. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Alaca Höyük

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Alaca Höyük begins with a bounded claim. For the material described here as rich burials and monumental architecture, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Alaca Höyük using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Khirokitia and Boncuklu Höyük using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Troy (Hisarlık)

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 151 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Bronze Age-Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Troy (Hisarlık) is a site or landscape associated with bronze age-roman research in Türkiye. The defining record highlighted in this atlas is multi-layered fortified settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	multi-layered fortified settlement
Chronological frame	Bronze Age-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are multi-layered fortified settlement linked to the cited phase of Bronze Age-Roman?
- What natural and human processes altered deposits at Troy (Hisarlık) after original use?
- Which nearby sites, especially Ephesus, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Troy (Hisarlık) with Ephesus, Enkomi and Çayönü. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Heinrich Schliemann's interventions damaged some contexts; later campaigns such as those led by Carl Blegen developed a more refined multi-level settlement sequence.

Troy (Hisarlık)

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Troy (Hisarlık) begins with a bounded claim. For the material described here as multi-layered fortified settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Troy (Hisarlık) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Enkomi and Çayönü using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Gordion

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 152 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Phrygian**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Gordion is a site or landscape associated with phrygian research in Türkiye. The defining record highlighted in this atlas is monumental tumuli and urban center. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental tumuli and urban center
Chronological frame	Phrygian; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental tumuli and urban center linked to the cited phase of Phrygian?
- What natural and human processes altered deposits at Gordion after original use?
- Which nearby sites, especially Aphrodisias, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Gordion with Aphrodisias, Göbekli Tepe and Nevalı Çori. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Gordion

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Gordion begins with a bounded claim. For the material described here as monumental tumuli and urban center, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Gordion using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Göbekli Tepe and Nevalı Çori using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sardis

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 153 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Lydian-Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Sardis is a site or landscape associated with lydian-roman research in Türkiye. The defining record highlighted in this atlas is coinage and urban excavations. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	coinage and urban excavations
Chronological frame	Lydian-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are coinage and urban excavations linked to the cited phase of Lydian-Roman?
- What natural and human processes altered deposits at Sardis after original use?
- Which nearby sites, especially Pergamon, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sardis with Pergamon, Karahan Tepe and Arslantepe. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sardis

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sardis begins with a bounded claim. For the material described here as coinage and urban excavations, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sardis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Karahan Tepe and Arslantepe using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ephesus

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 154 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Greek-Roman**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Ephesus is a site or landscape associated with greek-roman research in Türkiye. The defining record highlighted in this atlas is port city and monumental streets. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	port city and monumental streets
Chronological frame	Greek-Roman; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are port city and monumental streets linked to the cited phase of Greek-Roman?
- What natural and human processes altered deposits at Ephesus after original use?
- Which nearby sites, especially Kültepe (Kanesh), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ephesus with Kültepe (Kanesh), Çatalhöyük and Hattusa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ephesus

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ephesus begins with a bounded claim. For the material described here as port city and monumental streets, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ephesus using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Çatalhöyük and Hattusa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Aphrodisias

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 155 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Aphrodisias is a site or landscape associated with roman research in Türkiye. The defining record highlighted in this atlas is sculpture workshops and monumental city. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	sculpture workshops and monumental city
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are sculpture workshops and monumental city linked to the cited phase of Roman?
- What natural and human processes altered deposits at Aphrodisias after original use?
- Which nearby sites, especially Mersin-Yumuktepe, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Aphrodisias with Mersin-Yumuktepe, Aşıklı Höyük and Alaca Höyük. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Aphrodisias

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Aphrodisias begins with a bounded claim. For the material described here as sculpture workshops and monumental city, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Aphrodisias using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Aşıklı Höyük and Alaca Höyük using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pergamon

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 156 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Hellenistic-Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Pergamon is a site or landscape associated with hellenistic-roman research in Türkiye. The defining record highlighted in this atlas is acropolis urbanism and art. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	acropolis urbanism and art
Chronological frame	Hellenistic-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are acropolis urbanism and art linked to the cited phase of Hellenistic-Roman?
- What natural and human processes altered deposits at Pergamon after original use?
- Which nearby sites, especially Khirokitia, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pergamon with Khirokitia, Boncuklu Höyük and Troy (Hisarlık). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Pergamon

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Pergamon begins with a bounded claim. For the material described here as acropolis urbanism and art, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pergamon using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Boncuklu Höyük and Troy (Hisarlık) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kültepe (Kanesh)

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 157 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Bronze Age**	**Text**

01 WHAT SURVIVES AND WHY IT MATTERS

Kültepe (Kanesh) is a site or landscape associated with bronze age research in Türkiye. The defining record highlighted in this atlas is Assyrian merchant tablets and trade quarter. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Assyrian merchant tablets and trade quarter
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Assyrian merchant tablets and trade quarter linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Kültepe (Kanesh) after original use?
- Which nearby sites, especially Enkomi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kültepe (Kanesh) with Enkomi, Çayönü and Gordion. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kültepe (Kanesh)

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kültepe (Kanesh) begins with a bounded claim. For the material described here as Assyrian merchant tablets and trade quarter, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kültepe (Kanesh) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Çayönü and Gordion using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mersin-Yumuktepe

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 158 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Türkiye	**Neolithic onward**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Mersin-Yumuktepe is a site or landscape associated with neolithic onward research in Türkiye. The defining record highlighted in this atlas is long settlement mound sequence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long settlement mound sequence
Chronological frame	Neolithic onward; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long settlement mound sequence linked to the cited phase of Neolithic onward?
- What natural and human processes altered deposits at Mersin-Yumuktepe after original use?
- Which nearby sites, especially Göbekli Tepe, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mersin-Yumuktepe with Göbekli Tepe, Nevalı Çori and Sardis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mersin-Yumuktepe

Türkiye • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mersin-Yumuktepe begins with a bounded claim. For the material described here as long settlement mound sequence, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mersin-Yumuktepe using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nevalı Çori and Sardis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Khirokitia

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 159 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cyprus	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Khirokitia is a site or landscape associated with neolithic research in Cyprus. The defining record highlighted in this atlas is circular stone dwellings. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	circular stone dwellings
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are circular stone dwellings linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Khirokitia after original use?
- Which nearby sites, especially Karahan Tepe, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Khirokitia with Karahan Tepe, Arslantepe and Ephesus. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Khirokitia

Cyprus • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Khirokitia begins with a bounded claim. For the material described here as circular stone dwellings, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Khirokitia using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Arslantepe and Ephesus using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Enkomi

Anatolia, Cyprus, and Caucasian Fringe • Evidence dossier 160 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cyprus	**Late Bronze Age**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Enkomi is a site or landscape associated with late bronze age research in Cyprus. The defining record highlighted in this atlas is urban harbor and copper exchange. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in west asia: anatolia and cyprus is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	urban harbor and copper exchange
Chronological frame	Late Bronze Age; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are urban harbor and copper exchange linked to the cited phase of Late Bronze Age?
- What natural and human processes altered deposits at Enkomi after original use?
- Which nearby sites, especially Çatalhöyük, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Enkomi with Çatalhöyük, Hattusa and Aphrodisias. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Enkomi

Cyprus • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Enkomi begins with a bounded claim. For the material described here as urban harbor and copper exchange, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Enkomi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hattusa and Aphrodisias using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Knossos

Aegean and Ancient Greece • Evidence dossier 161 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Minoan Bronze Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Knossos is a site or landscape associated with minoan bronze age research in Greece. The defining record highlighted in this atlas is palatial complexes and Linear A. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palatial complexes and Linear A
Chronological frame	Minoan Bronze Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palatial complexes and Linear A linked to the cited phase of Minoan Bronze Age?
- What natural and human processes altered deposits at Knossos after original use?
- Which nearby sites, especially Zakros, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Knossos with Zakros, Lerna and Delphi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Arthur Evans' excavations brought Minoan palatial culture to prominence, although aspects of his visual reconstructions require cautious interpretation.

Knossos

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Knossos begins with a bounded claim. For the material described here as palatial complexes and Linear A, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Knossos using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lerna and Delphi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Phaistos

Aegean and Ancient Greece • Evidence dossier 162 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Minoan Bronze Age	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Phaistos is a site or landscape associated with minoan bronze age research in Greece. The defining record highlighted in this atlas is palace and inscribed disk. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palace and inscribed disk
Chronological frame	Minoan Bronze Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palace and inscribed disk linked to the cited phase of Minoan Bronze Age?
- What natural and human processes altered deposits at Phaistos after original use?
- Which nearby sites, especially Akrotiri (Thera), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Phaistos with Akrotiri (Thera), Sesklo and Olympia. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Phaistos

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Phaistos begins with a bounded claim. For the material described here as palace and inscribed disk, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Phaistos using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sesklo and Olympia using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Malia

Aegean and Ancient Greece • Evidence dossier 163 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Minoan Bronze Age	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Malia is a site or landscape associated with minoan bronze age research in Greece. The defining record highlighted in this atlas is palace and town. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palace and town
Chronological frame	Minoan Bronze Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palace and town linked to the cited phase of Minoan Bronze Age?
- What natural and human processes altered deposits at Malia after original use?
- Which nearby sites, especially Mycenae, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Malia with Mycenae, Dimini and Athens Acropolis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Malia

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Malia begins with a bounded claim. For the material described here as palace and town, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Malia using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dimini and Athens Acropolis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Zakros

Aegean and Ancient Greece • Evidence dossier 164 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Minoan Bronze Age	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Zakros is a site or landscape associated with minoan bronze age research in Greece. The defining record highlighted in this atlas is coastal palace and workshops. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	coastal palace and workshops
Chronological frame	Minoan Bronze Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are coastal palace and workshops linked to the cited phase of Minoan Bronze Age?
- What natural and human processes altered deposits at Zakros after original use?
- Which nearby sites, especially Tiryns, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Zakros with Tiryns, Dispilio and Corinth. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Zakros

Greece • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Zakros begins with a bounded claim. For the material described here as coastal palace and workshops, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Zakros using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dispilio and Corinth using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Akrotiri (Thera)

Aegean and Ancient Greece • Evidence dossier 165 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Bronze Age	Art

01 WHAT SURVIVES AND WHY IT MATTERS

Akrotiri (Thera) is a site or landscape associated with bronze age research in Greece. The defining record highlighted in this atlas is volcanic-buried town and frescoes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	volcanic-buried town and frescoes
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are volcanic-buried town and frescoes linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Akrotiri (Thera) after original use?
- Which nearby sites, especially Pylos (Palace of Nestor), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Akrotiri (Thera) with Pylos (Palace of Nestor), Franchthi Cave and Delos. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Work led by Spyridon Marinatos revealed an Aegean town sealed by volcanic deposits, preserving exceptional architecture and frescoes.

Akrotiri (Thera)

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Akrotiri (Thera) begins with a bounded claim. For the material described here as volcanic-buried town and frescoes, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Akrotiri (Thera) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Franchthi Cave and Delos using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mycenae

Aegean and Ancient Greece • Evidence dossier 166 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Late Bronze Age	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Mycenae is a site or landscape associated with late bronze age research in Greece. The defining record highlighted in this atlas is citadel shaft graves and tholos tombs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	citadel shaft graves and tholos tombs
Chronological frame	Late Bronze Age; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are citadel shaft graves and tholos tombs linked to the cited phase of Late Bronze Age?
- What natural and human processes altered deposits at Mycenae after original use?
- Which nearby sites, especially Thebes, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mycenae with Thebes, Dikili Tash and Knossos. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Heinrich Schliemann's nineteenth-century excavations of shaft graves made Late Bronze Age elite material culture internationally prominent.

Mycenae

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mycenae begins with a bounded claim. For the material described here as citadel shaft graves and tholos tombs, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mycenae using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dikili Tash and Knossos using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tiryns

Aegean and Ancient Greece • Evidence dossier 167 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Late Bronze Age	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Tiryns is a site or landscape associated with late bronze age research in Greece. The defining record highlighted in this atlas is cyclopean fortification. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	cyclopean fortification
Chronological frame	Late Bronze Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are cyclopean fortification linked to the cited phase of Late Bronze Age?
- What natural and human processes altered deposits at Tiryns after original use?
- Which nearby sites, especially Lerna, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tiryns with Lerna, Delphi and Phaistos. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tiryns

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tiryns begins with a bounded claim. For the material described here as cyclopean fortification, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tiryns using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Delphi and Phaistos using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pylos (Palace of Nestor)

Aegean and Ancient Greece • Evidence dossier 168 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Late Bronze Age	Text

01 WHAT SURVIVES AND WHY IT MATTERS

Pylos (Palace of Nestor) is a site or landscape associated with late bronze age research in Greece. The defining record highlighted in this atlas is Linear B archive and palace. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Linear B archive and palace
Chronological frame	Late Bronze Age; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Linear B archive and palace linked to the cited phase of Late Bronze Age?
- What natural and human processes altered deposits at Pylos (Palace of Nestor) after original use?
- Which nearby sites, especially Sesklo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pylos (Palace of Nestor) with Sesklo, Olympia and Malia. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Carl Blegen's excavations brought to light a palace archive in Linear B, crucial for the study of Mycenaean administration.

Pylos (Palace of Nestor)

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Pylos (Palace of Nestor) begins with a bounded claim. For the material described here as Linear B archive and palace, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pylos (Palace of Nestor) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Olympia and Malia using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Thebes

Aegean and Ancient Greece • Evidence dossier 169 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Mycenaean-Classical	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Thebes is a site or landscape associated with mycenaean-classical research in Greece. The defining record highlighted in this atlas is palatial and urban layers. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palatial and urban layers
Chronological frame	Mycenaean-Classical; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palatial and urban layers linked to the cited phase of Mycenaean-Classical?
- What natural and human processes altered deposits at Thebes after original use?
- Which nearby sites, especially Dimini, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Thebes with Dimini, Athens Acropolis and Zakros. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Thebes

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Thebes begins with a bounded claim. For the material described here as palatial and urban layers, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Thebes using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Athens Acropolis and Zakros using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lerna

Aegean and Ancient Greece • Evidence dossier 170 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Early Bronze Age	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Lerna is a site or landscape associated with early bronze age research in Greece. The defining record highlighted in this atlas is corridor house architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	corridor house architecture
Chronological frame	Early Bronze Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are corridor house architecture linked to the cited phase of Early Bronze Age?
- What natural and human processes altered deposits at Lerna after original use?
- Which nearby sites, especially Dispilio, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lerna with Dispilio, Corinth and Akrotiri (Thera). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lerna

Greece • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Lerna begins with a bounded claim. For the material described here as corridor house architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lerna using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Corinth and Akrotiri (Thera) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sesklo

Aegean and Ancient Greece • Evidence dossier 171 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Sesklo is a site or landscape associated with neolithic research in Greece. The defining record highlighted in this atlas is early farming settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early farming settlement
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early farming settlement linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Sesklo after original use?
- Which nearby sites, especially Franchthi Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sesklo with Franchthi Cave, Delos and Mycenae. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sesklo

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sesklo begins with a bounded claim. For the material described here as early farming settlement, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sesklo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Delos and Mycenae using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dimini

Aegean and Ancient Greece • Evidence dossier 172 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Dimini is a site or landscape associated with neolithic research in Greece. The defining record highlighted in this atlas is settlement enclosure and domestic planning. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	settlement enclosure and domestic planning
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are settlement enclosure and domestic planning linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Dimini after original use?
- Which nearby sites, especially Dikili Tash, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dimini with Dikili Tash, Knossos and Tiryns. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dimini

Greece • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Dimini begins with a bounded claim. For the material described here as settlement enclosure and domestic planning, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dimini using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Knossos and Tiryns using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dispilio

Aegean and Ancient Greece • Evidence dossier 173 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Dispilio is a site or landscape associated with neolithic research in Greece. The defining record highlighted in this atlas is lakeshore pile settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	lakeshore pile settlement
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are lakeshore pile settlement linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Dispilio after original use?
- Which nearby sites, especially Delphi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dispilio with Delphi, Phaistos and Pylos (Palace of Nestor). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dispilio

Greece • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Dispilio begins with a bounded claim. For the material described here as lakeshore pile settlement, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dispilio using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Phaistos and Pylos (Palace of Nestor) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Franchthi Cave

Aegean and Ancient Greece • Evidence dossier 174 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Paleolithic-Neolithic	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Franchthi Cave is a site or landscape associated with paleolithic-neolithic research in Greece. The defining record highlighted in this atlas is long cave sequence and plant domestication. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long cave sequence and plant domestication
Chronological frame	Paleolithic-Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long cave sequence and plant domestication linked to the cited phase of Paleolithic-Neolithic?
- What natural and human processes altered deposits at Franchthi Cave after original use?
- Which nearby sites, especially Olympia, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Franchthi Cave with Olympia, Malia and Thebes. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Franchthi Cave

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Franchthi Cave begins with a bounded claim. For the material described here as long cave sequence and plant domestication, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Franchthi Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Malia and Thebes using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dikili Tash

Aegean and Ancient Greece • Evidence dossier 175 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Neolithic-Bronze	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Dikili Tash is a site or landscape associated with neolithic-bronze research in Greece. The defining record highlighted in this atlas is settlement stratigraphy and agriculture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	settlement stratigraphy and agriculture
Chronological frame	Neolithic-Bronze; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are settlement stratigraphy and agriculture linked to the cited phase of Neolithic-Bronze?
- What natural and human processes altered deposits at Dikili Tash after original use?
- Which nearby sites, especially Athens Acropolis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dikili Tash with Athens Acropolis, Zakros and Lerna. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dikili Tash

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dikili Tash begins with a bounded claim. For the material described here as settlement stratigraphy and agriculture, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dikili Tash using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Zakros and Lerna using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Delphi

Aegean and Ancient Greece • Evidence dossier 176 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Archaic-Roman	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Delphi is a site or landscape associated with archaic-roman research in Greece. The defining record highlighted in this atlas is pan-Hellenic sanctuary. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	pan-Hellenic sanctuary
Chronological frame	Archaic-Roman; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are pan-Hellenic sanctuary linked to the cited phase of Archaic-Roman?
- What natural and human processes altered deposits at Delphi after original use?
- Which nearby sites, especially Corinth, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Delphi with Corinth, Akrotiri (Thera) and Sesklo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Delphi

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Delphi begins with a bounded claim. For the material described here as pan-Hellenic sanctuary, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Delphi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Akrotiri (Thera) and Sesklo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Olympia

Aegean and Ancient Greece • Evidence dossier 177 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Archaic-Roman	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Olympia is a site or landscape associated with archaic-roman research in Greece. The defining record highlighted in this atlas is sanctuary and athletic institutions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	sanctuary and athletic institutions
Chronological frame	Archaic-Roman; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are sanctuary and athletic institutions linked to the cited phase of Archaic-Roman?
- What natural and human processes altered deposits at Olympia after original use?
- Which nearby sites, especially Delos, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Olympia with Delos, Mycenae and Dimini. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Olympia

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Olympia begins with a bounded claim. For the material described here as sanctuary and athletic institutions, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Olympia using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mycenae and Dimini using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Athens Acropolis

Aegean and Ancient Greece • Evidence dossier 178 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Bronze Age-Classical	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Athens Acropolis is a site or landscape associated with bronze age-classical research in Greece. The defining record highlighted in this atlas is monumental sanctuary. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental sanctuary
Chronological frame	Bronze Age-Classical; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental sanctuary linked to the cited phase of Bronze Age-Classical?
- What natural and human processes altered deposits at Athens Acropolis after original use?
- Which nearby sites, especially Knossos, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Athens Acropolis with Knossos, Tiryns and Dispilio. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Athens Acropolis

Greece • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Athens Acropolis begins with a bounded claim. For the material described here as monumental sanctuary, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Athens Acropolis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tiryns and Dispilio using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Corinth

Aegean and Ancient Greece • Evidence dossier 179 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Greek-Roman	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Corinth is a site or landscape associated with greek-roman research in Greece. The defining record highlighted in this atlas is urban commerce and religious monuments. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	urban commerce and religious monuments
Chronological frame	Greek-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are urban commerce and religious monuments linked to the cited phase of Greek-Roman?
- What natural and human processes altered deposits at Corinth after original use?
- Which nearby sites, especially Phaistos, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Corinth with Phaistos, Pylos (Palace of Nestor) and Franchthi Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Corinth

Greece • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Corinth begins with a bounded claim. For the material described here as urban commerce and religious monuments, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Corinth using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pylos (Palace of Nestor) and Franchthi Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Delos

Aegean and Ancient Greece • Evidence dossier 180 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Greece	Hellenistic-Roman	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Delos is a site or landscape associated with hellenistic-roman research in Greece. The defining record highlighted in this atlas is cosmopolitan commercial island. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: aegean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	cosmopolitan commercial island
Chronological frame	Hellenistic-Roman; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are cosmopolitan commercial island linked to the cited phase of Hellenistic-Roman?
- What natural and human processes altered deposits at Delos after original use?
- Which nearby sites, especially Malia, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Delos with Malia, Thebes and Dikili Tash. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Delos

Greece • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Delos begins with a bounded claim. For the material described here as cosmopolitan commercial island, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Delos using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Thebes and Dikili Tash using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Roman Forum

Classical Rome and Mediterranean • Evidence dossier 181 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	Roman Republic-Empire	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Roman Forum is a site or landscape associated with roman republic-empire research in Italy. The defining record highlighted in this atlas is civic center and political monuments. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	civic center and political monuments
Chronological frame	Roman Republic-Empire; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are civic center and political monuments linked to the cited phase of Roman Republic-Empire?
- What natural and human processes altered deposits at Roman Forum after original use?
- Which nearby sites, especially Herculaneum, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Roman Forum with Herculaneum, Tarquinia Necropolis and Piazza Armerina. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Roman Forum

Italy • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Roman Forum begins with a bounded claim. For the material described here as civic center and political monuments, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Roman Forum using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tarquinia Necropolis and Piazza Armerina using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Palatine Hill

Classical Rome and Mediterranean • Evidence dossier 182 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	Iron Age-Roman	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Palatine Hill is a site or landscape associated with iron age-roman research in Italy. The defining record highlighted in this atlas is palatial architecture and early settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palatial architecture and early settlement
Chronological frame	Iron Age-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palatial architecture and early settlement linked to the cited phase of Iron Age-Roman?
- What natural and human processes altered deposits at Palatine Hill after original use?
- Which nearby sites, especially Ostia Antica, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Palatine Hill with Ostia Antica, Cerveteri Necropolis and Timgad. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Palatine Hill

Italy • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Palatine Hill begins with a bounded claim. For the material described here as palatial architecture and early settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Palatine Hill using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cerveteri Necropolis and Timgad using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pompeii

Classical Rome and Mediterranean • Evidence dossier 183 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Pompeii is a site or landscape associated with roman research in Italy. The defining record highlighted in this atlas is volcanically preserved urban lifeways. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	volcanically preserved urban lifeways
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are volcanically preserved urban lifeways linked to the cited phase of Roman?
- What natural and human processes altered deposits at Pompeii after original use?
- Which nearby sites, especially Portus, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pompeii with Portus, Veii and Djemila. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

From early treasure digging to Giuseppe Fiorelli's recording reforms and modern conservation, Pompeii demonstrates why a destroyed city is not a frozen complete snapshot.

Pompeii

Italy • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Pompeii begins with a bounded claim. For the material described here as volcanically preserved urban lifeways, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pompeii using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Veii and Djemila using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Herculaneum

Classical Rome and Mediterranean • Evidence dossier 184 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Roman**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Herculaneum is a site or landscape associated with roman research in Italy. The defining record highlighted in this atlas is volcanically sealed houses and organic remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	volcanically sealed houses and organic remains
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are volcanically sealed houses and organic remains linked to the cited phase of Roman?
- What natural and human processes altered deposits at Herculaneum after original use?
- Which nearby sites, especially Hadrian's Villa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Herculaneum with Hadrian's Villa, Cumae and Baalbek. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Exceptional preservation of organic materials provides information on buildings and ordinary life rarely visible in other Roman settlements.

Herculaneum

Italy • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Herculaneum begins with a bounded claim. For the material described here as volcanically sealed houses and organic remains, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Herculaneum using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cumae and Baalbek using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ostia Antica

Classical Rome and Mediterranean • Evidence dossier 185 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Roman**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Ostia Antica is a site or landscape associated with roman research in Italy. The defining record highlighted in this atlas is port city warehouses and apartment blocks. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	port city warehouses and apartment blocks
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are port city warehouses and apartment blocks linked to the cited phase of Roman?
- What natural and human processes altered deposits at Ostia Antica after original use?
- Which nearby sites, especially Villa of the Quintilii, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ostia Antica with Villa of the Quintilii, Baiae and Palmyra. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ostia Antica

Italy • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ostia Antica begins with a bounded claim. For the material described here as port city warehouses and apartment blocks, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ostia Antica using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Baiae and Palmyra using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Portus

Classical Rome and Mediterranean • Evidence dossier 186 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Roman**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Portus is a site or landscape associated with roman research in Italy. The defining record highlighted in this atlas is imperial harbor infrastructure. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	imperial harbor infrastructure
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are imperial harbor infrastructure linked to the cited phase of Roman?
- What natural and human processes altered deposits at Portus after original use?
- Which nearby sites, especially Paestum, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Portus with Paestum, Aquileia and Roman Forum. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Portus

Italy • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Portus begins with a bounded claim. For the material described here as imperial harbor infrastructure, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Portus using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Aquileia and Roman Forum using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hadrian's Villa

Classical Rome and Mediterranean • Evidence dossier 187 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Roman**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Hadrian's Villa is a site or landscape associated with roman research in Italy. The defining record highlighted in this atlas is imperial residential landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	imperial residential landscape
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are imperial residential landscape linked to the cited phase of Roman?
- What natural and human processes altered deposits at Hadrian's Villa after original use?
- Which nearby sites, especially Tarquinia Necropolis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hadrian's Villa with Tarquinia Necropolis, Piazza Armerina and Palatine Hill. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hadrian's Villa

Italy • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hadrian's Villa begins with a bounded claim. For the material described here as imperial residential landscape, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hadrian's Villa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Piazza Armerina and Palatine Hill using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Villa of the Quintilii

Classical Rome and Mediterranean • Evidence dossier 188 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Villa of the Quintilii is a site or landscape associated with roman research in Italy. The defining record highlighted in this atlas is elite estate and roadside architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	elite estate and roadside architecture
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are elite estate and roadside architecture linked to the cited phase of Roman?
- What natural and human processes altered deposits at Villa of the Quintilii after original use?
- Which nearby sites, especially Cerveteri Necropolis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Villa of the Quintilii with Cerveteri Necropolis, Timgad and Pompeii. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Villa of the Quintilii

Italy • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Villa of the Quintilii begins with a bounded claim. For the material described here as elite estate and roadside architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Villa of the Quintilii using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Timgad and Pompeii using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Paestum

Classical Rome and Mediterranean • Evidence dossier 189 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	Greek-Roman	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Paestum is a site or landscape associated with greek-roman research in Italy. The defining record highlighted in this atlas is temples and urban grid. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temples and urban grid
Chronological frame	Greek-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temples and urban grid linked to the cited phase of Greek-Roman?
- What natural and human processes altered deposits at Paestum after original use?
- Which nearby sites, especially Veii, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Paestum with Veii, Djemila and Herculaneum. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Paestum

Italy • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Paestum begins with a bounded claim. For the material described here as temples and urban grid, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Paestum using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Djemila and Herculaneum using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tarquinia Necropolis

Classical Rome and Mediterranean • Evidence dossier 190 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	Etruscan	Art

01 WHAT SURVIVES AND WHY IT MATTERS

Tarquinia Necropolis is a site or landscape associated with etruscan research in Italy. The defining record highlighted in this atlas is painted chamber tombs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	painted chamber tombs
Chronological frame	Etruscan; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are painted chamber tombs linked to the cited phase of Etruscan?
- What natural and human processes altered deposits at Tarquinia Necropolis after original use?
- Which nearby sites, especially Cumae, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tarquinia Necropolis with Cumae, Baalbek and Ostia Antica. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tarquinia Necropolis

Italy • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Tarquinia Necropolis begins with a bounded claim. For the material described here as painted chamber tombs, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tarquinia Necropolis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Baalbek and Ostia Antica using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cerveteri Necropolis

Classical Rome and Mediterranean • Evidence dossier 191 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Etruscan**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Cerveteri Necropolis is a site or landscape associated with etruscan research in Italy. The defining record highlighted in this atlas is tumuli and rock-cut tombs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	tumuli and rock-cut tombs
Chronological frame	Etruscan; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are tumuli and rock-cut tombs linked to the cited phase of Etruscan?
- What natural and human processes altered deposits at Cerveteri Necropolis after original use?
- Which nearby sites, especially Baiae, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cerveteri Necropolis with Baiae, Palmyra and Portus. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Cerveteri Necropolis

Italy • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Cerveteri Necropolis begins with a bounded claim. For the material described here as tumuli and rock-cut tombs, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cerveteri Necropolis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Palmyra and Portus using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Veii

Classical Rome and Mediterranean • Evidence dossier 192 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	Etruscan-Roman	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Veii is a site or landscape associated with etruscan-roman research in Italy. The defining record highlighted in this atlas is sanctuaries and urban occupation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	sanctuaries and urban occupation
Chronological frame	Etruscan-Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are sanctuaries and urban occupation linked to the cited phase of Etruscan-Roman?
- What natural and human processes altered deposits at Veii after original use?
- Which nearby sites, especially Aquileia, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Veii with Aquileia, Roman Forum and Hadrian's Villa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Veii

Italy • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Veii begins with a bounded claim. For the material described here as sanctuaries and urban occupation, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Veii using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Roman Forum and Hadrian's Villa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cumae

Classical Rome and Mediterranean • Evidence dossier 193 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	Greek-Roman	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Cumae is a site or landscape associated with greek-roman research in Italy. The defining record highlighted in this atlas is colonial town and cave sanctuaries. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	colonial town and cave sanctuaries
Chronological frame	Greek-Roman; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are colonial town and cave sanctuaries linked to the cited phase of Greek-Roman?
- What natural and human processes altered deposits at Cumae after original use?
- Which nearby sites, especially Piazza Armerina, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cumae with Piazza Armerina, Palatine Hill and Villa of the Quintilii. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Cumae

Italy • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Cumae begins with a bounded claim. For the material described here as colonial town and cave sanctuaries, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cumae using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Palatine Hill and Villa of the Quintilii using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Baiae

Classical Rome and Mediterranean • Evidence dossier 194 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Roman**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Baiae is a site or landscape associated with roman research in Italy. The defining record highlighted in this atlas is submerged coastal luxury landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	submerged coastal luxury landscape
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are submerged coastal luxury landscape linked to the cited phase of Roman?
- What natural and human processes altered deposits at Baiae after original use?
- Which nearby sites, especially Timgad, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Baiae with Timgad, Pompeii and Paestum. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Baiae

Italy • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Baiae begins with a bounded claim. For the material described here as submerged coastal luxury landscape, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Baiae using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pompeii and Paestum using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Aquileia

Classical Rome and Mediterranean • Evidence dossier 195 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	**Roman**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Aquileia is a site or landscape associated with roman research in Italy. The defining record highlighted in this atlas is northern Adriatic urban port. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	northern Adriatic urban port
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are northern Adriatic urban port linked to the cited phase of Roman?
- What natural and human processes altered deposits at Aquileia after original use?
- Which nearby sites, especially Djemila, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Aquileia with Djemila, Herculaneum and Tarquinia Necropolis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Aquileia

Italy • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Aquileia begins with a bounded claim. For the material described here as northern Adriatic urban port, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Aquileia using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Herculaneum and Tarquinia Necropolis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Piazza Armerina

Classical Rome and Mediterranean • Evidence dossier 196 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Italy	Late Roman	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Piazza Armerina is a site or landscape associated with late roman research in Italy. The defining record highlighted in this atlas is villa mosaics and elite residence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	villa mosaics and elite residence
Chronological frame	Late Roman; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are villa mosaics and elite residence linked to the cited phase of Late Roman?
- What natural and human processes altered deposits at Piazza Armerina after original use?
- Which nearby sites, especially Baalbek, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Piazza Armerina with Baalbek, Ostia Antica and Cerveteri Necropolis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Piazza Armerina

Italy • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Piazza Armerina begins with a bounded claim. For the material described here as villa mosaics and elite residence, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Piazza Armerina using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ostia Antica and Cerveteri Necropolis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Timgad

Classical Rome and Mediterranean • Evidence dossier 197 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Algeria	**Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Timgad is a site or landscape associated with roman research in Algeria. The defining record highlighted in this atlas is planned colonial town and street grid. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	planned colonial town and street grid
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are planned colonial town and street grid linked to the cited phase of Roman?
- What natural and human processes altered deposits at Timgad after original use?
- Which nearby sites, especially Palmyra, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Timgad with Palmyra, Portus and Veii. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Timgad

Algeria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Timgad begins with a bounded claim. For the material described here as planned colonial town and street grid, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Timgad using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Portus and Veii using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Djemila

Classical Rome and Mediterranean • Evidence dossier 198 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Algeria	Roman	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Djemila is a site or landscape associated with roman research in Algeria. The defining record highlighted in this atlas is provincial city in mountain terrain. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	provincial city in mountain terrain
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are provincial city in mountain terrain linked to the cited phase of Roman?
- What natural and human processes altered deposits at Djemila after original use?
- Which nearby sites, especially Roman Forum, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Djemila with Roman Forum, Hadrian's Villa and Cumae. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Djemila

Algeria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Djemila begins with a bounded claim. For the material described here as provincial city in mountain terrain, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Djemila using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hadrian's Villa and Cumae using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Baalbek

Classical Rome and Mediterranean • Evidence dossier 199 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Lebanon	Roman	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Baalbek is a site or landscape associated with roman research in Lebanon. The defining record highlighted in this atlas is monumental temple architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental temple architecture
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental temple architecture linked to the cited phase of Roman?
- What natural and human processes altered deposits at Baalbek after original use?
- Which nearby sites, especially Palatine Hill, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Baalbek with Palatine Hill, Villa of the Quintilii and Baiae. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Baalbek

Lebanon • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Baalbek begins with a bounded claim. For the material described here as monumental temple architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Baalbek using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Villa of the Quintilii and Baiae using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Palmyra

Classical Rome and Mediterranean • Evidence dossier 200 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Syria	Roman-era	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Palmyra is a site or landscape associated with roman-era research in Syria. The defining record highlighted in this atlas is caravan city and funerary towers. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: rome and mediterranean is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	caravan city and funerary towers
Chronological frame	Roman-era; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are caravan city and funerary towers linked to the cited phase of Roman-era?
- What natural and human processes altered deposits at Palmyra after original use?
- Which nearby sites, especially Pompeii, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Palmyra with Pompeii, Paestum and Aquileia. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Palmyra

Syria • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Palmyra begins with a bounded claim. For the material described here as caravan city and funerary towers, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Palmyra using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Paestum and Aquileia using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Stonehenge

Atlantic and Northern European Prehistory • Evidence dossier 201 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Neolithic-Bronze Age**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Stonehenge is a site or landscape associated with neolithic-bronze age research in United Kingdom. The defining record highlighted in this atlas is megalithic monument and landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	megalithic monument and landscape
Chronological frame	Neolithic-Bronze Age; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are megalithic monument and landscape linked to the cited phase of Neolithic-Bronze Age?
- What natural and human processes altered deposits at Stonehenge after original use?
- Which nearby sites, especially Skara Brae, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Stonehenge with Skara Brae, Carrowmore and Altamira. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Its multi-phase construction and changing landscape context are better established than any single definitive explanation of its symbolic purpose.

Stonehenge

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Stonehenge begins with a bounded claim. For the material described here as megalithic monument and landscape, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Stonehenge using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Carrowmore and Altamira using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Avebury

Atlantic and Northern European Prehistory • Evidence dossier 202 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Avebury is a site or landscape associated with neolithic research in United Kingdom. The defining record highlighted in this atlas is large henge and stone circles. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large henge and stone circles
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large henge and stone circles linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Avebury after original use?
- Which nearby sites, especially Maeshowe, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Avebury with Maeshowe, Gavrinis and El Castillo Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Avebury

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Avebury begins with a bounded claim. For the material described here as large henge and stone circles, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Avebury using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Gavrinis and El Castillo Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Durrington Walls

Atlantic and Northern European Prehistory • Evidence dossier 203 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Durrington Walls is a site or landscape associated with neolithic research in United Kingdom. The defining record highlighted in this atlas is large timber-and-earthwork complex. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large timber-and-earthwork complex
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large timber-and-earthwork complex linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Durrington Walls after original use?
- Which nearby sites, especially Ness of Brodgar, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Durrington Walls with Ness of Brodgar, Carnac Alignments and Atapuerca. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Durrington Walls

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Durrington Walls begins with a bounded claim. For the material described here as large timber-and-earthwork complex, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Durrington Walls using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Carnac Alignments and Atapuerca using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Skara Brae

Atlantic and Northern European Prehistory • Evidence dossier 204 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Skara Brae is a site or landscape associated with neolithic research in United Kingdom. The defining record highlighted in this atlas is stone-built village interiors. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	stone-built village interiors
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are stone-built village interiors linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Skara Brae after original use?
- Which nearby sites, especially Newgrange, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Skara Brae with Newgrange, Barnenez and La Draga. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Skara Brae

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Skara Brae begins with a bounded claim. For the material described here as stone-built village interiors, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Skara Brae using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Barnenez and La Draga using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Maeshowe

Atlantic and Northern European Prehistory • Evidence dossier 205 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Neolithic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Maeshowe is a site or landscape associated with neolithic research in United Kingdom. The defining record highlighted in this atlas is chambered passage tomb. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	chambered passage tomb
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are chambered passage tomb linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Maeshowe after original use?
- Which nearby sites, especially Knowth, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Maeshowe with Knowth, Lascaux Cave and Los Millares. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Maeshowe

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Maeshowe begins with a bounded claim. For the material described here as chambered passage tomb, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Maeshowe using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lascaux Cave and Los Millares using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ness of Brodgar

Atlantic and Northern European Prehistory • Evidence dossier 206 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Ness of Brodgar is a site or landscape associated with neolithic research in United Kingdom. The defining record highlighted in this atlas is monumental building complex. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental building complex
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental building complex linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Ness of Brodgar after original use?
- Which nearby sites, especially Poulnabrone, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ness of Brodgar with Poulnabrone, Chauvet Cave and Stonehenge. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ness of Brodgar

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ness of Brodgar begins with a bounded claim. For the material described here as monumental building complex, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ness of Brodgar using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chauvet Cave and Stonehenge using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Newgrange

Atlantic and Northern European Prehistory • Evidence dossier 207 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ireland	**Neolithic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Newgrange is a site or landscape associated with neolithic research in Ireland. The defining record highlighted in this atlas is passage tomb and solstitial alignment. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	passage tomb and solstitial alignment
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are passage tomb and solstitial alignment linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Newgrange after original use?
- Which nearby sites, especially Carrowmore, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Newgrange with Carrowmore, Altamira and Avebury. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Newgrange

Ireland • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Newgrange begins with a bounded claim. For the material described here as passage tomb and solstitial alignment, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Newgrange using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Altamira and Avebury using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Knowth

Atlantic and Northern European Prehistory • Evidence dossier 208 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ireland	**Neolithic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Knowth is a site or landscape associated with neolithic research in Ireland. The defining record highlighted in this atlas is passage mound and megalithic art. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	passage mound and megalithic art
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are passage mound and megalithic art linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Knowth after original use?
- Which nearby sites, especially Gavrinis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Knowth with Gavrinis, El Castillo Cave and Durrington Walls. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Knowth

Ireland • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Knowth begins with a bounded claim. For the material described here as passage mound and megalithic art, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Knowth using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with El Castillo Cave and Durrington Walls using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Poulnabrone

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ireland	**Neolithic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Poulnabrone is a site or landscape associated with neolithic research in Ireland. The defining record highlighted in this atlas is portal tomb burials. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	portal tomb burials
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are portal tomb burials linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Poulnabrone after original use?
- Which nearby sites, especially Carnac Alignments, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Poulnabrone with Carnac Alignments, Atapuerca and Skara Brae. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Poulnabrone

Ireland • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Poulnabrone begins with a bounded claim. For the material described here as portal tomb burials, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Poulnabrone using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Atapuerca and Skara Brae using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Carrowmore

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ireland	**Neolithic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Carrowmore is a site or landscape associated with neolithic research in Ireland. The defining record highlighted in this atlas is megalithic cemetery. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	megalithic cemetery
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are megalithic cemetery linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Carrowmore after original use?
- Which nearby sites, especially Barnenez, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Carrowmore with Barnenez, La Draga and Maeshowe. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Carrowmore

Ireland • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Carrowmore begins with a bounded claim. For the material described here as megalithic cemetery, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Carrowmore using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with La Draga and Maeshowe using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Gavrinis

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
France	**Neolithic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Gavrinis is a site or landscape associated with neolithic research in France. The defining record highlighted in this atlas is decorated passage tomb stones. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	decorated passage tomb stones
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are decorated passage tomb stones linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Gavrinis after original use?
- Which nearby sites, especially Lascaux Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Gavrinis with Lascaux Cave, Los Millares and Ness of Brodgar. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Gavrinis

France • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Gavrinis begins with a bounded claim. For the material described here as decorated passage tomb stones, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Gavrinis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Los Millares and Ness of Brodgar using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Carnac Alignments

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
France	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Carnac Alignments is a site or landscape associated with neolithic research in France. The defining record highlighted in this atlas is standing stone arrays. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	standing stone arrays
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are standing stone arrays linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Carnac Alignments after original use?
- Which nearby sites, especially Chauvet Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Carnac Alignments with Chauvet Cave, Stonehenge and Newgrange. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Carnac Alignments

France • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Carnac Alignments begins with a bounded claim. For the material described here as standing stone arrays, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Carnac Alignments using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Stonehenge and Newgrange using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Barnenez

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
France	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Barnenez is a site or landscape associated with neolithic research in France. The defining record highlighted in this atlas is multi-chambered cairn. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	multi-chambered cairn
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are multi-chambered cairn linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Barnenez after original use?
- Which nearby sites, especially Altamira, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Barnenez with Altamira, Avebury and Knowth. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Barnenez

France • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Barnenez begins with a bounded claim. For the material described here as multi-chambered cairn, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Barnenez using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Avebury and Knowth using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lascaux Cave

Atlantic and Northern European Prehistory • Evidence dossier 214 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
France	Upper Paleolithic	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Lascaux Cave is a site or landscape associated with upper paleolithic research in France. The defining record highlighted in this atlas is painted animals and cave art. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	painted animals and cave art
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are painted animals and cave art linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Lascaux Cave after original use?
- Which nearby sites, especially El Castillo Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lascaux Cave with El Castillo Cave, Durrington Walls and Poulnabrone. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The paintings were brought to wider attention in 1940, and the original cave's conservation problems later motivated strict visitor controls.

Lascaux Cave

France • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Lascaux Cave begins with a bounded claim. For the material described here as painted animals and cave art, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 **Osteological morphology and CT imaging**

Record sample/context identifiers and validate the chain of custody.

02 **Context-controlled geochronology**

Choose independent controls, repeat measurements and quantify uncertainty.

03 **Ancient DNA or proteomics where preservation allows**

Relate results to the local stratigraphic sequence and regional chronology.

04 **Taphonomic and paleoenvironmental analysis**

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lascaux Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Durrington Walls and Poulnabrone using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chauvet Cave

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
France	**Upper Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Chauvet Cave is a site or landscape associated with upper paleolithic research in France. The defining record highlighted in this atlas is early figurative cave imagery. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early figurative cave imagery
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early figurative cave imagery linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Chauvet Cave after original use?
- Which nearby sites, especially Atapuerca, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chauvet Cave with Atapuerca, Skara Brae and Carrowmore. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Discovered in 1994, the cave has important early figurative art; access restrictions emphasize preservation of the fragile environment.

Chauvet Cave

France • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Chauvet Cave begins with a bounded claim. For the material described here as early figurative cave imagery, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chauvet Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Skara Brae and Carrowmore using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Altamira

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Spain	**Upper Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Altamira is a site or landscape associated with upper paleolithic research in Spain. The defining record highlighted in this atlas is polychrome cave paintings. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	polychrome cave paintings
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are polychrome cave paintings linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Altamira after original use?
- Which nearby sites, especially La Draga, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Altamira with La Draga, Maeshowe and Gavrinis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Altamira

Spain • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Altamira begins with a bounded claim. For the material described here as polychrome cave paintings, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Altamira using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Maeshowe and Gavrinis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

El Castillo Cave

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Spain	**Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

El Castillo Cave is a site or landscape associated with paleolithic research in Spain. The defining record highlighted in this atlas is hand stencils and red motifs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	hand stencils and red motifs
Chronological frame	Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are hand stencils and red motifs linked to the cited phase of Paleolithic?
- What natural and human processes altered deposits at El Castillo Cave after original use?
- Which nearby sites, especially Los Millares, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare El Castillo Cave with Los Millares, Ness of Brodgar and Carnac Alignments. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

El Castillo Cave

Spain • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of El Castillo Cave begins with a bounded claim. For the material described here as hand stencils and red motifs, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for El Castillo Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ness of Brodgar and Carnac Alignments using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Atapuerca

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Spain	**Pleistocene**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Atapuerca is a site or landscape associated with pleistocene research in Spain. The defining record highlighted in this atlas is hominin fossils and cave sequences. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	hominin fossils and cave sequences
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are hominin fossils and cave sequences linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Atapuerca after original use?
- Which nearby sites, especially Stonehenge, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Atapuerca with Stonehenge, Newgrange and Barnenez. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Long-term excavations by multidisciplinary Spanish teams have yielded crucial Pleistocene human remains and environmental records.

Atapuerca

Spain • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Atapuerca begins with a bounded claim. For the material described here as hominin fossils and cave sequences, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Atapuerca using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Newgrange and Barnenez using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

La Draga

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Spain	**Neolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

La Draga is a site or landscape associated with neolithic research in Spain. The defining record highlighted in this atlas is waterlogged timber and farming village. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	waterlogged timber and farming village
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are waterlogged timber and farming village linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at La Draga after original use?
- Which nearby sites, especially Avebury, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare La Draga with Avebury, Knowth and Lascaux Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

La Draga

Spain • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of La Draga begins with a bounded claim. For the material described here as waterlogged timber and farming village, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for La Draga using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Knowth and Lascaux Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Los Millares

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Spain	Chalcolithic	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Los Millares is a site or landscape associated with chalcolithic research in Spain. The defining record highlighted in this atlas is fortified settlement and collective tombs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: atlantic and north is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortified settlement and collective tombs
Chronological frame	Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortified settlement and collective tombs linked to the cited phase of Chalcolithic?
- What natural and human processes altered deposits at Los Millares after original use?
- Which nearby sites, especially Durrington Walls, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Los Millares with Durrington Walls, Poulnabrone and Chauvet Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Los Millares

Spain • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Los Millares begins with a bounded claim. For the material described here as fortified settlement and collective tombs, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Los Millares using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Poulnabrone and Chauvet Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hallstatt Cemetery

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 221 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Austria	**Bronze-Iron Age**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Hallstatt Cemetery is a site or landscape associated with bronze-iron age research in Austria. The defining record highlighted in this atlas is salt mining and grave goods. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	salt mining and grave goods
Chronological frame	Bronze-Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are salt mining and grave goods linked to the cited phase of Bronze-Iron Age?
- What natural and human processes altered deposits at Hallstatt Cemetery after original use?
- Which nearby sites, especially Heuneburg, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hallstatt Cemetery with Heuneburg, Vindolanda and Oseberg Ship Burial. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hallstatt Cemetery

Austria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hallstatt Cemetery begins with a bounded claim. For the material described here as salt mining and grave goods, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hallstatt Cemetery using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Vindolanda and Oseberg Ship Burial using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dürrnberg

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 222 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Austria	Iron Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Dürrnberg is a site or landscape associated with iron age research in Austria. The defining record highlighted in this atlas is salt-mining Celtic occupation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	salt-mining Celtic occupation
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are salt-mining Celtic occupation linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Dürrnberg after original use?
- Which nearby sites, especially Hochdorf Burial, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dürrnberg with Hochdorf Burial, Hadrian's Wall and Gokstad Ship Burial. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dürrnberg

Austria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dürrnberg begins with a bounded claim. For the material described here as salt-mining Celtic occupation, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dürrnberg using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hadrian's Wall and Gokstad Ship Burial using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

La Tène

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 223 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Switzerland	**Iron Age**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

La Tène is a site or landscape associated with iron age research in Switzerland. The defining record highlighted in this atlas is material culture type-site. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	material culture type-site
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are material culture type-site linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at La Tène after original use?
- Which nearby sites, especially Glauberg, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare La Tène with Glauberg, Jorvik (York) and Varna Necropolis. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

La Tène

Switzerland • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of La Tène begins with a bounded claim. For the material described here as material culture type-site, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for La Tène using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Jorvik (York) and Varna Necropolis using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Heuneburg

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 224 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Germany	Iron Age	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Heuneburg is a site or landscape associated with iron age research in Germany. The defining record highlighted in this atlas is early urban hillfort and elite contacts. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early urban hillfort and elite contacts
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early urban hillfort and elite contacts linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Heuneburg after original use?
- Which nearby sites, especially Biskupin, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Heuneburg with Biskupin, Birka and Cucuteni-Trypillia Talianki. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Heuneburg

Germany • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Heuneburg begins with a bounded claim. For the material described here as early urban hillfort and elite contacts, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Heuneburg using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Birka and Cucuteni-Trypillia Talianki using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hochdorf Burial

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 225 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Germany	Iron Age	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Hochdorf Burial is a site or landscape associated with iron age research in Germany. The defining record highlighted in this atlas is elite wagon burial and banquet equipment. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	elite wagon burial and banquet equipment
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are elite wagon burial and banquet equipment linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Hochdorf Burial after original use?
- Which nearby sites, especially Sutton Hoo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hochdorf Burial with Sutton Hoo, Hedeby and Arkaim. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hochdorf Burial

Germany • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hochdorf Burial begins with a bounded claim. For the material described here as elite wagon burial and banquet equipment, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hochdorf Burial using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hedeby and Arkaim using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Glauberg

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 226 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Germany	Iron Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Glauberg is a site or landscape associated with iron age research in Germany. The defining record highlighted in this atlas is Celtic elite grave and statue. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Celtic elite grave and statue
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Celtic elite grave and statue linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Glauberg after original use?
- Which nearby sites, especially Staffordshire Hoard findspot, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Glauberg with Staffordshire Hoard findspot, Borre Mounds and Hallstatt Cemetery. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Glauberg

Germany • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Glauberg begins with a bounded claim. For the material described here as Celtic elite grave and statue, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Glauberg using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Borre Mounds and Hallstatt Cemetery using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Biskupin

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 227 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Poland	**Iron Age**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Biskupin is a site or landscape associated with iron age research in Poland. The defining record highlighted in this atlas is timber-fortified lakeside settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	timber-fortified lakeside settlement
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are timber-fortified lakeside settlement linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Biskupin after original use?
- Which nearby sites, especially Vindolanda, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Biskupin with Vindolanda, Oseberg Ship Burial and Dürrnberg. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Biskupin

Poland • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Biskupin begins with a bounded claim. For the material described here as timber-fortified lakeside settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Biskupin using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Oseberg Ship Burial and Dürrnberg using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sutton Hoo

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 228 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**early medieval**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Sutton Hoo is a site or landscape associated with early medieval research in United Kingdom. The defining record highlighted in this atlas is ship burial and elite grave goods. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ship burial and elite grave goods
Chronological frame	early medieval; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ship burial and elite grave goods linked to the cited phase of early medieval?
- What natural and human processes altered deposits at Sutton Hoo after original use?
- Which nearby sites, especially Hadrian's Wall, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sutton Hoo with Hadrian's Wall, Gokstad Ship Burial and La Tène. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Basil Brown's 1939 excavation of the ship burial revealed an exceptional Anglo-Saxon elite assemblage.

Sutton Hoo

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sutton Hoo begins with a bounded claim. For the material described here as ship burial and elite grave goods, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sutton Hoo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Gokstad Ship Burial and La Tène using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Staffordshire Hoard findspot

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 229 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	early medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Staffordshire Hoard findspot is a site or landscape associated with early medieval research in United Kingdom. The defining record highlighted in this atlas is Anglo-Saxon metalwork deposit. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Anglo-Saxon metalwork deposit
Chronological frame	early medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Anglo-Saxon metalwork deposit linked to the cited phase of early medieval?
- What natural and human processes altered deposits at Staffordshire Hoard findspot after original use?
- Which nearby sites, especially Jorvik (York), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Staffordshire Hoard findspot with Jorvik (York), Varna Necropolis and Heuneburg. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Staffordshire Hoard findspot

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Staffordshire Hoard findspot begins with a bounded claim. For the material described here as Anglo-Saxon metalwork deposit, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Staffordshire Hoard findspot using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Varna Necropolis and Heuneburg using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Vindolanda

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 230 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Roman**	**Text**

01 WHAT SURVIVES AND WHY IT MATTERS

Vindolanda is a site or landscape associated with roman research in United Kingdom. The defining record highlighted in this atlas is wooden writing tablets and frontier life. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	wooden writing tablets and frontier life
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are wooden writing tablets and frontier life linked to the cited phase of Roman?
- What natural and human processes altered deposits at Vindolanda after original use?
- Which nearby sites, especially Birka, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Vindolanda with Birka, Cucuteni-Trypillia Talianki and Hochdorf Burial. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Writing tablets recovered at the Roman frontier site provide rare personal and administrative documentary evidence.

Vindolanda

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Vindolanda begins with a bounded claim. For the material described here as wooden writing tablets and frontier life, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Vindolanda using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cucuteni-Trypillia Talianki and Hochdorf Burial using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hadrian's Wall

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 231 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Roman**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Hadrian's Wall is a site or landscape associated with roman research in United Kingdom. The defining record highlighted in this atlas is frontier forts and military infrastructure. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	frontier forts and military infrastructure
Chronological frame	Roman; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are frontier forts and military infrastructure linked to the cited phase of Roman?
- What natural and human processes altered deposits at Hadrian's Wall after original use?
- Which nearby sites, especially Hedeby, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hadrian's Wall with Hedeby, Arkaim and Glauberg. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hadrian's Wall

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hadrian's Wall begins with a bounded claim. For the material described here as frontier forts and military infrastructure, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hadrian's Wall using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Arkaim and Glauberg using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Jorvik (York)

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 232 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United Kingdom	**Viking Age**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Jorvik (York) is a site or landscape associated with viking age research in United Kingdom. The defining record highlighted in this atlas is waterlogged urban archaeology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	waterlogged urban archaeology
Chronological frame	Viking Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are waterlogged urban archaeology linked to the cited phase of Viking Age?
- What natural and human processes altered deposits at Jorvik (York) after original use?
- Which nearby sites, especially Borre Mounds, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Jorvik (York) with Borre Mounds, Hallstatt Cemetery and Biskupin. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Jorvik (York)

United Kingdom • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Jorvik (York) begins with a bounded claim. For the material described here as waterlogged urban archaeology, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Jorvik (York) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hallstatt Cemetery and Biskupin using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Birka

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 233 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sweden	**Viking Age**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Birka is a site or landscape associated with viking age research in Sweden. The defining record highlighted in this atlas is trade settlement and grave fields. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	trade settlement and grave fields
Chronological frame	Viking Age; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are trade settlement and grave fields linked to the cited phase of Viking Age?
- What natural and human processes altered deposits at Birka after original use?
- Which nearby sites, especially Oseberg Ship Burial, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Birka with Oseberg Ship Burial, Dürrnberg and Sutton Hoo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Birka

Sweden • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Birka begins with a bounded claim. For the material described here as trade settlement and grave fields, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Birka using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dürrnberg and Sutton Hoo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hedeby

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 234 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Germany	**Viking Age**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Hedeby is a site or landscape associated with viking age research in Germany. The defining record highlighted in this atlas is early medieval port and trading town. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early medieval port and trading town
Chronological frame	Viking Age; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early medieval port and trading town linked to the cited phase of Viking Age?
- What natural and human processes altered deposits at Hedeby after original use?
- Which nearby sites, especially Gokstad Ship Burial, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hedeby with Gokstad Ship Burial, La Tène and Staffordshire Hoard findspot. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hedeby

Germany • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hedeby begins with a bounded claim. For the material described here as early medieval port and trading town, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hedeby using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with La Tène and Staffordshire Hoard findspot using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Borre Mounds

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 235 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Norway	**Viking Age**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Borre Mounds is a site or landscape associated with viking age research in Norway. The defining record highlighted in this atlas is monumental burial mounds. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental burial mounds
Chronological frame	Viking Age; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental burial mounds linked to the cited phase of Viking Age?
- What natural and human processes altered deposits at Borre Mounds after original use?
- Which nearby sites, especially Varna Necropolis, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Borre Mounds with Varna Necropolis, Heuneburg and Vindolanda. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Borre Mounds

Norway • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Borre Mounds begins with a bounded claim. For the material described here as monumental burial mounds, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Borre Mounds using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Heuneburg and Vindolanda using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Oseberg Ship Burial

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 236 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Norway	**Viking Age**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Oseberg Ship Burial is a site or landscape associated with viking age research in Norway. The defining record highlighted in this atlas is ship and wooden furnishings. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ship and wooden furnishings
Chronological frame	Viking Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ship and wooden furnishings linked to the cited phase of Viking Age?
- What natural and human processes altered deposits at Oseberg Ship Burial after original use?
- Which nearby sites, especially Cucuteni-Trypillia Talianki, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Oseberg Ship Burial with Cucuteni-Trypillia Talianki, Hochdorf Burial and Hadrian's Wall. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The Oseberg ship excavation in 1904 preserved elaborate woodwork while creating challenging long-term conservation responsibilities.

Oseberg Ship Burial

Norway • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Oseberg Ship Burial begins with a bounded claim. For the material described here as ship and wooden furnishings, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Oseberg Ship Burial using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hochdorf Burial and Hadrian's Wall using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Gokstad Ship Burial

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 237 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Norway	**Viking Age**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Gokstad Ship Burial is a site or landscape associated with viking age research in Norway. The defining record highlighted in this atlas is ship grave and maritime technology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ship grave and maritime technology
Chronological frame	Viking Age; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ship grave and maritime technology linked to the cited phase of Viking Age?
- What natural and human processes altered deposits at Gokstad Ship Burial after original use?
- Which nearby sites, especially Arkaim, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Gokstad Ship Burial with Arkaim, Glauberg and Jorvik (York). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Gokstad Ship Burial

Norway • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Gokstad Ship Burial begins with a bounded claim. For the material described here as ship grave and maritime technology, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Gokstad Ship Burial using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Glauberg and Jorvik (York) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Varna Necropolis

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 238 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Bulgaria	**Chalcolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Varna Necropolis is a site or landscape associated with chalcolithic research in Bulgaria. The defining record highlighted in this atlas is early gold artifacts and social differentiation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early gold artifacts and social differentiation
Chronological frame	Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early gold artifacts and social differentiation linked to the cited phase of Chalcolithic?
- What natural and human processes altered deposits at Varna Necropolis after original use?
- Which nearby sites, especially Hallstatt Cemetery, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Varna Necropolis with Hallstatt Cemetery, Biskupin and Birka. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Varna Necropolis

Bulgaria • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Varna Necropolis begins with a bounded claim. For the material described here as early gold artifacts and social differentiation, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Varna Necropolis using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Biskupin and Birka using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cucuteni-Trypillia Talianki

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 239 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ukraine	**Chalcolithic**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Cucuteni-Trypillia Talianki is a site or landscape associated with chalcolithic research in Ukraine. The defining record highlighted in this atlas is large settlement plans and painted pottery. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large settlement plans and painted pottery
Chronological frame	Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large settlement plans and painted pottery linked to the cited phase of Chalcolithic?
- What natural and human processes altered deposits at Cucuteni-Trypillia Talianki after original use?
- Which nearby sites, especially Dürrnberg, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cucuteni-Trypillia Talianki with Dürrnberg, Sutton Hoo and Hedeby. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Cucuteni-Trypillia Talianki

Ukraine • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Cucuteni-Trypillia Talianki begins with a bounded claim. For the material described here as large settlement plans and painted pottery, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cucuteni-Trypillia Talianki using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sutton Hoo and Hedeby using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Arkaim

European Iron Age, Medieval, and Eastern Europe • Evidence dossier 240 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Bronze Age**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Arkaim is a site or landscape associated with bronze age research in Russia. The defining record highlighted in this atlas is circular fortified steppe settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in europe: central and eastern is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	circular fortified steppe settlement
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are circular fortified steppe settlement linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Arkaim after original use?
- Which nearby sites, especially La Tène, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Arkaim with La Tène, Staffordshire Hoard findspot and Borre Mounds. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Arkaim

Russia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Arkaim begins with a bounded claim. For the material described here as circular fortified steppe settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Arkaim using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Staffordshire Hoard findspot and Borre Mounds using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Harappa

South Asia: Indus and Deep History • Evidence dossier 241 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Indus Civilization**	**Text**

01 WHAT SURVIVES AND WHY IT MATTERS

Harappa is a site or landscape associated with indus civilization research in Pakistan. The defining record highlighted in this atlas is urban layout seals and craft production. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	urban layout seals and craft production
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are urban layout seals and craft production linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Harappa after original use?
- Which nearby sites, especially Rakhigarhi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Harappa with Rakhigarhi, Kot Diji and Bhimbetka Rock Shelters. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Daya Ram Sahni's excavations were foundational to the recognition of the Indus Civilization as a major South Asian urban tradition.

Harappa

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Harappa begins with a bounded claim. For the material described here as urban layout seals and craft production, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Harappa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kot Diji and Bhimbetka Rock Shelters using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mohenjo-daro

South Asia: Indus and Deep History • Evidence dossier 242 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Indus Civilization**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Mohenjo-daro is a site or landscape associated with indus civilization research in Pakistan. The defining record highlighted in this atlas is street grids drainage and Great Bath. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	street grids drainage and Great Bath
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are street grids drainage and Great Bath linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Mohenjo-daro after original use?
- Which nearby sites, especially Ganeriwala, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mohenjo-daro with Ganeriwala, Mehrgarh and Bagor. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Rakhaldas Bandyopadhyay's early work and subsequent large excavations revealed a substantial Indus city with elaborate water management.

Mohenjo-daro

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mohenjo-daro begins with a bounded claim. For the material described here as street grids drainage and Great Bath, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mohenjo-daro using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mehrgarh and Bagor using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dholavira

South Asia: Indus and Deep History • Evidence dossier 243 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Indus Civilization	Network

01 WHAT SURVIVES AND WHY IT MATTERS

Dholavira is a site or landscape associated with indus civilization research in India. The defining record highlighted in this atlas is water-management reservoirs and city divisions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	water-management reservoirs and city divisions
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are water-management reservoirs and city divisions linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Dholavira after original use?
- Which nearby sites, especially Kalibangan, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dholavira with Kalibangan, Bhirrana and Attirampakkam. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

R. S. Bisht's work helped demonstrate the complexity of the site's stone architecture, city divisions and water-management infrastructure.

Dholavira

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dholavira begins with a bounded claim. For the material described here as water-management reservoirs and city divisions, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dholavira using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bhirrana and Attirampakkam using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Rakhigarhi

South Asia: Indus and Deep History • Evidence dossier 244 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Indus Civilization	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Rakhigarhi is a site or landscape associated with indus civilization research in India. The defining record highlighted in this atlas is urban settlement and cemetery archaeology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	urban settlement and cemetery archaeology
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are urban settlement and cemetery archaeology linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Rakhigarhi after original use?
- Which nearby sites, especially Lothal, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Rakhigarhi with Lothal, Burzahom and Pattanam. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Rakhigarhi

India • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Rakhigarhi begins with a bounded claim. For the material described here as urban settlement and cemetery archaeology, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Rakhigarhi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Burzahom and Pattanam using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ganeriwala

South Asia: Indus and Deep History • Evidence dossier 245 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Indus Civilization**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Ganeriwala is a site or landscape associated with indus civilization research in Pakistan. The defining record highlighted in this atlas is major unexcavated or limited-excavation settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	major unexcavated or limited-excavation settlement
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are major unexcavated or limited-excavation settlement linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Ganeriwala after original use?
- Which nearby sites, especially Banawali, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ganeriwala with Banawali, Chirand and Arikamedu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ganeriwala

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ganeriwala begins with a bounded claim. For the material described here as major unexcavated or limited-excavation settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ganeriwala using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chirand and Arikamedu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kalibangan

South Asia: Indus and Deep History • Evidence dossier 246 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Indus Civilization	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Kalibangan is a site or landscape associated with indus civilization research in India. The defining record highlighted in this atlas is ploughed fields and town planning. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ploughed fields and town planning
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ploughed fields and town planning linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Kalibangan after original use?
- Which nearby sites, especially Chanhudaro, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kalibangan with Chanhudaro, Inamgaon and Harappa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kalibangan

India • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Kalibangan begins with a bounded claim. For the material described here as ploughed fields and town planning, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kalibangan using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Inamgaon and Harappa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lothal

South Asia: Indus and Deep History • Evidence dossier 247 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Indus Civilization	Network

01 WHAT SURVIVES AND WHY IT MATTERS

Lothal is a site or landscape associated with indus civilization research in India. The defining record highlighted in this atlas is craft manufacturing and maritime trade debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	craft manufacturing and maritime trade debates
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are craft manufacturing and maritime trade debates linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Lothal after original use?
- Which nearby sites, especially Kot Diji, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lothal with Kot Diji, Bhimbetka Rock Shelters and Mohenjo-daro. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lothal

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Lothal begins with a bounded claim. For the material described here as craft manufacturing and maritime trade debates, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lothal using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bhimbetka Rock Shelters and Mohenjo-daro using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Banawali

South Asia: Indus and Deep History • Evidence dossier 248 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Indus Civilization	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Banawali is a site or landscape associated with indus civilization research in India. The defining record highlighted in this atlas is fortified urban phases. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortified urban phases
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortified urban phases linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Banawali after original use?
- Which nearby sites, especially Mehrgarh, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Banawali with Mehrgarh, Bagor and Dholavira. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Banawali

India • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Banawali begins with a bounded claim. For the material described here as fortified urban phases, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Banawali using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bagor and Dholavira using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chanhudaro

South Asia: Indus and Deep History • Evidence dossier 249 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Indus Civilization**	**Text**

01 WHAT SURVIVES AND WHY IT MATTERS

Chanhudaro is a site or landscape associated with indus civilization research in Pakistan. The defining record highlighted in this atlas is craft production and seals. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	craft production and seals
Chronological frame	Indus Civilization; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are craft production and seals linked to the cited phase of Indus Civilization?
- What natural and human processes altered deposits at Chanhudaro after original use?
- Which nearby sites, especially Bhirrana, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chanhudaro with Bhirrana, Attirampakkam and Rakhigarhi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Chanhudaro

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Chanhudaro begins with a bounded claim. For the material described here as craft production and seals, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chanhudaro using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Attirampakkam and Rakhigarhi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kot Diji

South Asia: Indus and Deep History • Evidence dossier 250 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Early Harappan**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Kot Diji is a site or landscape associated with early harappan research in Pakistan. The defining record highlighted in this atlas is fortified settlement and ceramic phases. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortified settlement and ceramic phases
Chronological frame	Early Harappan; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortified settlement and ceramic phases linked to the cited phase of Early Harappan?
- What natural and human processes altered deposits at Kot Diji after original use?
- Which nearby sites, especially Burzahom, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kot Diji with Burzahom, Pattanam and Ganeriwala. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kot Diji

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kot Diji begins with a bounded claim. For the material described here as fortified settlement and ceramic phases, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kot Diji using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pattanam and Ganeriwala using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mehrgarh

South Asia: Indus and Deep History • Evidence dossier 251 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Neolithic-Chalcolithic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Mehrgarh is a site or landscape associated with neolithic-chalcolithic research in Pakistan. The defining record highlighted in this atlas is early cultivation and domestic animals. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early cultivation and domestic animals
Chronological frame	Neolithic-Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early cultivation and domestic animals linked to the cited phase of Neolithic-Chalcolithic?
- What natural and human processes altered deposits at Mehrgarh after original use?
- Which nearby sites, especially Chirand, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mehrgarh with Chirand, Arikamedu and Kalibangan. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Jean-François Jarrige and collaborators documented a long sequence central to understanding early food production in South Asia.

Mehrgarh

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mehrgarh begins with a bounded claim. For the material described here as early cultivation and domestic animals, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mehrgarh using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Arikamedu and Kalibangan using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bhirrana

South Asia: Indus and Deep History • Evidence dossier 252 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Early Harappan	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Bhirrana is a site or landscape associated with early harappan research in India. The defining record highlighted in this atlas is early settlement stratigraphy. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early settlement stratigraphy
Chronological frame	Early Harappan; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early settlement stratigraphy linked to the cited phase of Early Harappan?
- What natural and human processes altered deposits at Bhirrana after original use?
- Which nearby sites, especially Inamgaon, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bhirrana with Inamgaon, Harappa and Lothal. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Bhirrana

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Bhirrana begins with a bounded claim. For the material described here as early settlement stratigraphy, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bhirrana using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Harappa and Lothal using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Burzahom

South Asia: Indus and Deep History • Evidence dossier 253 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Burzahom is a site or landscape associated with neolithic research in India. The defining record highlighted in this atlas is pit dwellings and settlement archaeology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	pit dwellings and settlement archaeology
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are pit dwellings and settlement archaeology linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Burzahom after original use?
- Which nearby sites, especially Bhimbetka Rock Shelters, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Burzahom with Bhimbetka Rock Shelters, Mohenjo-daro and Banawali. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Burzahom

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Burzahom begins with a bounded claim. For the material described here as pit dwellings and settlement archaeology, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Burzahom using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mohenjo-daro and Banawali using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chirand

South Asia: Indus and Deep History • Evidence dossier 254 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Neolithic-Chalcolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Chirand is a site or landscape associated with neolithic-chalcolithic research in India. The defining record highlighted in this atlas is Ganges plain farming settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Ganges plain farming settlement
Chronological frame	Neolithic-Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Ganges plain farming settlement linked to the cited phase of Neolithic-Chalcolithic?
- What natural and human processes altered deposits at Chirand after original use?
- Which nearby sites, especially Bagor, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chirand with Bagor, Dholavira and Chanhudaro. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Chirand

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Chirand begins with a bounded claim. For the material described here as Ganges plain farming settlement, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chirand using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dholavira and Chanhudaro using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Inamgaon

South Asia: Indus and Deep History • Evidence dossier 255 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Chalcolithic	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Inamgaon is a site or landscape associated with chalcolithic research in India. The defining record highlighted in this atlas is domestic layouts and burials. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	domestic layouts and burials
Chronological frame	Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are domestic layouts and burials linked to the cited phase of Chalcolithic?
- What natural and human processes altered deposits at Inamgaon after original use?
- Which nearby sites, especially Attirampakkam, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Inamgaon with Attirampakkam, Rakhigarhi and Kot Diji. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Inamgaon

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Inamgaon begins with a bounded claim. For the material described here as domestic layouts and burials, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Inamgaon using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Rakhigarhi and Kot Diji using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bhimbetka Rock Shelters

South Asia: Indus and Deep History • Evidence dossier 256 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Paleolithic-Holocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Bhimbetka Rock Shelters is a site or landscape associated with paleolithic-holocene research in India. The defining record highlighted in this atlas is rock art and occupation shelters. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock art and occupation shelters
Chronological frame	Paleolithic-Holocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock art and occupation shelters linked to the cited phase of Paleolithic-Holocene?
- What natural and human processes altered deposits at Bhimbetka Rock Shelters after original use?
- Which nearby sites, especially Pattanam, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bhimbetka Rock Shelters with Pattanam, Ganeriwala and Mehrgarh. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

V. S. Wakankar drew attention to the rich painted rock-shelter complex and its relationship to much longer occupation histories.

Bhimbetka Rock Shelters

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Bhimbetka Rock Shelters begins with a bounded claim. For the material described here as rock art and occupation shelters, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bhimbetka Rock Shelters using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ganeriwala and Mehrgarh using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bagor

South Asia: Indus and Deep History • Evidence dossier 257 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Mesolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Bagor is a site or landscape associated with mesolithic research in India. The defining record highlighted in this atlas is hunter-gatherer settlement and animal use. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	hunter-gatherer settlement and animal use
Chronological frame	Mesolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are hunter-gatherer settlement and animal use linked to the cited phase of Mesolithic?
- What natural and human processes altered deposits at Bagor after original use?
- Which nearby sites, especially Arikamedu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bagor with Arikamedu, Kalibangan and Bhirrana. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Bagor

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Bagor begins with a bounded claim. For the material described here as hunter-gatherer settlement and animal use, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bagor using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kalibangan and Bhirrana using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Attirampakkam

South Asia: Indus and Deep History • Evidence dossier 258 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Paleolithic	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Attirampakkam is a site or landscape associated with paleolithic research in India. The defining record highlighted in this atlas is Acheulean and Middle Paleolithic sequences. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Acheulean and Middle Paleolithic sequences
Chronological frame	Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Acheulean and Middle Paleolithic sequences linked to the cited phase of Paleolithic?
- What natural and human processes altered deposits at Attirampakkam after original use?
- Which nearby sites, especially Harappa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Attirampakkam with Harappa, Lothal and Burzahom. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Attirampakkam

India • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Attirampakkam begins with a bounded claim. For the material described here as Acheulean and Middle Paleolithic sequences, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Attirampakkam using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lothal and Burzahom using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pattanam

South Asia: Indus and Deep History • Evidence dossier 259 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Early Historic	Network

01 WHAT SURVIVES AND WHY IT MATTERS

Pattanam is a site or landscape associated with early historic research in India. The defining record highlighted in this atlas is Indian Ocean trade-associated settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Indian Ocean trade-associated settlement
Chronological frame	Early Historic; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Indian Ocean trade-associated settlement linked to the cited phase of Early Historic?
- What natural and human processes altered deposits at Pattanam after original use?
- Which nearby sites, especially Mohenjo-daro, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pattanam with Mohenjo-daro, Banawali and Chirand. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Pattanam

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Pattanam begins with a bounded claim. For the material described here as Indian Ocean trade-associated settlement, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pattanam using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Banawali and Chirand using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Arikamedu

South Asia: Indus and Deep History • Evidence dossier 260 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Early Historic	Network

01 WHAT SURVIVES AND WHY IT MATTERS

Arikamedu is a site or landscape associated with early historic research in India. The defining record highlighted in this atlas is Indo-Roman trade ceramics and crafts. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Indo-Roman trade ceramics and crafts
Chronological frame	Early Historic; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Indo-Roman trade ceramics and crafts linked to the cited phase of Early Historic?
- What natural and human processes altered deposits at Arikamedu after original use?
- Which nearby sites, especially Dholavira, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Arikamedu with Dholavira, Chanhudaro and Inamgaon. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Arikamedu

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Arikamedu begins with a bounded claim. For the material described here as Indo-Roman trade ceramics and crafts, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Arikamedu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chanhudaro and Inamgaon using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sanchi

South Asia: Buddhist and Imperial Centers • Evidence dossier 261 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Mauryan onward	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Sanchi is a site or landscape associated with mauryan onward research in India. The defining record highlighted in this atlas is Buddhist stupas and narrative reliefs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Buddhist stupas and narrative reliefs
Chronological frame	Mauryan onward; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Buddhist stupas and narrative reliefs linked to the cited phase of Mauryan onward?
- What natural and human processes altered deposits at Sanchi after original use?
- Which nearby sites, especially Taxila, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sanchi with Taxila, Hampi and Tilaurakot. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The stupas and sculpted gateways permit study of Buddhism, patronage and reuse across several centuries rather than a single construction event.

Sanchi

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sanchi begins with a bounded claim. For the material described here as Buddhist stupas and narrative reliefs, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sanchi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hampi and Tilaurakot using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bodh Gaya

South Asia: Buddhist and Imperial Centers • Evidence dossier 262 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	ancient-medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Bodh Gaya is a site or landscape associated with ancient-medieval research in India. The defining record highlighted in this atlas is Buddhist sacred architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Buddhist sacred architecture
Chronological frame	ancient-medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Buddhist sacred architecture linked to the cited phase of ancient-medieval?
- What natural and human processes altered deposits at Bodh Gaya after original use?
- Which nearby sites, especially Takht-i-Bahi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bodh Gaya with Takht-i-Bahi, Pattadakal and Anuradhapura. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Bodh Gaya

India • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Bodh Gaya begins with a bounded claim. For the material described here as Buddhist sacred architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bodh Gaya using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pattadakal and Anuradhapura using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nalanda

South Asia: Buddhist and Imperial Centers • Evidence dossier 263 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Gupta-medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Nalanda is a site or landscape associated with gupta-medieval research in India. The defining record highlighted in this atlas is monastic university complex. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monastic university complex
Chronological frame	Gupta-medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monastic university complex linked to the cited phase of Gupta-medieval?
- What natural and human processes altered deposits at Nalanda after original use?
- Which nearby sites, especially Sirkap, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nalanda with Sirkap, Mahabalipuram and Sigiriya. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nalanda

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Nalanda begins with a bounded claim. For the material described here as monastic university complex, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nalanda using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mahabalipuram and Sigiriya using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Taxila

South Asia: Buddhist and Imperial Centers • Evidence dossier 264 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Achaemenid-Kushan**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Taxila is a site or landscape associated with achaemenid-kushan research in Pakistan. The defining record highlighted in this atlas is multi-period urban and Buddhist remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	multi-period urban and Buddhist remains
Chronological frame	Achaemenid-Kushan; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are multi-period urban and Buddhist remains linked to the cited phase of Achaemenid-Kushan?
- What natural and human processes altered deposits at Taxila after original use?
- Which nearby sites, especially Ajanta Caves, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Taxila with Ajanta Caves, Konark Sun Temple and Polonnaruwa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

John Marshall's excavations contributed to the study of overlapping urban traditions and Buddhist institutions in the northwest subcontinent.

Taxila

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Taxila begins with a bounded claim. For the material described here as multi-period urban and Buddhist remains, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Taxila using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Konark Sun Temple and Polonnaruwa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Takht-i-Bahi

South Asia: Buddhist and Imperial Centers • Evidence dossier 265 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Kushan**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Takht-i-Bahi is a site or landscape associated with kushan research in Pakistan. The defining record highlighted in this atlas is Buddhist monastery remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Buddhist monastery remains
Chronological frame	Kushan; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Buddhist monastery remains linked to the cited phase of Kushan?
- What natural and human processes altered deposits at Takht-i-Bahi after original use?
- Which nearby sites, especially Ellora Caves, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Takht-i-Bahi with Ellora Caves, Khajuraho and Mihintale. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Takht-i-Bahi

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Takht-i-Bahi begins with a bounded claim. For the material described here as Buddhist monastery remains, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Takht-i-Bahi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Khajuraho and Mihintale using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sirkap

South Asia: Buddhist and Imperial Centers • Evidence dossier 266 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Pakistan	**Indo-Greek-Kushan**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Sirkap is a site or landscape associated with indo-greek-kushan research in Pakistan. The defining record highlighted in this atlas is planned walled city. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	planned walled city
Chronological frame	Indo-Greek-Kushan; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are planned walled city linked to the cited phase of Indo-Greek-Kushan?
- What natural and human processes altered deposits at Sirkap after original use?
- Which nearby sites, especially Elephanta Caves, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sirkap with Elephanta Caves, Lumbini and Sanchi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sirkap

Pakistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sirkap begins with a bounded claim. For the material described here as planned walled city, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sirkap using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lumbini and Sanchi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ajanta Caves

South Asia: Buddhist and Imperial Centers • Evidence dossier 267 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	early centuries BCE-CE	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Ajanta Caves is a site or landscape associated with early centuries bce-ce research in India. The defining record highlighted in this atlas is rock-cut Buddhist art. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock-cut Buddhist art
Chronological frame	early centuries BCE-CE; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock-cut Buddhist art linked to the cited phase of early centuries BCE-CE?
- What natural and human processes altered deposits at Ajanta Caves after original use?
- Which nearby sites, especially Hampi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ajanta Caves with Hampi, Tilaurakot and Bodh Gaya. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ajanta Caves

India • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ajanta Caves begins with a bounded claim. For the material described here as rock-cut Buddhist art, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ajanta Caves using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tilaurakot and Bodh Gaya using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ellora Caves

South Asia: Buddhist and Imperial Centers • Evidence dossier 268 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	early medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Ellora Caves is a site or landscape associated with early medieval research in India. The defining record highlighted in this atlas is Buddhist Hindu and Jain rock-cut temples. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Buddhist Hindu and Jain rock-cut temples
Chronological frame	early medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Buddhist Hindu and Jain rock-cut temples linked to the cited phase of early medieval?
- What natural and human processes altered deposits at Ellora Caves after original use?
- Which nearby sites, especially Pattadakal, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ellora Caves with Pattadakal, Anuradhapura and Nalanda. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ellora Caves

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ellora Caves begins with a bounded claim. For the material described here as Buddhist Hindu and Jain rock-cut temples, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ellora Caves using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Anuradhapura and Nalanda using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Elephanta Caves

South Asia: Buddhist and Imperial Centers • Evidence dossier 269 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	early medieval	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Elephanta Caves is a site or landscape associated with early medieval research in India. The defining record highlighted in this atlas is Hindu cave sculpture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Hindu cave sculpture
Chronological frame	early medieval; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Hindu cave sculpture linked to the cited phase of early medieval?
- What natural and human processes altered deposits at Elephanta Caves after original use?
- Which nearby sites, especially Mahabalipuram, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Elephanta Caves with Mahabalipuram, Sigiriya and Taxila. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Elephanta Caves

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Elephanta Caves begins with a bounded claim. For the material described here as Hindu cave sculpture, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Elephanta Caves using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sigiriya and Taxila using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hampi

South Asia: Buddhist and Imperial Centers • Evidence dossier 270 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Vijayanagara	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Hampi is a site or landscape associated with vijayanagara research in India. The defining record highlighted in this atlas is imperial urban core and temple landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	imperial urban core and temple landscape
Chronological frame	Vijayanagara; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are imperial urban core and temple landscape linked to the cited phase of Vijayanagara?
- What natural and human processes altered deposits at Hampi after original use?
- Which nearby sites, especially Konark Sun Temple, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hampi with Konark Sun Temple, Polonnaruwa and Takht-i-Bahi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hampi

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hampi begins with a bounded claim. For the material described here as imperial urban core and temple landscape, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hampi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Polonnaruwa and Takht-i-Bahi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pattadakal

South Asia: Buddhist and Imperial Centers • Evidence dossier 271 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	early medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Pattadakal is a site or landscape associated with early medieval research in India. The defining record highlighted in this atlas is temple architectural experimentation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple architectural experimentation
Chronological frame	early medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple architectural experimentation linked to the cited phase of early medieval?
- What natural and human processes altered deposits at Pattadakal after original use?
- Which nearby sites, especially Khajuraho, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pattadakal with Khajuraho, Mihintale and Sirkap. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Pattadakal

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Pattadakal begins with a bounded claim. For the material described here as temple architectural experimentation, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pattadakal using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mihintale and Sirkap using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mahabalipuram

South Asia: Buddhist and Imperial Centers • Evidence dossier 272 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	Pallava	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Mahabalipuram is a site or landscape associated with pallava research in India. The defining record highlighted in this atlas is rock-cut and structural monuments. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock-cut and structural monuments
Chronological frame	Pallava; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock-cut and structural monuments linked to the cited phase of Pallava?
- What natural and human processes altered deposits at Mahabalipuram after original use?
- Which nearby sites, especially Lumbini, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mahabalipuram with Lumbini, Sanchi and Ajanta Caves. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mahabalipuram

India • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Mahabalipuram begins with a bounded claim. For the material described here as rock-cut and structural monuments, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mahabalipuram using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sanchi and Ajanta Caves using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Konark Sun Temple

South Asia: Buddhist and Imperial Centers • Evidence dossier 273 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	medieval	Art

01 WHAT SURVIVES AND WHY IT MATTERS

Konark Sun Temple is a site or landscape associated with medieval research in India. The defining record highlighted in this atlas is stone temple architecture and iconography. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	stone temple architecture and iconography
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are stone temple architecture and iconography linked to the cited phase of medieval?
- What natural and human processes altered deposits at Konark Sun Temple after original use?
- Which nearby sites, especially Tilaurakot, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Konark Sun Temple with Tilaurakot, Bodh Gaya and Ellora Caves. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Konark Sun Temple

India • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Konark Sun Temple begins with a bounded claim. For the material described here as stone temple architecture and iconography, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Konark Sun Temple using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bodh Gaya and Ellora Caves using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Khajuraho

South Asia: Buddhist and Imperial Centers • Evidence dossier 274 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
India	medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Khajuraho is a site or landscape associated with medieval research in India. The defining record highlighted in this atlas is Nagara temples and sculpture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Nagara temples and sculpture
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Nagara temples and sculpture linked to the cited phase of medieval?
- What natural and human processes altered deposits at Khajuraho after original use?
- Which nearby sites, especially Anuradhapura, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Khajuraho with Anuradhapura, Nalanda and Elephanta Caves. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Khajuraho

India • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Khajuraho begins with a bounded claim. For the material described here as Nagara temples and sculpture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Khajuraho using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nalanda and Elephanta Caves using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lumbini

South Asia: Buddhist and Imperial Centers • Evidence dossier 275 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nepal	ancient	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Lumbini is a site or landscape associated with ancient research in Nepal. The defining record highlighted in this atlas is Buddhist sacred landscape and early constructions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Buddhist sacred landscape and early constructions
Chronological frame	ancient; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Buddhist sacred landscape and early constructions linked to the cited phase of ancient?
- What natural and human processes altered deposits at Lumbini after original use?
- Which nearby sites, especially Sigiriya, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lumbini with Sigiriya, Taxila and Hampi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lumbini

Nepal • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Lumbini begins with a bounded claim. For the material described here as Buddhist sacred landscape and early constructions, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lumbini using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Taxila and Hampi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tilaurakot

South Asia: Buddhist and Imperial Centers • Evidence dossier 276 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Nepal	Early Historic	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Tilaurakot is a site or landscape associated with early historic research in Nepal. The defining record highlighted in this atlas is fortified urban site and Buddhist associations. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortified urban site and Buddhist associations
Chronological frame	Early Historic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortified urban site and Buddhist associations linked to the cited phase of Early Historic?
- What natural and human processes altered deposits at Tilaurakot after original use?
- Which nearby sites, especially Polonnaruwa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tilaurakot with Polonnaruwa, Takht-i-Bahi and Pattadakal. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tilaurakot

Nepal • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Tilaurakot begins with a bounded claim. For the material described here as fortified urban site and Buddhist associations, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tilaurakot using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Takht-i-Bahi and Pattadakal using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Anuradhapura

South Asia: Buddhist and Imperial Centers • Evidence dossier 277 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sri Lanka	Early Historic-medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Anuradhapura is a site or landscape associated with early historic-medieval research in Sri Lanka. The defining record highlighted in this atlas is monastic urban water systems. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monastic urban water systems
Chronological frame	Early Historic-medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monastic urban water systems linked to the cited phase of Early Historic-medieval?
- What natural and human processes altered deposits at Anuradhapura after original use?
- Which nearby sites, especially Mihintale, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Anuradhapura with Mihintale, Sirkap and Mahabalipuram. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Anuradhapura

Sri Lanka • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Anuradhapura begins with a bounded claim. For the material described here as monastic urban water systems, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Anuradhapura using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sirkap and Mahabalipuram using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sigiriya

South Asia: Buddhist and Imperial Centers • Evidence dossier 278 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sri Lanka	late antique	Art

01 WHAT SURVIVES AND WHY IT MATTERS

Sigiriya is a site or landscape associated with late antique research in Sri Lanka. The defining record highlighted in this atlas is rock fortress gardens and paintings. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock fortress gardens and paintings
Chronological frame	late antique; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock fortress gardens and paintings linked to the cited phase of late antique?
- What natural and human processes altered deposits at Sigiriya after original use?
- Which nearby sites, especially Sanchi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sigiriya with Sanchi, Ajanta Caves and Konark Sun Temple. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sigiriya

Sri Lanka • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Sigiriya begins with a bounded claim. For the material described here as rock fortress gardens and paintings, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sigiriya using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ajanta Caves and Konark Sun Temple using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Polonnaruwa

South Asia: Buddhist and Imperial Centers • Evidence dossier 279 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sri Lanka	medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Polonnaruwa is a site or landscape associated with medieval research in Sri Lanka. The defining record highlighted in this atlas is royal and monastic urban remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal and monastic urban remains
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal and monastic urban remains linked to the cited phase of medieval?
- What natural and human processes altered deposits at Polonnaruwa after original use?
- Which nearby sites, especially Bodh Gaya, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Polonnaruwa with Bodh Gaya, Ellora Caves and Khajuraho. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Polonnaruwa

Sri Lanka • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Polonnaruwa begins with a bounded claim. For the material described here as royal and monastic urban remains, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Polonnaruwa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ellora Caves and Khajuraho using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mihintale

South Asia: Buddhist and Imperial Centers • Evidence dossier 280 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Sri Lanka	ancient-medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Mihintale is a site or landscape associated with ancient-medieval research in Sri Lanka. The defining record highlighted in this atlas is Buddhist monastic landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south asia: historic is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Buddhist monastic landscape
Chronological frame	ancient-medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Buddhist monastic landscape linked to the cited phase of ancient-medieval?
- What natural and human processes altered deposits at Mihintale after original use?
- Which nearby sites, especially Nalanda, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mihintale with Nalanda, Elephanta Caves and Lumbini. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mihintale

Sri Lanka • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Mihintale begins with a bounded claim. For the material described here as Buddhist monastic landscape, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mihintale using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Elephanta Caves and Lumbini using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Zhoukoudian

East Asia: China, Korea, Japan • Evidence dossier 281 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Pleistocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Zhoukoudian is a site or landscape associated with pleistocene research in China. The defining record highlighted in this atlas is Homo erectus fossils and cave deposits. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Homo erectus fossils and cave deposits
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Homo erectus fossils and cave deposits linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Zhoukoudian after original use?
- Which nearby sites, especially Jiahu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Zhoukoudian with Jiahu, Taosi and Mawangdui. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The fossil locality known for Peking Man played a central role in early twentieth-century paleoanthropology and the history of fossil curation.

Zhoukoudian

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Zhoukoudian begins with a bounded claim. For the material described here as Homo erectus fossils and cave deposits, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Zhoukoudian using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Taosi and Mawangdui using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nihewan Basin

East Asia: China, Korea, Japan • Evidence dossier 282 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Early Pleistocene onward	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Nihewan Basin is a site or landscape associated with early pleistocene onward research in China. The defining record highlighted in this atlas is early hominin lithic assemblages. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early hominin lithic assemblages
Chronological frame	Early Pleistocene onward; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early hominin lithic assemblages linked to the cited phase of Early Pleistocene onward?
- What natural and human processes altered deposits at Nihewan Basin after original use?
- Which nearby sites, especially Peiligang, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nihewan Basin with Peiligang, Erlitou and Goguryeo Tomb Complex. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nihewan Basin

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Nihewan Basin begins with a bounded claim. For the material described here as early hominin lithic assemblages, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nihewan Basin using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Erlitou and Goguryeo Tomb Complex using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Shangchen

East Asia: China, Korea, Japan • Evidence dossier 283 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Early Pleistocene	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Shangchen is a site or landscape associated with early pleistocene research in China. The defining record highlighted in this atlas is early stone tools in loess deposits. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early stone tools in loess deposits
Chronological frame	Early Pleistocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early stone tools in loess deposits linked to the cited phase of Early Pleistocene?
- What natural and human processes altered deposits at Shangchen after original use?
- Which nearby sites, especially Banpo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Shangchen with Banpo, Yinxu (Anyang) and Gyeongju Tumuli. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Shangchen

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Shangchen begins with a bounded claim. For the material described here as early stone tools in loess deposits, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Shangchen using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Yinxu (Anyang) and Gyeongju Tumuli using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Jiahu

East Asia: China, Korea, Japan • Evidence dossier 284 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Neolithic	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Jiahu is a site or landscape associated with neolithic research in China. The defining record highlighted in this atlas is burials flutes and early symbolic signs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	burials flutes and early symbolic signs
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are burials flutes and early symbolic signs linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Jiahu after original use?
- Which nearby sites, especially Dadiwan, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Jiahu with Dadiwan, Sanxingdui and Yoshinogari. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Jiahu

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Jiahu begins with a bounded claim. For the material described here as burials flutes and early symbolic signs, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Jiahu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sanxingdui and Yoshinogari using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Peiligang

East Asia: China, Korea, Japan • Evidence dossier 285 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Peiligang is a site or landscape associated with neolithic research in China. The defining record highlighted in this atlas is early farming village. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early farming village
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early farming village linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Peiligang after original use?
- Which nearby sites, especially Liangzhu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Peiligang with Liangzhu, Jinsha and Sannai-Maruyama. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Peiligang

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Peiligang begins with a bounded claim. For the material described here as early farming village, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Peiligang using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Jinsha and Sannai-Maruyama using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Banpo

East Asia: China, Korea, Japan • Evidence dossier 286 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Banpo is a site or landscape associated with neolithic research in China. The defining record highlighted in this atlas is Yangshao houses and pottery. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Yangshao houses and pottery
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Yangshao houses and pottery linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Banpo after original use?
- Which nearby sites, especially Shimao, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Banpo with Shimao, Terracotta Army (Mausoleum of Qin Shi Huang) and Zhoukoudian. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Banpo

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Banpo begins with a bounded claim. For the material described here as Yangshao houses and pottery, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Banpo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Terracotta Army (Mausoleum of Qin Shi Huang) and Zhoukoudian using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dadiwan

East Asia: China, Korea, Japan • Evidence dossier 287 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Dadiwan is a site or landscape associated with neolithic research in China. The defining record highlighted in this atlas is early village occupation and architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early village occupation and architecture
Chronological frame	Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early village occupation and architecture linked to the cited phase of Neolithic?
- What natural and human processes altered deposits at Dadiwan after original use?
- Which nearby sites, especially Taosi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dadiwan with Taosi, Mawangdui and Nihewan Basin. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dadiwan

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dadiwan begins with a bounded claim. For the material described here as early village occupation and architecture, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dadiwan using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mawangdui and Nihewan Basin using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Liangzhu

East Asia: China, Korea, Japan • Evidence dossier 288 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Late Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Liangzhu is a site or landscape associated with late neolithic research in China. The defining record highlighted in this atlas is jade production city and hydraulic structures. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	jade production city and hydraulic structures
Chronological frame	Late Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are jade production city and hydraulic structures linked to the cited phase of Late Neolithic?
- What natural and human processes altered deposits at Liangzhu after original use?
- Which nearby sites, especially Erlitou, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Liangzhu with Erlitou, Goguryeo Tomb Complex and Shangchen. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Liangzhu

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Liangzhu begins with a bounded claim. For the material described here as jade production city and hydraulic structures, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Liangzhu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Goguryeo Tomb Complex and Shangchen using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Shimao

East Asia: China, Korea, Japan • Evidence dossier 289 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Late Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Shimao is a site or landscape associated with late neolithic research in China. The defining record highlighted in this atlas is fortified stone city and ritual remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortified stone city and ritual remains
Chronological frame	Late Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortified stone city and ritual remains linked to the cited phase of Late Neolithic?
- What natural and human processes altered deposits at Shimao after original use?
- Which nearby sites, especially Yinxu (Anyang), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Shimao with Yinxu (Anyang), Gyeongju Tumuli and Jiahu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Shimao

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Shimao begins with a bounded claim. For the material described here as fortified stone city and ritual remains, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Shimao using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Gyeongju Tumuli and Jiahu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Taosi

East Asia: China, Korea, Japan • Evidence dossier 290 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Late Neolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Taosi is a site or landscape associated with late neolithic research in China. The defining record highlighted in this atlas is large settlement and elite differentiation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large settlement and elite differentiation
Chronological frame	Late Neolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large settlement and elite differentiation linked to the cited phase of Late Neolithic?
- What natural and human processes altered deposits at Taosi after original use?
- Which nearby sites, especially Sanxingdui, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Taosi with Sanxingdui, Yoshinogari and Peiligang. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Taosi

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Taosi begins with a bounded claim. For the material described here as large settlement and elite differentiation, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Taosi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Yoshinogari and Peiligang using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Erlitou

East Asia: China, Korea, Japan • Evidence dossier 291 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Bronze Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Erlitou is a site or landscape associated with bronze age research in China. The defining record highlighted in this atlas is early palatial planning and bronze manufacture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early palatial planning and bronze manufacture
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early palatial planning and bronze manufacture linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Erlitou after original use?
- Which nearby sites, especially Jinsha, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Erlitou with Jinsha, Sannai-Maruyama and Banpo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Erlitou

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Erlitou begins with a bounded claim. For the material described here as early palatial planning and bronze manufacture, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Erlitou using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sannai-Maruyama and Banpo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Yinxu (Anyang)

East Asia: China, Korea, Japan • Evidence dossier 292 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Late Shang	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Yinxu (Anyang) is a site or landscape associated with late shang research in China. The defining record highlighted in this atlas is oracle bones and royal burials. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	oracle bones and royal burials
Chronological frame	Late Shang; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are oracle bones and royal burials linked to the cited phase of Late Shang?
- What natural and human processes altered deposits at Yinxu (Anyang) after original use?
- Which nearby sites, especially Terracotta Army (Mausoleum of Qin Shi Huang), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Yinxu (Anyang) with Terracotta Army (Mausoleum of Qin Shi Huang), Zhoukoudian and Dadiwan. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The Shang royal cemetery and oracle-bone corpus can be cross-read for military action, ritual authority and political administration.

Yinxu (Anyang)

China • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Yinxu (Anyang) begins with a bounded claim. For the material described here as oracle bones and royal burials, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Yinxu (Anyang) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Zhoukoudian and Dadiwan using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sanxingdui

East Asia: China, Korea, Japan • Evidence dossier 293 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Bronze Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Sanxingdui is a site or landscape associated with bronze age research in China. The defining record highlighted in this atlas is bronze masks and sacrificial deposits. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	bronze masks and sacrificial deposits
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are bronze masks and sacrificial deposits linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Sanxingdui after original use?
- Which nearby sites, especially Mawangdui, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sanxingdui with Mawangdui, Nihewan Basin and Liangzhu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sanxingdui

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sanxingdui begins with a bounded claim. For the material described here as bronze masks and sacrificial deposits, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sanxingdui using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nihewan Basin and Liangzhu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Jinsha

East Asia: China, Korea, Japan • Evidence dossier 294 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Bronze Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Jinsha is a site or landscape associated with bronze age research in China. The defining record highlighted in this atlas is ritual deposits and gold artifacts. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ritual deposits and gold artifacts
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ritual deposits and gold artifacts linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Jinsha after original use?
- Which nearby sites, especially Goguryeo Tomb Complex, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Jinsha with Goguryeo Tomb Complex, Shangchen and Shimao. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Jinsha

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Jinsha begins with a bounded claim. For the material described here as ritual deposits and gold artifacts, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Jinsha using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Shangchen and Shimao using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Terracotta Army (Mausoleum of Qin Shi Huang)

East Asia: China, Korea, Japan • Evidence dossier 295 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Qin	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Terracotta Army (Mausoleum of Qin Shi Huang) is a site or landscape associated with qin research in China. The defining record highlighted in this atlas is life-size ceramic warriors and imperial burial. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	life-size ceramic warriors and imperial burial
Chronological frame	Qin; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are life-size ceramic warriors and imperial burial linked to the cited phase of Qin?
- What natural and human processes altered deposits at Terracotta Army (Mausoleum of Qin Shi Huang) after original use?
- Which nearby sites, especially Gyeongju Tumuli, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Terracotta Army (Mausoleum of Qin Shi Huang) with Gyeongju Tumuli, Jiahu and Taosi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Discovered in the 1970s, the vast terracotta figures provide evidence for imperial workshop organization, scale and highly controlled burial design.

Terracotta Army (Mausoleum of Qin Shi Huang)

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Terracotta Army (Mausoleum of Qin Shi Huang) begins with a bounded claim. For the material described here as life-size ceramic warriors and imperial burial, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Terracotta Army (Mausoleum of Qin Shi Huang) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Jiahu and Taosi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mawangdui

East Asia: China, Korea, Japan • Evidence dossier 296 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
China	Han	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Mawangdui is a site or landscape associated with han research in China. The defining record highlighted in this atlas is elite tombs silk manuscripts and preserved remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	elite tombs silk manuscripts and preserved remains
Chronological frame	Han; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are elite tombs silk manuscripts and preserved remains linked to the cited phase of Han?
- What natural and human processes altered deposits at Mawangdui after original use?
- Which nearby sites, especially Yoshinogari, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mawangdui with Yoshinogari, Peiligang and Erlitou. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mawangdui

China • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mawangdui begins with a bounded claim. For the material described here as elite tombs silk manuscripts and preserved remains, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mawangdui using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Peiligang and Erlitou using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Goguryeo Tomb Complex

East Asia: China, Korea, Japan • Evidence dossier 297 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
North Korea	ancient	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Goguryeo Tomb Complex is a site or landscape associated with ancient research in North Korea. The defining record highlighted in this atlas is mural tomb chambers. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	mural tomb chambers
Chronological frame	ancient; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are mural tomb chambers linked to the cited phase of ancient?
- What natural and human processes altered deposits at Goguryeo Tomb Complex after original use?
- Which nearby sites, especially Sannai-Maruyama, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Goguryeo Tomb Complex with Sannai-Maruyama, Banpo and Yinxu (Anyang). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Goguryeo Tomb Complex

North Korea • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Goguryeo Tomb Complex begins with a bounded claim. For the material described here as mural tomb chambers, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Goguryeo Tomb Complex using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Banpo and Yinxu (Anyang) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Gyeongju Tumuli

East Asia: China, Korea, Japan • Evidence dossier 298 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
South Korea	**Silla**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Gyeongju Tumuli is a site or landscape associated with silla research in South Korea. The defining record highlighted in this atlas is elite mounded tombs and grave goods. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	elite mounded tombs and grave goods
Chronological frame	Silla; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are elite mounded tombs and grave goods linked to the cited phase of Silla?
- What natural and human processes altered deposits at Gyeongju Tumuli after original use?
- Which nearby sites, especially Zhoukoudian, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Gyeongju Tumuli with Zhoukoudian, Dadiwan and Sanxingdui. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Gyeongju Tumuli

South Korea • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Gyeongju Tumuli begins with a bounded claim. For the material described here as elite mounded tombs and grave goods, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Gyeongju Tumuli using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dadiwan and Sanxingdui using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Yoshinogari

East Asia: China, Korea, Japan • Evidence dossier 299 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Japan	**Yayoi**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Yoshinogari is a site or landscape associated with yayoi research in Japan. The defining record highlighted in this atlas is moated settlement and timber structures. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	moated settlement and timber structures
Chronological frame	Yayoi; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are moated settlement and timber structures linked to the cited phase of Yayoi?
- What natural and human processes altered deposits at Yoshinogari after original use?
- Which nearby sites, especially Nihewan Basin, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Yoshinogari with Nihewan Basin, Liangzhu and Jinsha. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Yoshinogari

Japan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Yoshinogari begins with a bounded claim. For the material described here as moated settlement and timber structures, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Yoshinogari using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Liangzhu and Jinsha using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sannai-Maruyama

East Asia: China, Korea, Japan • Evidence dossier 300 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Japan	**Jomon**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Sannai-Maruyama is a site or landscape associated with jomon research in Japan. The defining record highlighted in this atlas is large long-lived settlement and pit dwellings. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in east asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large long-lived settlement and pit dwellings
Chronological frame	Jomon; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large long-lived settlement and pit dwellings linked to the cited phase of Jomon?
- What natural and human processes altered deposits at Sannai-Maruyama after original use?
- Which nearby sites, especially Shangchen, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sannai-Maruyama with Shangchen, Shimao and Terracotta Army (Mausoleum of Qin Shi Huang). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sannai-Maruyama

Japan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sannai-Maruyama begins with a bounded claim. For the material described here as large long-lived settlement and pit dwellings, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sannai-Maruyama using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Shimao and Terracotta Army (Mausoleum of Qin Shi Huang) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Angkor Wat

Southeast Asia • Evidence dossier 301 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cambodia	**Khmer**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Angkor Wat is a site or landscape associated with khmer research in Cambodia. The defining record highlighted in this atlas is temple and cosmological architectural plan. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple and cosmological architectural plan
Chronological frame	Khmer; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple and cosmological architectural plan linked to the cited phase of Khmer?
- What natural and human processes altered deposits at Angkor Wat after original use?
- Which nearby sites, especially Koh Ker, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Angkor Wat with Koh Ker, Ban Chiang and My Son. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The temple is one part of an extensive Khmer urban and hydraulic landscape, not an isolated stone monument.

RESEARCH POINTER: https://whc.unesco.org/en/list/668/

Angkor Wat

Cambodia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Angkor Wat begins with a bounded claim. For the material described here as temple and cosmological architectural plan, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Angkor Wat using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ban Chiang and My Son using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Angkor Thom

Southeast Asia • Evidence dossier 302 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cambodia	**Khmer**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Angkor Thom is a site or landscape associated with khmer research in Cambodia. The defining record highlighted in this atlas is walled royal city and Bayon. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	walled royal city and Bayon
Chronological frame	Khmer; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are walled royal city and Bayon linked to the cited phase of Khmer?
- What natural and human processes altered deposits at Angkor Thom after original use?
- Which nearby sites, especially Preah Vihear, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Angkor Thom with Preah Vihear, Non Nok Tha and Oc Eo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Remote sensing and landscape archaeology reveal the importance of embankments, waterways, domestic mounds and routes around its famous temples.

RESEARCH POINTER: https://whc.unesco.org/en/list/668/

Angkor Thom

Cambodia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Angkor Thom begins with a bounded claim. For the material described here as walled royal city and Bayon, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Angkor Thom using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Non Nok Tha and Oc Eo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ta Prohm

Southeast Asia • Evidence dossier 303 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cambodia	**Khmer**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Ta Prohm is a site or landscape associated with khmer research in Cambodia. The defining record highlighted in this atlas is temple-monastery and environmental conservation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple-monastery and environmental conservation
Chronological frame	Khmer; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple-monastery and environmental conservation linked to the cited phase of Khmer?
- What natural and human processes altered deposits at Ta Prohm after original use?
- Which nearby sites, especially Sambor Prei Kuk, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ta Prohm with Sambor Prei Kuk, Phimai and Borobudur. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ta Prohm

Cambodia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ta Prohm begins with a bounded claim. For the material described here as temple-monastery and environmental conservation, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ta Prohm using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Phimai and Borobudur using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Koh Ker

Southeast Asia • Evidence dossier 304 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cambodia	**Khmer**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Koh Ker is a site or landscape associated with khmer research in Cambodia. The defining record highlighted in this atlas is short-lived capital and pyramid temple. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	short-lived capital and pyramid temple
Chronological frame	Khmer; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are short-lived capital and pyramid temple linked to the cited phase of Khmer?
- What natural and human processes altered deposits at Koh Ker after original use?
- Which nearby sites, especially Banteay Chhmar, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Koh Ker with Banteay Chhmar, Si Thep and Prambanan. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Koh Ker

Cambodia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Koh Ker begins with a bounded claim. For the material described here as short-lived capital and pyramid temple, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Koh Ker using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Si Thep and Prambanan using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Preah Vihear

Southeast Asia • Evidence dossier 305 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cambodia	**Khmer**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Preah Vihear is a site or landscape associated with khmer research in Cambodia. The defining record highlighted in this atlas is mountain temple and processional layout. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	mountain temple and processional layout
Chronological frame	Khmer; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are mountain temple and processional layout linked to the cited phase of Khmer?
- What natural and human processes altered deposits at Preah Vihear after original use?
- Which nearby sites, especially Wat Phou, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Preah Vihear with Wat Phou, Dong Son sites and Gunung Padang. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Preah Vihear

Cambodia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Preah Vihear begins with a bounded claim. For the material described here as mountain temple and processional layout, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Preah Vihear using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Dong Son sites and Gunung Padang using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sambor Prei Kuk

Southeast Asia • Evidence dossier 306 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cambodia	**pre-Angkorian**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Sambor Prei Kuk is a site or landscape associated with pre-angkorian research in Cambodia. The defining record highlighted in this atlas is brick temple complexes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	brick temple complexes
Chronological frame	pre-Angkorian; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are brick temple complexes linked to the cited phase of pre-Angkorian?
- What natural and human processes altered deposits at Sambor Prei Kuk after original use?
- Which nearby sites, especially Plain of Jars, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sambor Prei Kuk with Plain of Jars, Co Loa and Angkor Wat. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sambor Prei Kuk

Cambodia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Sambor Prei Kuk begins with a bounded claim. For the material described here as brick temple complexes, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sambor Prei Kuk using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Co Loa and Angkor Wat using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Banteay Chhmar

Southeast Asia • Evidence dossier 307 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Cambodia	**Khmer**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Banteay Chhmar is a site or landscape associated with khmer research in Cambodia. The defining record highlighted in this atlas is large provincial temple and relief sculpture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large provincial temple and relief sculpture
Chronological frame	Khmer; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large provincial temple and relief sculpture linked to the cited phase of Khmer?
- What natural and human processes altered deposits at Banteay Chhmar after original use?
- Which nearby sites, especially Ban Chiang, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Banteay Chhmar with Ban Chiang, My Son and Angkor Thom. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Banteay Chhmar

Cambodia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Banteay Chhmar begins with a bounded claim. For the material described here as large provincial temple and relief sculpture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Banteay Chhmar using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with My Son and Angkor Thom using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Wat Phou

Southeast Asia • Evidence dossier 308 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Laos	Khmer	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Wat Phou is a site or landscape associated with khmer research in Laos. The defining record highlighted in this atlas is temple landscape aligned with sacred mountain. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple landscape aligned with sacred mountain
Chronological frame	Khmer; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple landscape aligned with sacred mountain linked to the cited phase of Khmer?
- What natural and human processes altered deposits at Wat Phou after original use?
- Which nearby sites, especially Non Nok Tha, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Wat Phou with Non Nok Tha, Oc Eo and Ta Prohm. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Wat Phou

Laos • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Wat Phou begins with a bounded claim. For the material described here as temple landscape aligned with sacred mountain, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Wat Phou using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Oc Eo and Ta Prohm using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Plain of Jars

Southeast Asia • Evidence dossier 309 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Laos	Iron Age	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Plain of Jars is a site or landscape associated with iron age research in Laos. The defining record highlighted in this atlas is megalithic stone jars and mortuary practice. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	megalithic stone jars and mortuary practice
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are megalithic stone jars and mortuary practice linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Plain of Jars after original use?
- Which nearby sites, especially Phimai, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Plain of Jars with Phimai, Borobudur and Koh Ker. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Plain of Jars

Laos • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Plain of Jars begins with a bounded claim. For the material described here as megalithic stone jars and mortuary practice, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Plain of Jars using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Borobudur and Koh Ker using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ban Chiang

Southeast Asia • Evidence dossier 310 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Thailand	**prehistoric**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Ban Chiang is a site or landscape associated with prehistoric research in Thailand. The defining record highlighted in this atlas is burials ceramics and metalworking chronology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	burials ceramics and metalworking chronology
Chronological frame	prehistoric; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are burials ceramics and metalworking chronology linked to the cited phase of prehistoric?
- What natural and human processes altered deposits at Ban Chiang after original use?
- Which nearby sites, especially Si Thep, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ban Chiang with Si Thep, Prambanan and Preah Vihear. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ban Chiang

Thailand • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ban Chiang begins with a bounded claim. For the material described here as burials ceramics and metalworking chronology, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ban Chiang using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Prambanan and Preah Vihear using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Non Nok Tha

Southeast Asia • Evidence dossier 311 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Thailand	**prehistoric**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Non Nok Tha is a site or landscape associated with prehistoric research in Thailand. The defining record highlighted in this atlas is burial and bronze chronology debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	burial and bronze chronology debates
Chronological frame	prehistoric; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are burial and bronze chronology debates linked to the cited phase of prehistoric?
- What natural and human processes altered deposits at Non Nok Tha after original use?
- Which nearby sites, especially Dong Son sites, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Non Nok Tha with Dong Son sites, Gunung Padang and Sambor Prei Kuk. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Non Nok Tha

Thailand • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Non Nok Tha begins with a bounded claim. For the material described here as burial and bronze chronology debates, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Non Nok Tha using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Gunung Padang and Sambor Prei Kuk using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Phimai

Southeast Asia • Evidence dossier 312 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Thailand	**Khmer**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Phimai is a site or landscape associated with khmer research in Thailand. The defining record highlighted in this atlas is temple and regional road network. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple and regional road network
Chronological frame	Khmer; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple and regional road network linked to the cited phase of Khmer?
- What natural and human processes altered deposits at Phimai after original use?
- Which nearby sites, especially Co Loa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Phimai with Co Loa, Angkor Wat and Banteay Chhmar. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Phimai

Thailand • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Phimai begins with a bounded claim. For the material described here as temple and regional road network, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Phimai using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Angkor Wat and Banteay Chhmar using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Si Thep

Southeast Asia • Evidence dossier 313 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Thailand	**Dvaravati onward**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Si Thep is a site or landscape associated with dvaravati onward research in Thailand. The defining record highlighted in this atlas is moated ancient city and monuments. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	moated ancient city and monuments
Chronological frame	Dvaravati onward; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are moated ancient city and monuments linked to the cited phase of Dvaravati onward?
- What natural and human processes altered deposits at Si Thep after original use?
- Which nearby sites, especially My Son, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Si Thep with My Son, Angkor Thom and Wat Phou. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Si Thep

Thailand • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Si Thep begins with a bounded claim. For the material described here as moated ancient city and monuments, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Si Thep using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Angkor Thom and Wat Phou using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Dong Son sites

Southeast Asia • Evidence dossier 314 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Vietnam	**Iron Age**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Dong Son sites is a site or landscape associated with iron age research in Vietnam. The defining record highlighted in this atlas is bronze drums and regional exchange. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	bronze drums and regional exchange
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are bronze drums and regional exchange linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Dong Son sites after original use?
- Which nearby sites, especially Oc Eo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Dong Son sites with Oc Eo, Ta Prohm and Plain of Jars. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Dong Son sites

Vietnam • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Dong Son sites begins with a bounded claim. For the material described here as bronze drums and regional exchange, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Dong Son sites using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ta Prohm and Plain of Jars using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Co Loa

Southeast Asia • Evidence dossier 315 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Vietnam	**Iron Age**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Co Loa is a site or landscape associated with iron age research in Vietnam. The defining record highlighted in this atlas is multiple concentric earthworks. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	multiple concentric earthworks
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are multiple concentric earthworks linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Co Loa after original use?
- Which nearby sites, especially Borobudur, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Co Loa with Borobudur, Koh Ker and Ban Chiang. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Co Loa

Vietnam • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Co Loa begins with a bounded claim. For the material described here as multiple concentric earthworks, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Co Loa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Koh Ker and Ban Chiang using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

My Son

Southeast Asia • Evidence dossier 316 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Vietnam	**Champa**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

My Son is a site or landscape associated with champa research in Vietnam. The defining record highlighted in this atlas is brick sanctuaries and sculptural traditions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	brick sanctuaries and sculptural traditions
Chronological frame	Champa; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are brick sanctuaries and sculptural traditions linked to the cited phase of Champa?
- What natural and human processes altered deposits at My Son after original use?
- Which nearby sites, especially Prambanan, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare My Son with Prambanan, Preah Vihear and Non Nok Tha. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

My Son

Vietnam • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of My Son begins with a bounded claim. For the material described here as brick sanctuaries and sculptural traditions, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for My Son using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Preah Vihear and Non Nok Tha using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Oc Eo

Southeast Asia • Evidence dossier 317 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Vietnam	**early centuries CE**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Oc Eo is a site or landscape associated with early centuries ce research in Vietnam. The defining record highlighted in this atlas is Funan-era trade town and canals. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Funan-era trade town and canals
Chronological frame	early centuries CE; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Funan-era trade town and canals linked to the cited phase of early centuries CE?
- What natural and human processes altered deposits at Oc Eo after original use?
- Which nearby sites, especially Gunung Padang, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Oc Eo with Gunung Padang, Sambor Prei Kuk and Phimai. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Oc Eo

Vietnam • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Oc Eo begins with a bounded claim. For the material described here as Funan-era trade town and canals, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Oc Eo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sambor Prei Kuk and Phimai using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Borobudur

Southeast Asia • Evidence dossier 318 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Indonesia	**Sailendra**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Borobudur is a site or landscape associated with sailendra research in Indonesia. The defining record highlighted in this atlas is Buddhist monument and narrative reliefs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Buddhist monument and narrative reliefs
Chronological frame	Sailendra; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Buddhist monument and narrative reliefs linked to the cited phase of Sailendra?
- What natural and human processes altered deposits at Borobudur after original use?
- Which nearby sites, especially Angkor Wat, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Borobudur with Angkor Wat, Banteay Chhmar and Si Thep. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Restoration history is part of the archaeological record: the monument's imagery and fabric must be interpreted alongside earlier disassembly and conservation.

Borobudur

Indonesia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Borobudur begins with a bounded claim. For the material described here as Buddhist monument and narrative reliefs, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Borobudur using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Banteay Chhmar and Si Thep using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Prambanan

Southeast Asia • Evidence dossier 319 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Indonesia	**Mataram**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Prambanan is a site or landscape associated with mataram research in Indonesia. The defining record highlighted in this atlas is Hindu temple ensemble. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Hindu temple ensemble
Chronological frame	Mataram; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Hindu temple ensemble linked to the cited phase of Mataram?
- What natural and human processes altered deposits at Prambanan after original use?
- Which nearby sites, especially Angkor Thom, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Prambanan with Angkor Thom, Wat Phou and Dong Son sites. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Prambanan

Indonesia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Prambanan begins with a bounded claim. For the material described here as Hindu temple ensemble, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Prambanan using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Wat Phou and Dong Son sites using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Gunung Padang

Southeast Asia • Evidence dossier 320 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Indonesia	**prehistoric-historic**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Gunung Padang is a site or landscape associated with prehistoric-historic research in Indonesia. The defining record highlighted in this atlas is megalithic terraces and disputed deep dates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in southeast asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	megalithic terraces and disputed deep dates
Chronological frame	prehistoric-historic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are megalithic terraces and disputed deep dates linked to the cited phase of prehistoric-historic?
- What natural and human processes altered deposits at Gunung Padang after original use?
- Which nearby sites, especially Ta Prohm, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Gunung Padang with Ta Prohm, Plain of Jars and Co Loa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Gunung Padang

Indonesia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Gunung Padang begins with a bounded claim. For the material described here as megalithic terraces and disputed deep dates, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Gunung Padang using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Plain of Jars and Co Loa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Denisova Cave

Inner Asia, Steppe, and Siberia • Evidence dossier 321 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Denisova Cave is a site or landscape associated with paleolithic research in Russia. The defining record highlighted in this atlas is Denisovan remains and ancient DNA. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Denisovan remains and ancient DNA
Chronological frame	Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Denisovan remains and ancient DNA linked to the cited phase of Paleolithic?
- What natural and human processes altered deposits at Denisova Cave after original use?
- Which nearby sites, especially Afontova Gora, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Denisova Cave with Afontova Gora, Pazyryk Burials and Tashbulak. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Ancient biomolecular work transformed interpretation of fragmentary finds and showed why genetic lineages need not correspond to simple artifact cultures.

Denisova Cave

Russia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Denisova Cave begins with a bounded claim. For the material described here as Denisovan remains and ancient DNA, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Denisova Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pazyryk Burials and Tashbulak using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chagyrskaya Cave

Inner Asia, Steppe, and Siberia • Evidence dossier 322 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Middle Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Chagyrskaya Cave is a site or landscape associated with middle paleolithic research in Russia. The defining record highlighted in this atlas is Neanderthal fossils and lithic technology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Neanderthal fossils and lithic technology
Chronological frame	Middle Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Neanderthal fossils and lithic technology linked to the cited phase of Middle Paleolithic?
- What natural and human processes altered deposits at Chagyrskaya Cave after original use?
- Which nearby sites, especially Yana Rhinoceros Horn Site, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chagyrskaya Cave with Yana Rhinoceros Horn Site, Ukok Plateau and Samarkand (Afrasiab). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Chagyrskaya Cave

Russia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Chagyrskaya Cave begins with a bounded claim. For the material described here as Neanderthal fossils and lithic technology, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chagyrskaya Cave using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ukok Plateau and Samarkand (Afrasiab) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mal'ta

Inner Asia, Steppe, and Siberia • Evidence dossier 323 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Upper Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Mal'ta is a site or landscape associated with upper paleolithic research in Russia. The defining record highlighted in this atlas is burials and carved ivory figurines. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	burials and carved ivory figurines
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are burials and carved ivory figurines linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Mal'ta after original use?
- Which nearby sites, especially Ust'-Ishim, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mal'ta with Ust'-Ishim, Issyk Kurgan and Merv. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mal'ta

Russia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mal'ta begins with a bounded claim. For the material described here as burials and carved ivory figurines, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mal'ta using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Issyk Kurgan and Merv using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Afontova Gora

Inner Asia, Steppe, and Siberia • Evidence dossier 324 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Upper Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Afontova Gora is a site or landscape associated with upper paleolithic research in Russia. The defining record highlighted in this atlas is human remains and cultural deposits. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	human remains and cultural deposits
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are human remains and cultural deposits linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Afontova Gora after original use?
- Which nearby sites, especially Kostenki, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Afontova Gora with Kostenki, Tamgaly Petroglyphs and Nisa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Afontova Gora

Russia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Afontova Gora begins with a bounded claim. For the material described here as human remains and cultural deposits, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Afontova Gora using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tamgaly Petroglyphs and Nisa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Yana Rhinoceros Horn Site

Inner Asia, Steppe, and Siberia • Evidence dossier 325 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Upper Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Yana Rhinoceros Horn Site is a site or landscape associated with upper paleolithic research in Russia. The defining record highlighted in this atlas is high-latitude settlement evidence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	high-latitude settlement evidence
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are high-latitude settlement evidence linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Yana Rhinoceros Horn Site after original use?
- Which nearby sites, especially Sungir, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Yana Rhinoceros Horn Site with Sungir, Botai and Karakorum. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Yana Rhinoceros Horn Site

Russia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Yana Rhinoceros Horn Site begins with a bounded claim. For the material described here as high-latitude settlement evidence, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Yana Rhinoceros Horn Site using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Botai and Karakorum using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ust'-Ishim

Inner Asia, Steppe, and Siberia • Evidence dossier 326 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Upper Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Ust'-Ishim is a site or landscape associated with upper paleolithic research in Russia. The defining record highlighted in this atlas is early modern human fossil and genetics. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early modern human fossil and genetics
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early modern human fossil and genetics linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Ust'-Ishim after original use?
- Which nearby sites, especially Arzhan, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ust'-Ishim with Arzhan, Begash and Denisova Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ust'-Ishim

Russia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ust'-Ishim begins with a bounded claim. For the material described here as early modern human fossil and genetics, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ust'-Ishim using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Begash and Denisova Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kostenki

Inner Asia, Steppe, and Siberia • Evidence dossier 327 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Upper Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Kostenki is a site or landscape associated with upper paleolithic research in Russia. The defining record highlighted in this atlas is settlements and mammoth-bone contexts. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	settlements and mammoth-bone contexts
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are settlements and mammoth-bone contexts linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Kostenki after original use?
- Which nearby sites, especially Pazyryk Burials, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kostenki with Pazyryk Burials, Tashbulak and Chagyrskaya Cave. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kostenki

Russia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Kostenki begins with a bounded claim. For the material described here as settlements and mammoth-bone contexts, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kostenki using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tashbulak and Chagyrskaya Cave using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sungir

Inner Asia, Steppe, and Siberia • Evidence dossier 328 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Upper Paleolithic**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Sungir is a site or landscape associated with upper paleolithic research in Russia. The defining record highlighted in this atlas is elaborate burials and ornaments. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	elaborate burials and ornaments
Chronological frame	Upper Paleolithic; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are elaborate burials and ornaments linked to the cited phase of Upper Paleolithic?
- What natural and human processes altered deposits at Sungir after original use?
- Which nearby sites, especially Ukok Plateau, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sungir with Ukok Plateau, Samarkand (Afrasiab) and Mal'ta. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sungir

Russia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sungir begins with a bounded claim. For the material described here as elaborate burials and ornaments, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sungir using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Samarkand (Afrasiab) and Mal'ta using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Arzhan

Inner Asia, Steppe, and Siberia • Evidence dossier 329 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	Iron Age	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Arzhan is a site or landscape associated with iron age research in Russia. The defining record highlighted in this atlas is early Scythian burial mounds. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Scythian burial mounds
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Scythian burial mounds linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Arzhan after original use?
- Which nearby sites, especially Issyk Kurgan, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Arzhan with Issyk Kurgan, Merv and Afontova Gora. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Arzhan

Russia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Arzhan begins with a bounded claim. For the material described here as early Scythian burial mounds, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Arzhan using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Merv and Afontova Gora using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pazyryk Burials

Inner Asia, Steppe, and Siberia • Evidence dossier 330 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	Iron Age	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Pazyryk Burials is a site or landscape associated with iron age research in Russia. The defining record highlighted in this atlas is frozen kurgans textiles and horses. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	frozen kurgans textiles and horses
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are frozen kurgans textiles and horses linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Pazyryk Burials after original use?
- Which nearby sites, especially Tamgaly Petroglyphs, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pazyryk Burials with Tamgaly Petroglyphs, Nisa and Yana Rhinoceros Horn Site. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Permafrost conditions preserved textiles, wooden objects and horse equipment that would ordinarily perish, making conservation and context essential.

Pazyryk Burials

Russia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Pazyryk Burials begins with a bounded claim. For the material described here as frozen kurgans textiles and horses, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pazyryk Burials using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nisa and Yana Rhinoceros Horn Site using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ukok Plateau

Inner Asia, Steppe, and Siberia • Evidence dossier 331 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Russia	**Iron Age**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Ukok Plateau is a site or landscape associated with iron age research in Russia. The defining record highlighted in this atlas is permafrost-preserved burials. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	permafrost-preserved burials
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are permafrost-preserved burials linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Ukok Plateau after original use?
- Which nearby sites, especially Botai, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ukok Plateau with Botai, Karakorum and Ust'-Ishim. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ukok Plateau

Russia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ukok Plateau begins with a bounded claim. For the material described here as permafrost-preserved burials, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ukok Plateau using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Karakorum and Ust'-Ishim using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Issyk Kurgan

Inner Asia, Steppe, and Siberia • Evidence dossier 332 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kazakhstan	**Iron Age**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Issyk Kurgan is a site or landscape associated with iron age research in Kazakhstan. The defining record highlighted in this atlas is elite Saka burial regalia. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	elite Saka burial regalia
Chronological frame	Iron Age; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are elite Saka burial regalia linked to the cited phase of Iron Age?
- What natural and human processes altered deposits at Issyk Kurgan after original use?
- Which nearby sites, especially Begash, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Issyk Kurgan with Begash, Denisova Cave and Kostenki. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Issyk Kurgan

Kazakhstan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Issyk Kurgan begins with a bounded claim. For the material described here as elite Saka burial regalia, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Issyk Kurgan using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Denisova Cave and Kostenki using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tamgaly Petroglyphs

Inner Asia, Steppe, and Siberia • Evidence dossier 333 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kazakhstan	**Bronze Age onward**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Tamgaly Petroglyphs is a site or landscape associated with bronze age onward research in Kazakhstan. The defining record highlighted in this atlas is rock art ritual landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock art ritual landscape
Chronological frame	Bronze Age onward; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock art ritual landscape linked to the cited phase of Bronze Age onward?
- What natural and human processes altered deposits at Tamgaly Petroglyphs after original use?
- Which nearby sites, especially Tashbulak, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tamgaly Petroglyphs with Tashbulak, Chagyrskaya Cave and Sungir. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tamgaly Petroglyphs

Kazakhstan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tamgaly Petroglyphs begins with a bounded claim. For the material described here as rock art ritual landscape, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tamgaly Petroglyphs using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chagyrskaya Cave and Sungir using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Botai

Inner Asia, Steppe, and Siberia • Evidence dossier 334 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kazakhstan	Chalcolithic	Lithic

01 WHAT SURVIVES AND WHY IT MATTERS

Botai is a site or landscape associated with chalcolithic research in Kazakhstan. The defining record highlighted in this atlas is equid exploitation and domestication debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	equid exploitation and domestication debates
Chronological frame	Chalcolithic; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are equid exploitation and domestication debates linked to the cited phase of Chalcolithic?
- What natural and human processes altered deposits at Botai after original use?
- Which nearby sites, especially Samarkand (Afrasiab), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Botai with Samarkand (Afrasiab), Mal'ta and Arzhan. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Botai

Kazakhstan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Botai begins with a bounded claim. For the material described here as equid exploitation and domestication debates, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Botai using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mal'ta and Arzhan using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Begash

Inner Asia, Steppe, and Siberia • Evidence dossier 335 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Kazakhstan	Bronze Age	Network

01 WHAT SURVIVES AND WHY IT MATTERS

Begash is a site or landscape associated with bronze age research in Kazakhstan. The defining record highlighted in this atlas is pastoral trade and crops. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	pastoral trade and crops
Chronological frame	Bronze Age; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are pastoral trade and crops linked to the cited phase of Bronze Age?
- What natural and human processes altered deposits at Begash after original use?
- Which nearby sites, especially Merv, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Begash with Merv, Afontova Gora and Pazyryk Burials. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Begash

Kazakhstan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Begash begins with a bounded claim. For the material described here as pastoral trade and crops, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Begash using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Afontova Gora and Pazyryk Burials using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tashbulak

Inner Asia, Steppe, and Siberia • Evidence dossier 336 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Uzbekistan	**medieval**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Tashbulak is a site or landscape associated with medieval research in Uzbekistan. The defining record highlighted in this atlas is mountain town along regional trade routes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	mountain town along regional trade routes
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are mountain town along regional trade routes linked to the cited phase of medieval?
- What natural and human processes altered deposits at Tashbulak after original use?
- Which nearby sites, especially Nisa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tashbulak with Nisa, Yana Rhinoceros Horn Site and Ukok Plateau. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tashbulak

Uzbekistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tashbulak begins with a bounded claim. For the material described here as mountain town along regional trade routes, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tashbulak using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Yana Rhinoceros Horn Site and Ukok Plateau using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Samarkand (Afrasiab)

Inner Asia, Steppe, and Siberia • Evidence dossier 337 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Uzbekistan	**antique-medieval**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Samarkand (Afrasiab) is a site or landscape associated with antique-medieval research in Uzbekistan. The defining record highlighted in this atlas is urban stratigraphy and murals. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	urban stratigraphy and murals
Chronological frame	antique-medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are urban stratigraphy and murals linked to the cited phase of antique-medieval?
- What natural and human processes altered deposits at Samarkand (Afrasiab) after original use?
- Which nearby sites, especially Karakorum, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Samarkand (Afrasiab) with Karakorum, Ust'-Ishim and Issyk Kurgan. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Samarkand (Afrasiab)

Uzbekistan • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Samarkand (Afrasiab) begins with a bounded claim. For the material described here as urban stratigraphy and murals, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Samarkand (Afrasiab) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ust'-Ishim and Issyk Kurgan using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Merv

Inner Asia, Steppe, and Siberia • Evidence dossier 338 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Turkmenistan	**antique-medieval**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Merv is a site or landscape associated with antique-medieval research in Turkmenistan. The defining record highlighted in this atlas is multi-city oasis occupation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	multi-city oasis occupation
Chronological frame	antique-medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are multi-city oasis occupation linked to the cited phase of antique-medieval?
- What natural and human processes altered deposits at Merv after original use?
- Which nearby sites, especially Denisova Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Merv with Denisova Cave, Kostenki and Tamgaly Petroglyphs. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Merv

Turkmenistan • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Merv begins with a bounded claim. For the material described here as multi-city oasis occupation, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Merv using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kostenki and Tamgaly Petroglyphs using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nisa

Inner Asia, Steppe, and Siberia • Evidence dossier 339 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Turkmenistan	**Parthian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Nisa is a site or landscape associated with parthian research in Turkmenistan. The defining record highlighted in this atlas is royal complex and ivory rhyta. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal complex and ivory rhyta
Chronological frame	Parthian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal complex and ivory rhyta linked to the cited phase of Parthian?
- What natural and human processes altered deposits at Nisa after original use?
- Which nearby sites, especially Chagyrskaya Cave, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nisa with Chagyrskaya Cave, Sungir and Botai. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nisa

Turkmenistan • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nisa begins with a bounded claim. For the material described here as royal complex and ivory rhyta, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nisa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sungir and Botai using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Karakorum

Inner Asia, Steppe, and Siberia • Evidence dossier 340 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mongolia	**Mongol Empire**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Karakorum is a site or landscape associated with mongol empire research in Mongolia. The defining record highlighted in this atlas is imperial capital and cosmopolitan exchange. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in central and north asia is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	imperial capital and cosmopolitan exchange
Chronological frame	Mongol Empire; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are imperial capital and cosmopolitan exchange linked to the cited phase of Mongol Empire?
- What natural and human processes altered deposits at Karakorum after original use?
- Which nearby sites, especially Mal'ta, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Karakorum with Mal'ta, Arzhan and Begash. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Karakorum

Mongolia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Karakorum begins with a bounded claim. For the material described here as imperial capital and cosmopolitan exchange, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Karakorum using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Arzhan and Begash using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Clovis (Blackwater Draw)

North America • Evidence dossier 341 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Late Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Clovis (Blackwater Draw) is a site or landscape associated with late pleistocene research in United States. The defining record highlighted in this atlas is fluted projectile points and mammoth hunting. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fluted projectile points and mammoth hunting
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fluted projectile points and mammoth hunting linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Clovis (Blackwater Draw) after original use?
- Which nearby sites, especially Paisley Caves, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Clovis (Blackwater Draw) with Paisley Caves, Watson Brake and Bandelier. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Clovis (Blackwater Draw)

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Clovis (Blackwater Draw) begins with a bounded claim. For the material described here as fluted projectile points and mammoth hunting, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Clovis (Blackwater Draw) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Watson Brake and Bandelier using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Folsom Site

North America • Evidence dossier 342 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Late Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Folsom Site is a site or landscape associated with late pleistocene research in United States. The defining record highlighted in this atlas is bison kill with fluted projectile point. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	bison kill with fluted projectile point
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are bison kill with fluted projectile point linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Folsom Site after original use?
- Which nearby sites, especially Cooper's Ferry, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Folsom Site with Cooper's Ferry, Serpent Mound and Casa Grande Ruins. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Folsom Site

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Folsom Site begins with a bounded claim. For the material described here as bison kill with fluted projectile point, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Folsom Site using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Serpent Mound and Casa Grande Ruins using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Meadowcroft Rockshelter

North America • Evidence dossier 343 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Late Pleistocene-Holocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Meadowcroft Rockshelter is a site or landscape associated with late pleistocene-holocene research in United States. The defining record highlighted in this atlas is long occupation and early dating debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long occupation and early dating debates
Chronological frame	Late Pleistocene-Holocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long occupation and early dating debates linked to the cited phase of Late Pleistocene-Holocene?
- What natural and human processes altered deposits at Meadowcroft Rockshelter after original use?
- Which nearby sites, especially Manis Mastodon, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Meadowcroft Rockshelter with Manis Mastodon, Hopewell Mound Group and Snaketown. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Meadowcroft Rockshelter

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Meadowcroft Rockshelter begins with a bounded claim. For the material described here as long occupation and early dating debates, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Meadowcroft Rockshelter using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Hopewell Mound Group and Snaketown using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Paisley Caves

North America • Evidence dossier 344 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Late Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Paisley Caves is a site or landscape associated with late pleistocene research in United States. The defining record highlighted in this atlas is human coprolites and ancient DNA. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	human coprolites and ancient DNA
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are human coprolites and ancient DNA linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Paisley Caves after original use?
- Which nearby sites, especially Cactus Hill, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Paisley Caves with Cactus Hill, Newark Earthworks and Etowah Mounds. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Paisley Caves

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Paisley Caves begins with a bounded claim. For the material described here as human coprolites and ancient DNA, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Paisley Caves using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Newark Earthworks and Etowah Mounds using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cooper's Ferry

North America • Evidence dossier 345 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Late Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Cooper's Ferry is a site or landscape associated with late pleistocene research in United States. The defining record highlighted in this atlas is early occupation along inland migration route. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early occupation along inland migration route
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early occupation along inland migration route linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Cooper's Ferry after original use?
- Which nearby sites, especially Cahokia, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cooper's Ferry with Cahokia, Chaco Canyon and Moundville. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Cooper's Ferry

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Cooper's Ferry begins with a bounded claim. For the material described here as early occupation along inland migration route, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cooper's Ferry using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chaco Canyon and Moundville using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Manis Mastodon

North America • Evidence dossier 346 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Late Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Manis Mastodon is a site or landscape associated with late pleistocene research in United States. The defining record highlighted in this atlas is bone point evidence associated with mastodon remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	bone point evidence associated with mastodon remains
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are bone point evidence associated with mastodon remains linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Manis Mastodon after original use?
- Which nearby sites, especially Poverty Point, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Manis Mastodon with Poverty Point, Mesa Verde and Clovis (Blackwater Draw). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Manis Mastodon

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Manis Mastodon begins with a bounded claim. For the material described here as bone point evidence associated with mastodon remains, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Manis Mastodon using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mesa Verde and Clovis (Blackwater Draw) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cactus Hill

North America • Evidence dossier 347 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Late Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Cactus Hill is a site or landscape associated with late pleistocene research in United States. The defining record highlighted in this atlas is proposed pre-Clovis levels and debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	proposed pre-Clovis levels and debates
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are proposed pre-Clovis levels and debates linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Cactus Hill after original use?
- Which nearby sites, especially Watson Brake, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cactus Hill with Watson Brake, Bandelier and Folsom Site. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Cactus Hill

United States • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Cactus Hill begins with a bounded claim. For the material described here as proposed pre-Clovis levels and debates, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cactus Hill using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bandelier and Folsom Site using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cahokia

North America • Evidence dossier 348 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Mississippian**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Cahokia is a site or landscape associated with mississippian research in United States. The defining record highlighted in this atlas is large platform mounds and urban community. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large platform mounds and urban community
Chronological frame	Mississippian; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large platform mounds and urban community linked to the cited phase of Mississippian?
- What natural and human processes altered deposits at Cahokia after original use?
- Which nearby sites, especially Serpent Mound, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cahokia with Serpent Mound, Casa Grande Ruins and Meadowcroft Rockshelter. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

The vast Mississippian mound-and-plaza complex is evidence of a major Indigenous population center, with multiple phases of construction and social change.

RESEARCH POINTER: https://whc.unesco.org/en/list/198

Cahokia

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Cahokia begins with a bounded claim. For the material described here as large platform mounds and urban community, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cahokia using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Casa Grande Ruins and Meadowcroft Rockshelter using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Poverty Point

North America • Evidence dossier 349 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Late Archaic**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Poverty Point is a site or landscape associated with late archaic research in United States. The defining record highlighted in this atlas is large earthworks and exchange networks. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large earthworks and exchange networks
Chronological frame	Late Archaic; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large earthworks and exchange networks linked to the cited phase of Late Archaic?
- What natural and human processes altered deposits at Poverty Point after original use?
- Which nearby sites, especially Hopewell Mound Group, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Poverty Point with Hopewell Mound Group, Snaketown and Paisley Caves. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Poverty Point

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Poverty Point begins with a bounded claim. For the material described here as large earthworks and exchange networks, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Poverty Point using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Snaketown and Paisley Caves using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Watson Brake

North America • Evidence dossier 350 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Middle Archaic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Watson Brake is a site or landscape associated with middle archaic research in United States. The defining record highlighted in this atlas is early mound complex. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early mound complex
Chronological frame	Middle Archaic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early mound complex linked to the cited phase of Middle Archaic?
- What natural and human processes altered deposits at Watson Brake after original use?
- Which nearby sites, especially Newark Earthworks, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Watson Brake with Newark Earthworks, Etowah Mounds and Cooper's Ferry. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Watson Brake

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Watson Brake begins with a bounded claim. For the material described here as early mound complex, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Watson Brake using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Etowah Mounds and Cooper's Ferry using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Serpent Mound

North America • Evidence dossier 351 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**prehistoric**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Serpent Mound is a site or landscape associated with prehistoric research in United States. The defining record highlighted in this atlas is effigy earthwork and chronology debates. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	effigy earthwork and chronology debates
Chronological frame	prehistoric; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are effigy earthwork and chronology debates linked to the cited phase of prehistoric?
- What natural and human processes altered deposits at Serpent Mound after original use?
- Which nearby sites, especially Chaco Canyon, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Serpent Mound with Chaco Canyon, Moundville and Manis Mastodon. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Serpent Mound

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Serpent Mound begins with a bounded claim. For the material described here as effigy earthwork and chronology debates, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Serpent Mound using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Moundville and Manis Mastodon using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Hopewell Mound Group

North America • Evidence dossier 352 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Hopewell**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Hopewell Mound Group is a site or landscape associated with hopewell research in United States. The defining record highlighted in this atlas is earthworks and interregional exchange. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	earthworks and interregional exchange
Chronological frame	Hopewell; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are earthworks and interregional exchange linked to the cited phase of Hopewell?
- What natural and human processes altered deposits at Hopewell Mound Group after original use?
- Which nearby sites, especially Mesa Verde, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Hopewell Mound Group with Mesa Verde, Clovis (Blackwater Draw) and Cactus Hill. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Hopewell Mound Group

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Hopewell Mound Group begins with a bounded claim. For the material described here as earthworks and interregional exchange, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Hopewell Mound Group using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Clovis (Blackwater Draw) and Cactus Hill using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Newark Earthworks

North America • Evidence dossier 353 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Hopewell**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Newark Earthworks is a site or landscape associated with hopewell research in United States. The defining record highlighted in this atlas is geometric enclosures and ceremonial landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	geometric enclosures and ceremonial landscape
Chronological frame	Hopewell; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are geometric enclosures and ceremonial landscape linked to the cited phase of Hopewell?
- What natural and human processes altered deposits at Newark Earthworks after original use?
- Which nearby sites, especially Bandelier, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Newark Earthworks with Bandelier, Folsom Site and Cahokia. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Newark Earthworks

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Newark Earthworks begins with a bounded claim. For the material described here as geometric enclosures and ceremonial landscape, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Newark Earthworks using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Folsom Site and Cahokia using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chaco Canyon

North America • Evidence dossier 354 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Ancestral Puebloan**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Chaco Canyon is a site or landscape associated with ancestral puebloan research in United States. The defining record highlighted in this atlas is great houses roads and timber procurement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	great houses roads and timber procurement
Chronological frame	Ancestral Puebloan; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are great houses roads and timber procurement linked to the cited phase of Ancestral Puebloan?
- What natural and human processes altered deposits at Chaco Canyon after original use?
- Which nearby sites, especially Casa Grande Ruins, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chaco Canyon with Casa Grande Ruins, Meadowcroft Rockshelter and Poverty Point. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Large buildings, engineered roads and imported timbers have fueled competing accounts of pilgrimage, residence, hierarchy and regional integration.

Chaco Canyon

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Chaco Canyon begins with a bounded claim. For the material described here as great houses roads and timber procurement, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chaco Canyon using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Meadowcroft Rockshelter and Poverty Point using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mesa Verde

North America • Evidence dossier 355 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Ancestral Puebloan**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Mesa Verde is a site or landscape associated with ancestral puebloan research in United States. The defining record highlighted in this atlas is cliff dwellings and community histories. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	cliff dwellings and community histories
Chronological frame	Ancestral Puebloan; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are cliff dwellings and community histories linked to the cited phase of Ancestral Puebloan?
- What natural and human processes altered deposits at Mesa Verde after original use?
- Which nearby sites, especially Snaketown, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mesa Verde with Snaketown, Paisley Caves and Watson Brake. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mesa Verde

United States • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Mesa Verde begins with a bounded claim. For the material described here as cliff dwellings and community histories, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mesa Verde using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Paisley Caves and Watson Brake using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bandelier

North America • Evidence dossier 356 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Ancestral Puebloan**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Bandelier is a site or landscape associated with ancestral puebloan research in United States. The defining record highlighted in this atlas is canyon settlements and masonry remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	canyon settlements and masonry remains
Chronological frame	Ancestral Puebloan; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are canyon settlements and masonry remains linked to the cited phase of Ancestral Puebloan?
- What natural and human processes altered deposits at Bandelier after original use?
- Which nearby sites, especially Etowah Mounds, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bandelier with Etowah Mounds, Cooper's Ferry and Serpent Mound. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Bandelier

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Bandelier begins with a bounded claim. For the material described here as canyon settlements and masonry remains, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bandelier using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cooper's Ferry and Serpent Mound using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Casa Grande Ruins

North America • Evidence dossier 357 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Hohokam**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Casa Grande Ruins is a site or landscape associated with hohokam research in United States. The defining record highlighted in this atlas is large adobe building and canal networks. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large adobe building and canal networks
Chronological frame	Hohokam; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large adobe building and canal networks linked to the cited phase of Hohokam?
- What natural and human processes altered deposits at Casa Grande Ruins after original use?
- Which nearby sites, especially Moundville, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Casa Grande Ruins with Moundville, Manis Mastodon and Hopewell Mound Group. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Casa Grande Ruins

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Casa Grande Ruins begins with a bounded claim. For the material described here as large adobe building and canal networks, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Casa Grande Ruins using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Manis Mastodon and Hopewell Mound Group using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Snaketown

North America • Evidence dossier 358 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Hohokam**	**Network**

01 WHAT SURVIVES AND WHY IT MATTERS

Snaketown is a site or landscape associated with hohokam research in United States. The defining record highlighted in this atlas is irrigation and desert village archaeology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	irrigation and desert village archaeology
Chronological frame	Hohokam; exact ranges depend on excavated contexts
Analytical priority	test spatial infrastructure, material provenance and economic links against site-wide chronology
Important caveat	A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are irrigation and desert village archaeology linked to the cited phase of Hohokam?
- What natural and human processes altered deposits at Snaketown after original use?
- Which nearby sites, especially Clovis (Blackwater Draw), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Snaketown with Clovis (Blackwater Draw), Cactus Hill and Newark Earthworks. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Snaketown

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Snaketown begins with a bounded claim. For the material described here as irrigation and desert village archaeology, the immediate objective is to test spatial infrastructure, material provenance and economic links against site-wide chronology. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 GIS catchment and network modelling

Record sample/context identifiers and validate the chain of custody.

02 Petrography and isotopic sourcing

Choose independent controls, repeat measurements and quantify uncertainty.

03 Remote sensing of routes and hydraulic works

Relate results to the local stratigraphic sequence and regional chronology.

04 Stratified assemblage and cargo comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did exchange, ecology and built infrastructure change through securely dated phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. A foreign object is evidence of movement but not necessarily direct contact or a particular political relationship. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Snaketown using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cactus Hill and Newark Earthworks using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Etowah Mounds

North America • Evidence dossier 359 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Mississippian**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Etowah Mounds is a site or landscape associated with mississippian research in United States. The defining record highlighted in this atlas is platform mounds and elite burials. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	platform mounds and elite burials
Chronological frame	Mississippian; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are platform mounds and elite burials linked to the cited phase of Mississippian?
- What natural and human processes altered deposits at Etowah Mounds after original use?
- Which nearby sites, especially Folsom Site, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Etowah Mounds with Folsom Site, Cahokia and Chaco Canyon. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Etowah Mounds

United States • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Etowah Mounds begins with a bounded claim. For the material described here as platform mounds and elite burials, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Etowah Mounds using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cahokia and Chaco Canyon using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Moundville

North America • Evidence dossier 360 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
United States	**Mississippian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Moundville is a site or landscape associated with mississippian research in United States. The defining record highlighted in this atlas is civic ceremonial center and art. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in north america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	civic ceremonial center and art
Chronological frame	Mississippian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are civic ceremonial center and art linked to the cited phase of Mississippian?
- What natural and human processes altered deposits at Moundville after original use?
- Which nearby sites, especially Meadowcroft Rockshelter, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Moundville with Meadowcroft Rockshelter, Poverty Point and Mesa Verde. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Moundville

United States • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Moundville begins with a bounded claim. For the material described here as civic ceremonial center and art, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Moundville using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Poverty Point and Mesa Verde using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

San Lorenzo Tenochtitlán

Mesoamerica • Evidence dossier 361 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	**Olmec**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

San Lorenzo Tenochtitlán is a site or landscape associated with olmec research in Mexico. The defining record highlighted in this atlas is colossal heads and early monumental center. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	colossal heads and early monumental center
Chronological frame	Olmec; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are colossal heads and early monumental center linked to the cited phase of Olmec?
- What natural and human processes altered deposits at San Lorenzo Tenochtitlán after original use?
- Which nearby sites, especially El Manatí, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare San Lorenzo Tenochtitlán with El Manatí, Yagul and El Mirador. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

San Lorenzo Tenochtitlán

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of San Lorenzo Tenochtitlán begins with a bounded claim. For the material described here as colossal heads and early monumental center, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for San Lorenzo Tenochtitlán using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Yagul and El Mirador using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

La Venta

Mesoamerica • Evidence dossier 362 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Olmec	Material

01 WHAT SURVIVES AND WHY IT MATTERS

La Venta is a site or landscape associated with olmec research in Mexico. The defining record highlighted in this atlas is earthen pyramid and ritual offerings. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	earthen pyramid and ritual offerings
Chronological frame	Olmec; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are earthen pyramid and ritual offerings linked to the cited phase of Olmec?
- What natural and human processes altered deposits at La Venta after original use?
- Which nearby sites, especially Teotihuacan, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare La Venta with Teotihuacan, Palenque and Copán. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

La Venta

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of La Venta begins with a bounded claim. For the material described here as earthen pyramid and ritual offerings, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for La Venta using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Palenque and Copán using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tres Zapotes

Mesoamerica • Evidence dossier 363 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	**Formative**	**Text**

01 WHAT SURVIVES AND WHY IT MATTERS

Tres Zapotes is a site or landscape associated with formative research in Mexico. The defining record highlighted in this atlas is colossal head and Long Count inscription. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	colossal head and Long Count inscription
Chronological frame	Formative; exact ranges depend on excavated contexts
Analytical priority	place inscriptions or documents within excavated architecture, administrative context and paleography
Important caveat	Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are colossal head and Long Count inscription linked to the cited phase of Formative?
- What natural and human processes altered deposits at Tres Zapotes after original use?
- Which nearby sites, especially Tenochtitlan Templo Mayor, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tres Zapotes with Tenochtitlan Templo Mayor, Calakmul and Quiriguá. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tres Zapotes

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tres Zapotes begins with a bounded claim. For the material described here as colossal head and Long Count inscription, the immediate objective is to place inscriptions or documents within excavated architecture, administrative context and paleography. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Epigraphy and sign-level documentation

Record sample/context identifiers and validate the chain of custody.

02 Context mapping of tablets or inscriptions

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic and stratigraphic cross-dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Imaging and digital transcription

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which statements are contemporary firsthand records, later copies, or interpretive reconstructions? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Institutional texts represent selective recorded voices, and chronology derived from royal names needs independent checks. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tres Zapotes using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Calakmul and Quiriguá using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

El Manatí

Mesoamerica • Evidence dossier 364 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	**Formative**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

El Manatí is a site or landscape associated with formative research in Mexico. The defining record highlighted in this atlas is waterlogged ritual offerings. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	waterlogged ritual offerings
Chronological frame	Formative; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are waterlogged ritual offerings linked to the cited phase of Formative?
- What natural and human processes altered deposits at El Manatí after original use?
- Which nearby sites, especially Tlatelolco, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare El Manatí with Tlatelolco, Chichén Itzá and Caracol. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

El Manatí

Mexico • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of El Manatí begins with a bounded claim. For the material described here as waterlogged ritual offerings, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for El Manatí using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chichén Itzá and Caracol using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Teotihuacan

Mesoamerica • Evidence dossier 365 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Classic	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Teotihuacan is a site or landscape associated with classic research in Mexico. The defining record highlighted in this atlas is planned metropolis pyramids and apartment compounds. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	planned metropolis pyramids and apartment compounds
Chronological frame	Classic; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are planned metropolis pyramids and apartment compounds linked to the cited phase of Classic?
- What natural and human processes altered deposits at Teotihuacan after original use?
- Which nearby sites, especially Monte Albán, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Teotihuacan with Monte Albán, Uxmal and Lamanai. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Manuel Gamio's work helped establish the settlement as a major subject for modern Mexican archaeology; apartment compounds broaden analysis beyond pyramids.

Teotihuacan

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Teotihuacan begins with a bounded claim. For the material described here as planned metropolis pyramids and apartment compounds, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Teotihuacan using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Uxmal and Lamanai using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tenochtitlan Templo Mayor

Mesoamerica • Evidence dossier 366 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Late Postclassic	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Tenochtitlan Templo Mayor is a site or landscape associated with late postclassic research in Mexico. The defining record highlighted in this atlas is Aztec sacred precinct and offerings. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Aztec sacred precinct and offerings
Chronological frame	Late Postclassic; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Aztec sacred precinct and offerings linked to the cited phase of Late Postclassic?
- What natural and human processes altered deposits at Tenochtitlan Templo Mayor after original use?
- Which nearby sites, especially Mitla, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tenochtitlan Templo Mayor with Mitla, Tikal and San Lorenzo Tenochtitlán. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Eduardo Matos Moctezuma's excavation program brought stratified offerings and successive temple phases into the reconstruction of Mexica ceremonial life.

Tenochtitlan Templo Mayor

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tenochtitlan Templo Mayor begins with a bounded claim. For the material described here as Aztec sacred precinct and offerings, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tenochtitlan Templo Mayor using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tikal and San Lorenzo Tenochtitlán using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tlatelolco

Mesoamerica • Evidence dossier 367 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Late Postclassic	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Tlatelolco is a site or landscape associated with late postclassic research in Mexico. The defining record highlighted in this atlas is market city and colonial destruction layers. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	market city and colonial destruction layers
Chronological frame	Late Postclassic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are market city and colonial destruction layers linked to the cited phase of Late Postclassic?
- What natural and human processes altered deposits at Tlatelolco after original use?
- Which nearby sites, especially Yagul, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tlatelolco with Yagul, El Mirador and La Venta. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tlatelolco

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tlatelolco begins with a bounded claim. For the material described here as market city and colonial destruction layers, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tlatelolco using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with El Mirador and La Venta using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Monte Albán

Mesoamerica • Evidence dossier 368 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Zapotec	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Monte Albán is a site or landscape associated with zapotec research in Mexico. The defining record highlighted in this atlas is hilltop capital monuments and tombs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	hilltop capital monuments and tombs
Chronological frame	Zapotec; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are hilltop capital monuments and tombs linked to the cited phase of Zapotec?
- What natural and human processes altered deposits at Monte Albán after original use?
- Which nearby sites, especially Palenque, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Monte Albán with Palenque, Copán and Tres Zapotes. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Alfonso Caso's work helped establish the Zapotec capital's long history and its connections with surrounding valleys.

Monte Albán

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Monte Albán begins with a bounded claim. For the material described here as hilltop capital monuments and tombs, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Monte Albán using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Copán and Tres Zapotes using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mitla

Mesoamerica • Evidence dossier 369 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	**Postclassic**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Mitla is a site or landscape associated with postclassic research in Mexico. The defining record highlighted in this atlas is geometric mosaic architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	geometric mosaic architecture
Chronological frame	Postclassic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are geometric mosaic architecture linked to the cited phase of Postclassic?
- What natural and human processes altered deposits at Mitla after original use?
- Which nearby sites, especially Calakmul, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mitla with Calakmul, Quiriguá and El Manatí. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mitla

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Mitla begins with a bounded claim. For the material described here as geometric mosaic architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mitla using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Quiriguá and El Manatí using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Yagul

Mesoamerica • Evidence dossier 370 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	**Postclassic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Yagul is a site or landscape associated with postclassic research in Mexico. The defining record highlighted in this atlas is hilltop settlement and tombs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	hilltop settlement and tombs
Chronological frame	Postclassic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are hilltop settlement and tombs linked to the cited phase of Postclassic?
- What natural and human processes altered deposits at Yagul after original use?
- Which nearby sites, especially Chichén Itzá, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Yagul with Chichén Itzá, Caracol and Teotihuacan. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Yagul

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Yagul begins with a bounded claim. For the material described here as hilltop settlement and tombs, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Yagul using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Caracol and Teotihuacan using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Palenque

Mesoamerica • Evidence dossier 371 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Classic Maya	Burial

01 WHAT SURVIVES AND WHY IT MATTERS

Palenque is a site or landscape associated with classic maya research in Mexico. The defining record highlighted in this atlas is palaces royal tomb and inscriptions. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	palaces royal tomb and inscriptions
Chronological frame	Classic Maya; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are palaces royal tomb and inscriptions linked to the cited phase of Classic Maya?
- What natural and human processes altered deposits at Palenque after original use?
- Which nearby sites, especially Uxmal, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Palenque with Uxmal, Lamanai and Tenochtitlan Templo Mayor. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Alberto Ruz Lhuillier's excavation of the Temple of the Inscriptions tomb helped link Maya monumental texts and royal burial archaeology.

Palenque

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Palenque begins with a bounded claim. For the material described here as palaces royal tomb and inscriptions, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Palenque using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lamanai and Tenochtitlan Templo Mayor using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Calakmul

Mesoamerica • Evidence dossier 372 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Classic Maya	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Calakmul is a site or landscape associated with classic maya research in Mexico. The defining record highlighted in this atlas is dynastic city and forest landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	dynastic city and forest landscape
Chronological frame	Classic Maya; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are dynastic city and forest landscape linked to the cited phase of Classic Maya?
- What natural and human processes altered deposits at Calakmul after original use?
- Which nearby sites, especially Tikal, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Calakmul with Tikal, San Lorenzo Tenochtitlán and Tlatelolco. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Calakmul

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Calakmul begins with a bounded claim. For the material described here as dynastic city and forest landscape, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Calakmul using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with San Lorenzo Tenochtitlán and Tlatelolco using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chichén Itzá

Mesoamerica • Evidence dossier 373 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Terminal Classic-Postcla...	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Chichén Itzá is a site or landscape associated with terminal classic-postclassic research in Mexico. The defining record highlighted in this atlas is temple complex and regional power. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple complex and regional power
Chronological frame	Terminal Classic-Postclassic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple complex and regional power linked to the cited phase of Terminal Classic-Postclassic?
- What natural and human processes altered deposits at Chichén Itzá after original use?
- Which nearby sites, especially El Mirador, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chichén Itzá with El Mirador, La Venta and Monte Albán. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Chichén Itzá

Mexico • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Chichén Itzá begins with a bounded claim. For the material described here as temple complex and regional power, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chichén Itzá using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with La Venta and Monte Albán using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Uxmal

Mesoamerica • Evidence dossier 374 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Mexico	Classic Maya	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Uxmal is a site or landscape associated with classic maya research in Mexico. The defining record highlighted in this atlas is Puuc architecture and ceremonial urbanism. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Puuc architecture and ceremonial urbanism
Chronological frame	Classic Maya; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Puuc architecture and ceremonial urbanism linked to the cited phase of Classic Maya?
- What natural and human processes altered deposits at Uxmal after original use?
- Which nearby sites, especially Copán, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Uxmal with Copán, Tres Zapotes and Mitla. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Uxmal

Mexico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Uxmal begins with a bounded claim. For the material described here as Puuc architecture and ceremonial urbanism, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Uxmal using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tres Zapotes and Mitla using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tikal

Mesoamerica • Evidence dossier 375 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Guatemala	Classic Maya	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Tikal is a site or landscape associated with classic maya research in Guatemala. The defining record highlighted in this atlas is temple pyramids dynastic stelae and urban ecology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple pyramids dynastic stelae and urban ecology
Chronological frame	Classic Maya; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple pyramids dynastic stelae and urban ecology linked to the cited phase of Classic Maya?
- What natural and human processes altered deposits at Tikal after original use?
- Which nearby sites, especially Quiriguá, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tikal with Quiriguá, El Manatí and Yagul. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Long-term mapping and excavations revealed an extensive urban landscape, not only the monumental central temple precinct.

Tikal

Guatemala • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Tikal begins with a bounded claim. For the material described here as temple pyramids dynastic stelae and urban ecology, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tikal using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with El Manatí and Yagul using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

El Mirador

Mesoamerica • Evidence dossier 376 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Guatemala	Preclassic Maya	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

El Mirador is a site or landscape associated with preclassic maya research in Guatemala. The defining record highlighted in this atlas is large early monumental complexes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	large early monumental complexes
Chronological frame	Preclassic Maya; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are large early monumental complexes linked to the cited phase of Preclassic Maya?
- What natural and human processes altered deposits at El Mirador after original use?
- Which nearby sites, especially Caracol, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare El Mirador with Caracol, Teotihuacan and Palenque. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

El Mirador

Guatemala • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of El Mirador begins with a bounded claim. For the material described here as large early monumental complexes, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for El Mirador using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Teotihuacan and Palenque using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Copán

Mesoamerica • Evidence dossier 377 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Honduras	Classic Maya	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Copán is a site or landscape associated with classic maya research in Honduras. The defining record highlighted in this atlas is hieroglyphic stairway and sculpture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	hieroglyphic stairway and sculpture
Chronological frame	Classic Maya; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are hieroglyphic stairway and sculpture linked to the cited phase of Classic Maya?
- What natural and human processes altered deposits at Copán after original use?
- Which nearby sites, especially Lamanai, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Copán with Lamanai, Tenochtitlan Templo Mayor and Calakmul. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Copán

Honduras • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Copán begins with a bounded claim. For the material described here as hieroglyphic stairway and sculpture, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Copán using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tenochtitlan Templo Mayor and Calakmul using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Quiriguá

Mesoamerica • Evidence dossier 378 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Guatemala	Classic Maya	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Quiriguá is a site or landscape associated with classic maya research in Guatemala. The defining record highlighted in this atlas is monumental carved stelae. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental carved stelae
Chronological frame	Classic Maya; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental carved stelae linked to the cited phase of Classic Maya?
- What natural and human processes altered deposits at Quiriguá after original use?
- Which nearby sites, especially San Lorenzo Tenochtitlán, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Quiriguá with San Lorenzo Tenochtitlán, Tlatelolco and Chichén Itzá. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Quiriguá

Guatemala • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Quiriguá begins with a bounded claim. For the material described here as monumental carved stelae, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Quiriguá using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tlatelolco and Chichén Itzá using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Caracol

Mesoamerica • Evidence dossier 379 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Belize	**Classic Maya**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Caracol is a site or landscape associated with classic maya research in Belize. The defining record highlighted in this atlas is terrace agriculture and causeway networks. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	terrace agriculture and causeway networks
Chronological frame	Classic Maya; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are terrace agriculture and causeway networks linked to the cited phase of Classic Maya?
- What natural and human processes altered deposits at Caracol after original use?
- Which nearby sites, especially La Venta, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Caracol with La Venta, Monte Albán and Uxmal. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Caracol

Belize • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Caracol begins with a bounded claim. For the material described here as terrace agriculture and causeway networks, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Caracol using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Monte Albán and Uxmal using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lamanai

Mesoamerica • Evidence dossier 380 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Belize	**Preclassic-colonial**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Lamanai is a site or landscape associated with preclassic-colonial research in Belize. The defining record highlighted in this atlas is long habitation sequence and temples. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in mesoamerica is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	long habitation sequence and temples
Chronological frame	Preclassic-colonial; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are long habitation sequence and temples linked to the cited phase of Preclassic-colonial?
- What natural and human processes altered deposits at Lamanai after original use?
- Which nearby sites, especially Tres Zapotes, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lamanai with Tres Zapotes, Mitla and Tikal. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lamanai

Belize • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Lamanai begins with a bounded claim. For the material described here as long habitation sequence and temples, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lamanai using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mitla and Tikal using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Caral-Supe

South America • Evidence dossier 381 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	Late Archaic	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Caral-Supe is a site or landscape associated with late archaic research in Peru. The defining record highlighted in this atlas is monumental pyramids and urbanism. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental pyramids and urbanism
Chronological frame	Late Archaic; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental pyramids and urbanism linked to the cited phase of Late Archaic?
- What natural and human processes altered deposits at Caral-Supe after original use?
- Which nearby sites, especially Cerro Sechín, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Caral-Supe with Cerro Sechín, Chan Chan and Tiwanaku. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Research led by Ruth Shady highlighted early monumental architecture and social organization on Peru's central coast.

Caral-Supe

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Caral-Supe begins with a bounded claim. For the material described here as monumental pyramids and urbanism, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Caral-Supe using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chan Chan and Tiwanaku using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Huaca Prieta

South America • Evidence dossier 382 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	Preceramic	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Huaca Prieta is a site or landscape associated with preceramic research in Peru. The defining record highlighted in this atlas is early textiles and coastal food systems. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early textiles and coastal food systems
Chronological frame	Preceramic; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early textiles and coastal food systems linked to the cited phase of Preceramic?
- What natural and human processes altered deposits at Huaca Prieta after original use?
- Which nearby sites, especially Chavín de Huántar, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Huaca Prieta with Chavín de Huántar, Monte Verde and Puma Punku. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Huaca Prieta

Peru • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Huaca Prieta begins with a bounded claim. For the material described here as early textiles and coastal food systems, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Huaca Prieta using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Monte Verde and Puma Punku using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sechín Bajo

South America • Evidence dossier 383 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Preceramic-Formative**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Sechín Bajo is a site or landscape associated with preceramic-formative research in Peru. The defining record highlighted in this atlas is early ceremonial architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early ceremonial architecture
Chronological frame	Preceramic-Formative; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early ceremonial architecture linked to the cited phase of Preceramic-Formative?
- What natural and human processes altered deposits at Sechín Bajo after original use?
- Which nearby sites, especially Kotosh, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sechín Bajo with Kotosh, Pachacamac and Machu Picchu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sechín Bajo

Peru • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sechín Bajo begins with a bounded claim. For the material described here as early ceremonial architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sechín Bajo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Pachacamac and Machu Picchu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cerro Sechín

South America • Evidence dossier 384 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Formative**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Cerro Sechín is a site or landscape associated with formative research in Peru. The defining record highlighted in this atlas is carved stone imagery and ritual architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	carved stone imagery and ritual architecture
Chronological frame	Formative; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are carved stone imagery and ritual architecture linked to the cited phase of Formative?
- What natural and human processes altered deposits at Cerro Sechín after original use?
- Which nearby sites, especially Chankillo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cerro Sechín with Chankillo, Nazca Lines and Sacsayhuamán. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Cerro Sechín

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Cerro Sechín begins with a bounded claim. For the material described here as carved stone imagery and ritual architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cerro Sechín using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nazca Lines and Sacsayhuamán using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chavín de Huántar

South America • Evidence dossier 385 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	Early Horizon	Art

01 WHAT SURVIVES AND WHY IT MATTERS

Chavín de Huántar is a site or landscape associated with early horizon research in Peru. The defining record highlighted in this atlas is temple galleries and iconography. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple galleries and iconography
Chronological frame	Early Horizon; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple galleries and iconography linked to the cited phase of Early Horizon?
- What natural and human processes altered deposits at Chavín de Huántar after original use?
- Which nearby sites, especially Moche Huaca de la Luna, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chavín de Huántar with Moche Huaca de la Luna, Cahuachi and Ingapirca. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Chavín de Huántar

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Chavín de Huántar begins with a bounded claim. For the material described here as temple galleries and iconography, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chavín de Huántar using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cahuachi and Ingapirca using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kotosh

South America • Evidence dossier 386 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Preceramic-Formative**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Kotosh is a site or landscape associated with preceramic-formative research in Peru. The defining record highlighted in this atlas is temple of crossed hands. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	temple of crossed hands
Chronological frame	Preceramic-Formative; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are temple of crossed hands linked to the cited phase of Preceramic-Formative?
- What natural and human processes altered deposits at Kotosh after original use?
- Which nearby sites, especially Sipán, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kotosh with Sipán, Wari and Caral-Supe. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kotosh

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Kotosh begins with a bounded claim. For the material described here as temple of crossed hands, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kotosh using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Wari and Caral-Supe using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chankillo

South America • Evidence dossier 387 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	Early Horizon	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Chankillo is a site or landscape associated with early horizon research in Peru. The defining record highlighted in this atlas is thirteen towers and calendrical landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	thirteen towers and calendrical landscape
Chronological frame	Early Horizon; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are thirteen towers and calendrical landscape linked to the cited phase of Early Horizon?
- What natural and human processes altered deposits at Chankillo after original use?
- Which nearby sites, especially Chan Chan, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chankillo with Chan Chan, Tiwanaku and Huaca Prieta. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Chankillo

Peru • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Chankillo begins with a bounded claim. For the material described here as thirteen towers and calendrical landscape, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chankillo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tiwanaku and Huaca Prieta using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Moche Huaca de la Luna

South America • Evidence dossier 388 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	Early Intermediate	Art

01 WHAT SURVIVES AND WHY IT MATTERS

Moche Huaca de la Luna is a site or landscape associated with early intermediate research in Peru. The defining record highlighted in this atlas is painted temple and sacrifice evidence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	painted temple and sacrifice evidence
Chronological frame	Early Intermediate; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are painted temple and sacrifice evidence linked to the cited phase of Early Intermediate?
- What natural and human processes altered deposits at Moche Huaca de la Luna after original use?
- Which nearby sites, especially Monte Verde, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Moche Huaca de la Luna with Monte Verde, Puma Punku and Sechín Bajo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Moche Huaca de la Luna

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Moche Huaca de la Luna begins with a bounded claim. For the material described here as painted temple and sacrifice evidence, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Moche Huaca de la Luna using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Puma Punku and Sechín Bajo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sipán

South America • Evidence dossier 389 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Moche**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Sipán is a site or landscape associated with moche research in Peru. The defining record highlighted in this atlas is elite tombs and fine metalwork. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	elite tombs and fine metalwork
Chronological frame	Moche; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are elite tombs and fine metalwork linked to the cited phase of Moche?
- What natural and human processes altered deposits at Sipán after original use?
- Which nearby sites, especially Pachacamac, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sipán with Pachacamac, Machu Picchu and Cerro Sechín. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Walter Alva's excavation of Moche elite tombs showed how intact contextual recovery can transform interpretations of art and authority.

Sipán

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Sipán begins with a bounded claim. For the material described here as elite tombs and fine metalwork, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sipán using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Machu Picchu and Cerro Sechín using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Chan Chan

South America • Evidence dossier 390 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Chimú**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Chan Chan is a site or landscape associated with chimú research in Peru. The defining record highlighted in this atlas is adobe city and palace compounds. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	adobe city and palace compounds
Chronological frame	Chimú; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are adobe city and palace compounds linked to the cited phase of Chimú?
- What natural and human processes altered deposits at Chan Chan after original use?
- Which nearby sites, especially Nazca Lines, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Chan Chan with Nazca Lines, Sacsayhuamán and Chavín de Huántar. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Chan Chan

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Chan Chan begins with a bounded claim. For the material described here as adobe city and palace compounds, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Chan Chan using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sacsayhuamán and Chavín de Huántar using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Monte Verde

South America • Evidence dossier 391 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Chile	Late Pleistocene	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Monte Verde is a site or landscape associated with late pleistocene research in Chile. The defining record highlighted in this atlas is waterlogged evidence of early human settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	waterlogged evidence of early human settlement
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are waterlogged evidence of early human settlement linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Monte Verde after original use?
- Which nearby sites, especially Cahuachi, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Monte Verde with Cahuachi, Ingapirca and Kotosh. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Monte Verde

Chile • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Monte Verde begins with a bounded claim. For the material described here as waterlogged evidence of early human settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Monte Verde using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ingapirca and Kotosh using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Pachacamac

South America • Evidence dossier 392 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	Lima-Inca	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Pachacamac is a site or landscape associated with lima-inca research in Peru. The defining record highlighted in this atlas is pilgrimage sanctuary and oracles. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	pilgrimage sanctuary and oracles
Chronological frame	Lima-Inca; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are pilgrimage sanctuary and oracles linked to the cited phase of Lima-Inca?
- What natural and human processes altered deposits at Pachacamac after original use?
- Which nearby sites, especially Wari, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Pachacamac with Wari, Caral-Supe and Chankillo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Pachacamac

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Pachacamac begins with a bounded claim. For the material described here as pilgrimage sanctuary and oracles, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Pachacamac using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Caral-Supe and Chankillo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nazca Lines

South America • Evidence dossier 393 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Nasca**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Nazca Lines is a site or landscape associated with nasca research in Peru. The defining record highlighted in this atlas is desert geoglyphs and ritual landscapes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	desert geoglyphs and ritual landscapes
Chronological frame	Nasca; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are desert geoglyphs and ritual landscapes linked to the cited phase of Nasca?
- What natural and human processes altered deposits at Nazca Lines after original use?
- Which nearby sites, especially Tiwanaku, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nazca Lines with Tiwanaku, Huaca Prieta and Moche Huaca de la Luna. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Nazca Lines

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nazca Lines begins with a bounded claim. For the material described here as desert geoglyphs and ritual landscapes, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nazca Lines using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Huaca Prieta and Moche Huaca de la Luna using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Cahuachi

South America • Evidence dossier 394 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Nasca**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Cahuachi is a site or landscape associated with nasca research in Peru. The defining record highlighted in this atlas is ceremonial center and adobe mounds. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ceremonial center and adobe mounds
Chronological frame	Nasca; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ceremonial center and adobe mounds linked to the cited phase of Nasca?
- What natural and human processes altered deposits at Cahuachi after original use?
- Which nearby sites, especially Puma Punku, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Cahuachi with Puma Punku, Sechín Bajo and Sipán. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Cahuachi

Peru • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Cahuachi begins with a bounded claim. For the material described here as ceremonial center and adobe mounds, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Cahuachi using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sechín Bajo and Sipán using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Wari

South America • Evidence dossier 395 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	Middle Horizon	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Wari is a site or landscape associated with middle horizon research in Peru. The defining record highlighted in this atlas is planned imperial architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	planned imperial architecture
Chronological frame	Middle Horizon; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are planned imperial architecture linked to the cited phase of Middle Horizon?
- What natural and human processes altered deposits at Wari after original use?
- Which nearby sites, especially Machu Picchu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Wari with Machu Picchu, Cerro Sechín and Chan Chan. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Wari

Peru • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Wari begins with a bounded claim. For the material described here as planned imperial architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Wari using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Cerro Sechín and Chan Chan using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tiwanaku

South America • Evidence dossier 396 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Bolivia	**Middle Horizon**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Tiwanaku is a site or landscape associated with middle horizon research in Bolivia. The defining record highlighted in this atlas is monumental civic architecture and raised fields. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental civic architecture and raised fields
Chronological frame	Middle Horizon; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental civic architecture and raised fields linked to the cited phase of Middle Horizon?
- What natural and human processes altered deposits at Tiwanaku after original use?
- Which nearby sites, especially Sacsayhuamán, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tiwanaku with Sacsayhuamán, Chavín de Huántar and Monte Verde. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tiwanaku

Bolivia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Tiwanaku begins with a bounded claim. For the material described here as monumental civic architecture and raised fields, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tiwanaku using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chavín de Huántar and Monte Verde using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Puma Punku

South America • Evidence dossier 397 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Bolivia	**Middle Horizon**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Puma Punku is a site or landscape associated with middle horizon research in Bolivia. The defining record highlighted in this atlas is finely worked monumental stonework. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	finely worked monumental stonework
Chronological frame	Middle Horizon; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are finely worked monumental stonework linked to the cited phase of Middle Horizon?
- What natural and human processes altered deposits at Puma Punku after original use?
- Which nearby sites, especially Ingapirca, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Puma Punku with Ingapirca, Kotosh and Pachacamac. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Puma Punku

Bolivia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Puma Punku begins with a bounded claim. For the material described here as finely worked monumental stonework, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Puma Punku using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kotosh and Pachacamac using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Machu Picchu

South America • Evidence dossier 398 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Inca**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Machu Picchu is a site or landscape associated with inca research in Peru. The defining record highlighted in this atlas is mountain estate and terrace engineering. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	mountain estate and terrace engineering
Chronological frame	Inca; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are mountain estate and terrace engineering linked to the cited phase of Inca?
- What natural and human processes altered deposits at Machu Picchu after original use?
- Which nearby sites, especially Caral-Supe, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Machu Picchu with Caral-Supe, Chankillo and Nazca Lines. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Hiram Bingham made the site internationally prominent in 1911, but the place was known to local people; archaeology must not present colonial publicity as literal discovery.

Machu Picchu

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Machu Picchu begins with a bounded claim. For the material described here as mountain estate and terrace engineering, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Machu Picchu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Chankillo and Nazca Lines using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Sacsayhuamán

South America • Evidence dossier 399 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Inca**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Sacsayhuamán is a site or landscape associated with inca research in Peru. The defining record highlighted in this atlas is polygonal masonry and imperial landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	polygonal masonry and imperial landscape
Chronological frame	Inca; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are polygonal masonry and imperial landscape linked to the cited phase of Inca?
- What natural and human processes altered deposits at Sacsayhuamán after original use?
- Which nearby sites, especially Huaca Prieta, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sacsayhuamán with Huaca Prieta, Moche Huaca de la Luna and Cahuachi. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sacsayhuamán

Peru • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Sacsayhuamán begins with a bounded claim. For the material described here as polygonal masonry and imperial landscape, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sacsayhuamán using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Moche Huaca de la Luna and Cahuachi using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ingapirca

South America • Evidence dossier 400 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ecuador	**Inca**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Ingapirca is a site or landscape associated with inca research in Ecuador. The defining record highlighted in this atlas is provincial imperial architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in south america is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	provincial imperial architecture
Chronological frame	Inca; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are provincial imperial architecture linked to the cited phase of Inca?
- What natural and human processes altered deposits at Ingapirca after original use?
- Which nearby sites, especially Sechín Bajo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ingapirca with Sechín Bajo, Sipán and Wari. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ingapirca

Ecuador • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ingapirca begins with a bounded claim. For the material described here as provincial imperial architecture, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ingapirca using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sipán and Wari using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kuelap

Amazonia, Andes, Caribbean • Evidence dossier 401 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Chachapoya**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Kuelap is a site or landscape associated with chachapoya research in Peru. The defining record highlighted in this atlas is stone-walled mountain settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	stone-walled mountain settlement
Chronological frame	Chachapoya; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are stone-walled mountain settlement linked to the cited phase of Chachapoya?
- What natural and human processes altered deposits at Kuelap after original use?
- Which nearby sites, especially Huari (Wari) satellite Pikillacta, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kuelap with Huari (Wari) satellite Pikillacta, Ciudad Perdida (Teyuna) and Acre Geoglyphs. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kuelap

Peru • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kuelap begins with a bounded claim. For the material described here as stone-walled mountain settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kuelap using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ciudad Perdida (Teyuna) and Acre Geoglyphs using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Gran Pajatén

Amazonia, Andes, Caribbean • Evidence dossier 402 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Chachapoya**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Gran Pajatén is a site or landscape associated with chachapoya research in Peru. The defining record highlighted in this atlas is stone buildings and forest environment. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	stone buildings and forest environment
Chronological frame	Chachapoya; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are stone buildings and forest environment linked to the cited phase of Chachapoya?
- What natural and human processes altered deposits at Gran Pajatén after original use?
- Which nearby sites, especially Choquequirao, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Gran Pajatén with Choquequirao, El Infiernito and Upano Valley sites. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Gran Pajatén

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Gran Pajatén begins with a bounded claim. For the material described here as stone buildings and forest environment, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Gran Pajatén using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with El Infiernito and Upano Valley sites using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Túcume

Amazonia, Andes, Caribbean • Evidence dossier 403 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	Lambayeque-Chimú	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Túcume is a site or landscape associated with lambayeque-chimú research in Peru. The defining record highlighted in this atlas is monumental adobe pyramids. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental adobe pyramids
Chronological frame	Lambayeque-Chimú; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental adobe pyramids linked to the cited phase of Lambayeque-Chimú?
- What natural and human processes altered deposits at Túcume after original use?
- Which nearby sites, especially Ollantaytambo, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Túcume with Ollantaytambo, Valdivia sites and Caguana Ceremonial Park. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Túcume

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Túcume begins with a bounded claim. For the material described here as monumental adobe pyramids, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Túcume using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Valdivia sites and Caguana Ceremonial Park using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Huari (Wari) satellite Pikillacta

Amazonia, Andes, Caribbean • Evidence dossier 404 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Middle Horizon**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Huari (Wari) satellite Pikillacta is a site or landscape associated with middle horizon research in Peru. The defining record highlighted in this atlas is planned administrative settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	planned administrative settlement
Chronological frame	Middle Horizon; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are planned administrative settlement linked to the cited phase of Middle Horizon?
- What natural and human processes altered deposits at Huari (Wari) satellite Pikillacta after original use?
- Which nearby sites, especially Qorikancha, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Huari (Wari) satellite Pikillacta with Qorikancha, La Tolita and Tibees. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Huari (Wari) satellite Pikillacta

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Huari (Wari) satellite Pikillacta begins with a bounded claim. For the material described here as planned administrative settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Huari (Wari) satellite Pikillacta using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with La Tolita and Tibees using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Choquequirao

Amazonia, Andes, Caribbean • Evidence dossier 405 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Inca**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Choquequirao is a site or landscape associated with inca research in Peru. The defining record highlighted in this atlas is terraced mountain settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	terraced mountain settlement
Chronological frame	Inca; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are terraced mountain settlement linked to the cited phase of Inca?
- What natural and human processes altered deposits at Choquequirao after original use?
- Which nearby sites, especially San Agustín, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Choquequirao with San Agustín, Marajó Island Mounds and La Isabela. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Choquequirao

Peru • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Choquequirao begins with a bounded claim. For the material described here as terraced mountain settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Choquequirao using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Marajó Island Mounds and La Isabela using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ollantaytambo

Amazonia, Andes, Caribbean • Evidence dossier 406 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Inca**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Ollantaytambo is a site or landscape associated with inca research in Peru. The defining record highlighted in this atlas is fortified town and terraces. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortified town and terraces
Chronological frame	Inca; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortified town and terraces linked to the cited phase of Inca?
- What natural and human processes altered deposits at Ollantaytambo after original use?
- Which nearby sites, especially Tierradentro, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ollantaytambo with Tierradentro, Kuhikugu and Kuelap. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ollantaytambo

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ollantaytambo begins with a bounded claim. For the material described here as fortified town and terraces, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ollantaytambo using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kuhikugu and Kuelap using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Qorikancha

Amazonia, Andes, Caribbean • Evidence dossier 407 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Peru	**Inca-colonial**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Qorikancha is a site or landscape associated with inca-colonial research in Peru. The defining record highlighted in this atlas is sacred urban architecture and reuse. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	sacred urban architecture and reuse
Chronological frame	Inca-colonial; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are sacred urban architecture and reuse linked to the cited phase of Inca-colonial?
- What natural and human processes altered deposits at Qorikancha after original use?
- Which nearby sites, especially Ciudad Perdida (Teyuna), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Qorikancha with Ciudad Perdida (Teyuna), Acre Geoglyphs and Gran Pajatén. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Qorikancha

Peru • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Qorikancha begins with a bounded claim. For the material described here as sacred urban architecture and reuse, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Qorikancha using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Acre Geoglyphs and Gran Pajatén using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

San Agustín

Amazonia, Andes, Caribbean • Evidence dossier 408 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Colombia	**Pre-Hispanic**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

San Agustín is a site or landscape associated with pre-hispanic research in Colombia. The defining record highlighted in this atlas is monumental funerary sculpture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental funerary sculpture
Chronological frame	Pre-Hispanic; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental funerary sculpture linked to the cited phase of Pre-Hispanic?
- What natural and human processes altered deposits at San Agustín after original use?
- Which nearby sites, especially El Infiernito, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare San Agustín with El Infiernito, Upano Valley sites and Túcume. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

San Agustín

Colombia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of San Agustín begins with a bounded claim. For the material described here as monumental funerary sculpture, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for San Agustín using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Upano Valley sites and Túcume using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tierradentro

Amazonia, Andes, Caribbean • Evidence dossier 409 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Colombia	**Pre-Hispanic**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Tierradentro is a site or landscape associated with pre-hispanic research in Colombia. The defining record highlighted in this atlas is painted underground tombs. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	painted underground tombs
Chronological frame	Pre-Hispanic; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are painted underground tombs linked to the cited phase of Pre-Hispanic?
- What natural and human processes altered deposits at Tierradentro after original use?
- Which nearby sites, especially Valdivia sites, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tierradentro with Valdivia sites, Caguana Ceremonial Park and Huari (Wari) satellite Pikillacta. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tierradentro

Colombia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tierradentro begins with a bounded claim. For the material described here as painted underground tombs, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tierradentro using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Caguana Ceremonial Park and Huari (Wari) satellite Pikillacta using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ciudad Perdida (Teyuna)

Amazonia, Andes, Caribbean • Evidence dossier 410 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Colombia	**Tairona**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Ciudad Perdida (Teyuna) is a site or landscape associated with tairona research in Colombia. The defining record highlighted in this atlas is terraced mountain city. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	terraced mountain city
Chronological frame	Tairona; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are terraced mountain city linked to the cited phase of Tairona?
- What natural and human processes altered deposits at Ciudad Perdida (Teyuna) after original use?
- Which nearby sites, especially La Tolita, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ciudad Perdida (Teyuna) with La Tolita, Tibees and Choquequirao. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ciudad Perdida (Teyuna)

Colombia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ciudad Perdida (Teyuna) begins with a bounded claim. For the material described here as terraced mountain city, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ciudad Perdida (Teyuna) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tibees and Choquequirao using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

El Infiernito

Amazonia, Andes, Caribbean • Evidence dossier 411 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Colombia	**Muisca**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

El Infiernito is a site or landscape associated with muisca research in Colombia. The defining record highlighted in this atlas is standing stones and ceremonial landscape. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	standing stones and ceremonial landscape
Chronological frame	Muisca; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are standing stones and ceremonial landscape linked to the cited phase of Muisca?
- What natural and human processes altered deposits at El Infiernito after original use?
- Which nearby sites, especially Marajó Island Mounds, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare El Infiernito with Marajó Island Mounds, La Isabela and Ollantaytambo. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

El Infiernito

Colombia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of El Infiernito begins with a bounded claim. For the material described here as standing stones and ceremonial landscape, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for El Infiernito using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with La Isabela and Ollantaytambo using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Valdivia sites

Amazonia, Andes, Caribbean • Evidence dossier 412 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ecuador	**Formative**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Valdivia sites is a site or landscape associated with formative research in Ecuador. The defining record highlighted in this atlas is early ceramics and coastal settlement. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early ceramics and coastal settlement
Chronological frame	Formative; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early ceramics and coastal settlement linked to the cited phase of Formative?
- What natural and human processes altered deposits at Valdivia sites after original use?
- Which nearby sites, especially Kuhikugu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Valdivia sites with Kuhikugu, Kuelap and Qorikancha. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Valdivia sites

Ecuador • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Valdivia sites begins with a bounded claim. For the material described here as early ceramics and coastal settlement, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Valdivia sites using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kuelap and Qorikancha using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

La Tolita

Amazonia, Andes, Caribbean • Evidence dossier 413 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ecuador	Formative-Regional Devel...	Material

01 WHAT SURVIVES AND WHY IT MATTERS

La Tolita is a site or landscape associated with formative-regional development research in Ecuador. The defining record highlighted in this atlas is ceramics and metalwork. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ceramics and metalwork
Chronological frame	Formative-Regional Development; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ceramics and metalwork linked to the cited phase of Formative-Regional Development?
- What natural and human processes altered deposits at La Tolita after original use?
- Which nearby sites, especially Acre Geoglyphs, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare La Tolita with Acre Geoglyphs, Gran Pajatén and San Agustín. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

La Tolita

Ecuador • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of La Tolita begins with a bounded claim. For the material described here as ceramics and metalwork, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for La Tolita using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Gran Pajatén and San Agustín using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Marajó Island Mounds

Amazonia, Andes, Caribbean • Evidence dossier 414 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Brazil	**pre-Columbian**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Marajó Island Mounds is a site or landscape associated with pre-columbian research in Brazil. The defining record highlighted in this atlas is Amazonian mound settlements and ceramics. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Amazonian mound settlements and ceramics
Chronological frame	pre-Columbian; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Amazonian mound settlements and ceramics linked to the cited phase of pre-Columbian?
- What natural and human processes altered deposits at Marajó Island Mounds after original use?
- Which nearby sites, especially Upano Valley sites, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Marajó Island Mounds with Upano Valley sites, Túcume and Tierradentro. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Marajó Island Mounds

Brazil • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Marajó Island Mounds begins with a bounded claim. For the material described here as Amazonian mound settlements and ceramics, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Marajó Island Mounds using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Túcume and Tierradentro using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kuhikugu

Amazonia, Andes, Caribbean • Evidence dossier 415 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Brazil	**pre-Columbian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Kuhikugu is a site or landscape associated with pre-columbian research in Brazil. The defining record highlighted in this atlas is planned Amazonian villages and roads. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	planned Amazonian villages and roads
Chronological frame	pre-Columbian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are planned Amazonian villages and roads linked to the cited phase of pre-Columbian?
- What natural and human processes altered deposits at Kuhikugu after original use?
- Which nearby sites, especially Caguana Ceremonial Park, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kuhikugu with Caguana Ceremonial Park, Huari (Wari) satellite Pikillacta and Ciudad Perdida (Teyuna). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kuhikugu

Brazil • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Kuhikugu begins with a bounded claim. For the material described here as planned Amazonian villages and roads, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kuhikugu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Huari (Wari) satellite Pikillacta and Ciudad Perdida (Teyuna) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Acre Geoglyphs

Amazonia, Andes, Caribbean • Evidence dossier 416 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Brazil	**pre-Columbian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Acre Geoglyphs is a site or landscape associated with pre-columbian research in Brazil. The defining record highlighted in this atlas is ditched geometric earthworks. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ditched geometric earthworks
Chronological frame	pre-Columbian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ditched geometric earthworks linked to the cited phase of pre-Columbian?
- What natural and human processes altered deposits at Acre Geoglyphs after original use?
- Which nearby sites, especially Tibees, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Acre Geoglyphs with Tibees, Choquequirao and El Infiernito. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Acre Geoglyphs

Brazil • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Acre Geoglyphs begins with a bounded claim. For the material described here as ditched geometric earthworks, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Acre Geoglyphs using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Choquequirao and El Infiernito using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Upano Valley sites

Amazonia, Andes, Caribbean • Evidence dossier 417 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Ecuador	**pre-Columbian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Upano Valley sites is a site or landscape associated with pre-columbian research in Ecuador. The defining record highlighted in this atlas is earthen platform networks and roadways. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	earthen platform networks and roadways
Chronological frame	pre-Columbian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are earthen platform networks and roadways linked to the cited phase of pre-Columbian?
- What natural and human processes altered deposits at Upano Valley sites after original use?
- Which nearby sites, especially La Isabela, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Upano Valley sites with La Isabela, Ollantaytambo and Valdivia sites. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Upano Valley sites

Ecuador • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Upano Valley sites begins with a bounded claim. For the material described here as earthen platform networks and roadways, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Upano Valley sites using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ollantaytambo and Valdivia sites using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Caguana Ceremonial Park

Amazonia, Andes, Caribbean • Evidence dossier 418 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Puerto Rico	**Taíno**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Caguana Ceremonial Park is a site or landscape associated with taíno research in Puerto Rico. The defining record highlighted in this atlas is ball courts and carved stones. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ball courts and carved stones
Chronological frame	Taíno; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ball courts and carved stones linked to the cited phase of Taíno?
- What natural and human processes altered deposits at Caguana Ceremonial Park after original use?
- Which nearby sites, especially Kuelap, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Caguana Ceremonial Park with Kuelap, Qorikancha and La Tolita. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Caguana Ceremonial Park

Puerto Rico • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Caguana Ceremonial Park begins with a bounded claim. For the material described here as ball courts and carved stones, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Caguana Ceremonial Park using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Qorikancha and La Tolita using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Tibees

Amazonia, Andes, Caribbean • Evidence dossier 419 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Puerto Rico	**pre-Columbian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Tibees is a site or landscape associated with pre-columbian research in Puerto Rico. The defining record highlighted in this atlas is plazas and ceremonial courts. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	plazas and ceremonial courts
Chronological frame	pre-Columbian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are plazas and ceremonial courts linked to the cited phase of pre-Columbian?
- What natural and human processes altered deposits at Tibees after original use?
- Which nearby sites, especially Gran Pajatén, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Tibees with Gran Pajatén, San Agustín and Marajó Island Mounds. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Tibees

Puerto Rico • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Tibees begins with a bounded claim. For the material described here as plazas and ceremonial courts, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Tibees using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with San Agustín and Marajó Island Mounds using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

La Isabela

Amazonia, Andes, Caribbean • Evidence dossier 420 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Dominican Republic	**colonial**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

La Isabela is a site or landscape associated with colonial research in Dominican Republic. The defining record highlighted in this atlas is early Spanish colonial settlement remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in americas: wider landscapes is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Spanish colonial settlement remains
Chronological frame	colonial; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Spanish colonial settlement remains linked to the cited phase of colonial?
- What natural and human processes altered deposits at La Isabela after original use?
- Which nearby sites, especially Túcume, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare La Isabela with Túcume, Tierradentro and Kuhikugu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

La Isabela

Dominican Republic • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of La Isabela begins with a bounded claim. For the material described here as early Spanish colonial settlement remains, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for La Isabela using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Tierradentro and Kuhikugu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Niah Caves

Oceania and Pacific • Evidence dossier 421 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Malaysia	Pleistocene-Holocene	Bio

01 WHAT SURVIVES AND WHY IT MATTERS

Niah Caves is a site or landscape associated with pleistocene-holocene research in Malaysia. The defining record highlighted in this atlas is human remains and rainforest occupation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	human remains and rainforest occupation
Chronological frame	Pleistocene-Holocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are human remains and rainforest occupation linked to the cited phase of Pleistocene-Holocene?
- What natural and human processes altered deposits at Niah Caves after original use?
- Which nearby sites, especially Mata Menge, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Niah Caves with Mata Menge, Teouma and Nan Madol. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Niah Caves

Malaysia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Niah Caves begins with a bounded claim. For the material described here as human remains and rainforest occupation, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Niah Caves using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Teouma and Nan Madol using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Madai Caves

Oceania and Pacific • Evidence dossier 422 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Malaysia	prehistoric-historic	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Madai Caves is a site or landscape associated with prehistoric-historic research in Malaysia. The defining record highlighted in this atlas is rock shelter occupation and bird-nest economy. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	rock shelter occupation and bird-nest economy
Chronological frame	prehistoric-historic; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are rock shelter occupation and bird-nest economy linked to the cited phase of prehistoric-historic?
- What natural and human processes altered deposits at Madai Caves after original use?
- Which nearby sites, especially Leang Bulu' Sipong 4, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Madai Caves with Leang Bulu' Sipong 4, Talepakemalai and Lelu Ruins. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Madai Caves

Malaysia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Madai Caves begins with a bounded claim. For the material described here as rock shelter occupation and bird-nest economy, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Madai Caves using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Talepakemalai and Lelu Ruins using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Liang Bua

Oceania and Pacific • Evidence dossier 423 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Indonesia	**Late Pleistocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Liang Bua is a site or landscape associated with late pleistocene research in Indonesia. The defining record highlighted in this atlas is Homo floresiensis remains. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	Homo floresiensis remains
Chronological frame	Late Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are Homo floresiensis remains linked to the cited phase of Late Pleistocene?
- What natural and human processes altered deposits at Liang Bua after original use?
- Which nearby sites, especially Leang Tedongnge, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Liang Bua with Leang Tedongnge, Bourewa and Ha'amonga 'a Maui. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Liang Bua

Indonesia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Liang Bua begins with a bounded claim. For the material described here as Homo floresiensis remains, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Liang Bua using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Bourewa and Ha'amonga 'a Maui using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Mata Menge

Oceania and Pacific • Evidence dossier 424 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Indonesia	**Middle Pleistocene**	**Bio**

01 WHAT SURVIVES AND WHY IT MATTERS

Mata Menge is a site or landscape associated with middle pleistocene research in Indonesia. The defining record highlighted in this atlas is earlier small-bodied hominin fossils. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	earlier small-bodied hominin fossils
Chronological frame	Middle Pleistocene; exact ranges depend on excavated contexts
Analytical priority	establish specimen identity, geological position, comparative anatomy and age of associated deposits
Important caveat	Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are earlier small-bodied hominin fossils linked to the cited phase of Middle Pleistocene?
- What natural and human processes altered deposits at Mata Menge after original use?
- Which nearby sites, especially Kuk Swamp, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Mata Menge with Kuk Swamp, Sigatoka Sand Dunes and Talietumu. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Mata Menge

Indonesia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Mata Menge begins with a bounded claim. For the material described here as earlier small-bodied hominin fossils, the immediate objective is to establish specimen identity, geological position, comparative anatomy and age of associated deposits. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Osteological morphology and CT imaging

Record sample/context identifiers and validate the chain of custody.

02 Context-controlled geochronology

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ancient DNA or proteomics where preservation allows

Relate results to the local stratigraphic sequence and regional chronology.

04 Taphonomic and paleoenvironmental analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which skeletal traits and associations withstand independent measurement, dating, and taxonomic review? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Fossils are not automatically in primary depositional position; the discovery layer and the age of the individual need separate documentation. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Mata Menge using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Sigatoka Sand Dunes and Talietumu using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Leang Bulu' Sipong 4

Oceania and Pacific • Evidence dossier 425 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Indonesia	**Pleistocene**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Leang Bulu' Sipong 4 is a site or landscape associated with pleistocene research in Indonesia. The defining record highlighted in this atlas is figurative rock art and hunting scenes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	figurative rock art and hunting scenes
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are figurative rock art and hunting scenes linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Leang Bulu' Sipong 4 after original use?
- Which nearby sites, especially Buang Merabak, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Leang Bulu' Sipong 4 with Buang Merabak, Rapa Nui (Rano Raraku) and Lapaha. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Leang Bulu' Sipong 4

Indonesia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Leang Bulu' Sipong 4 begins with a bounded claim. For the material described here as figurative rock art and hunting scenes, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Leang Bulu' Sipong 4 using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Rapa Nui (Rano Raraku) and Lapaha using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Leang Tedongnge

Oceania and Pacific • Evidence dossier 426 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Indonesia	**Pleistocene**	**Art**

01 WHAT SURVIVES AND WHY IT MATTERS

Leang Tedongnge is a site or landscape associated with pleistocene research in Indonesia. The defining record highlighted in this atlas is painted pig image and dating research. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	painted pig image and dating research
Chronological frame	Pleistocene; exact ranges depend on excavated contexts
Analytical priority	record pigment or carved surface, setting, superposition and long-term exposure history
Important caveat	Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are painted pig image and dating research linked to the cited phase of Pleistocene?
- What natural and human processes altered deposits at Leang Tedongnge after original use?
- Which nearby sites, especially Watom Island, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Leang Tedongnge with Watom Island, Ahu Tongariki and Niah Caves. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Leang Tedongnge

Indonesia • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Leang Tedongnge begins with a bounded claim. For the material described here as painted pig image and dating research, the immediate objective is to record pigment or carved surface, setting, superposition and long-term exposure history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 High-resolution photogrammetry and raking light

Record sample/context identifiers and validate the chain of custody.

02 Pigment, mineral and surface analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Cross-cutting and superposition recording

Relate results to the local stratigraphic sequence and regional chronology.

04 Conservation-grade microenvironment monitoring

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which visual phases, production techniques and associations can be demonstrated without attributing untestable beliefs? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Symbolic meaning cannot be read directly from modern iconographic comparisons; a motif may have many possible functions. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Leang Tedongnge using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ahu Tongariki and Niah Caves using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Kuk Swamp

Oceania and Pacific • Evidence dossier 427 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Papua New Guinea	**Holocene**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Kuk Swamp is a site or landscape associated with holocene research in Papua New Guinea. The defining record highlighted in this atlas is early agriculture and wetlands engineering. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early agriculture and wetlands engineering
Chronological frame	Holocene; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early agriculture and wetlands engineering linked to the cited phase of Holocene?
- What natural and human processes altered deposits at Kuk Swamp after original use?
- Which nearby sites, especially Teouma, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Kuk Swamp with Teouma, Nan Madol and Madai Caves. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Kuk Swamp

Papua New Guinea • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Kuk Swamp begins with a bounded claim. For the material described here as early agriculture and wetlands engineering, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Kuk Swamp using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Nan Madol and Madai Caves using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Buang Merabak

Oceania and Pacific • Evidence dossier 428 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Papua New Guinea	**Pleistocene-Holocene**	**Lithic**

01 WHAT SURVIVES AND WHY IT MATTERS

Buang Merabak is a site or landscape associated with pleistocene-holocene research in Papua New Guinea. The defining record highlighted in this atlas is island cave occupation. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	island cave occupation
Chronological frame	Pleistocene-Holocene; exact ranges depend on excavated contexts
Analytical priority	document artifact associations, manufacturing sequences and post-depositional alteration
Important caveat	Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are island cave occupation linked to the cited phase of Pleistocene-Holocene?
- What natural and human processes altered deposits at Buang Merabak after original use?
- Which nearby sites, especially Talepakemalai, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Buang Merabak with Talepakemalai, Lelu Ruins and Liang Bua. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Buang Merabak

Papua New Guinea • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Buang Merabak begins with a bounded claim. For the material described here as island cave occupation, the immediate objective is to document artifact associations, manufacturing sequences and post-depositional alteration. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Lithic technological analysis and refitting

Record sample/context identifiers and validate the chain of custody.

02 Microscopic use-wear and residues

Choose independent controls, repeat measurements and quantify uncertainty.

03 Geological dating and sedimentology

Relate results to the local stratigraphic sequence and regional chronology.

04 Raw-material sourcing and spatial modelling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What activity patterns and manufacturing decisions can be reconstructed from refits and use-wear? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Stone-tool typologies are not biological species labels; technological similarity need not mean direct descent. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Buang Merabak using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lelu Ruins and Liang Bua using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Watom Island

Oceania and Pacific • Evidence dossier 429 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Papua New Guinea	**Lapita**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Watom Island is a site or landscape associated with lapita research in Papua New Guinea. The defining record highlighted in this atlas is decorated pottery and Pacific dispersal. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	decorated pottery and Pacific dispersal
Chronological frame	Lapita; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are decorated pottery and Pacific dispersal linked to the cited phase of Lapita?
- What natural and human processes altered deposits at Watom Island after original use?
- Which nearby sites, especially Bourewa, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Watom Island with Bourewa, Ha'amonga 'a Maui and Mata Menge. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Watom Island

Papua New Guinea • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Watom Island begins with a bounded claim. For the material described here as decorated pottery and Pacific dispersal, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Watom Island using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Ha'amonga 'a Maui and Mata Menge using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Teouma

Oceania and Pacific • Evidence dossier 430 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Vanuatu	**Lapita**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Teouma is a site or landscape associated with lapita research in Vanuatu. The defining record highlighted in this atlas is cemetery and first-settlement archaeology. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	cemetery and first-settlement archaeology
Chronological frame	Lapita; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are cemetery and first-settlement archaeology linked to the cited phase of Lapita?
- What natural and human processes altered deposits at Teouma after original use?
- Which nearby sites, especially Sigatoka Sand Dunes, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Teouma with Sigatoka Sand Dunes, Talietumu and Leang Bulu' Sipong 4. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Teouma

Vanuatu • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Teouma begins with a bounded claim. For the material described here as cemetery and first-settlement archaeology, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Teouma using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Talietumu and Leang Bulu' Sipong 4 using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Talepakemalai

Oceania and Pacific • Evidence dossier 431 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Papua New Guinea	**Lapita**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Talepakemalai is a site or landscape associated with lapita research in Papua New Guinea. The defining record highlighted in this atlas is stilt-house and pottery sequence. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	stilt-house and pottery sequence
Chronological frame	Lapita; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are stilt-house and pottery sequence linked to the cited phase of Lapita?
- What natural and human processes altered deposits at Talepakemalai after original use?
- Which nearby sites, especially Rapa Nui (Rano Raraku), provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Talepakemalai with Rapa Nui (Rano Raraku), Lapaha and Leang Tedongnge. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Talepakemalai

Papua New Guinea • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Talepakemalai begins with a bounded claim. For the material described here as stilt-house and pottery sequence, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Talepakemalai using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Lapaha and Leang Tedongnge using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Bourewa

Oceania and Pacific • Evidence dossier 432 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Fiji	**Lapita**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Bourewa is a site or landscape associated with lapita research in Fiji. The defining record highlighted in this atlas is early Fiji colonization site. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	early Fiji colonization site
Chronological frame	Lapita; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are early Fiji colonization site linked to the cited phase of Lapita?
- What natural and human processes altered deposits at Bourewa after original use?
- Which nearby sites, especially Ahu Tongariki, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Bourewa with Ahu Tongariki, Niah Caves and Kuk Swamp. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Bourewa

Fiji • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Bourewa begins with a bounded claim. For the material described here as early Fiji colonization site, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Bourewa using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Niah Caves and Kuk Swamp using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Sigatoka Sand Dunes

Oceania and Pacific • Evidence dossier 433 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Fiji	**Lapita onward**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Sigatoka Sand Dunes is a site or landscape associated with lapita onward research in Fiji. The defining record highlighted in this atlas is burials and pottery in coastal dunes. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	burials and pottery in coastal dunes
Chronological frame	Lapita onward; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are burials and pottery in coastal dunes linked to the cited phase of Lapita onward?
- What natural and human processes altered deposits at Sigatoka Sand Dunes after original use?
- Which nearby sites, especially Nan Madol, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Sigatoka Sand Dunes with Nan Madol, Madai Caves and Buang Merabak. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Sigatoka Sand Dunes

Fiji • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Sigatoka Sand Dunes begins with a bounded claim. For the material described here as burials and pottery in coastal dunes, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Sigatoka Sand Dunes using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Madai Caves and Buang Merabak using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

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Rapa Nui (Rano Raraku)

Oceania and Pacific • Evidence dossier 434 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Chile	**Polynesian**	**Material**

01 WHAT SURVIVES AND WHY IT MATTERS

Rapa Nui (Rano Raraku) is a site or landscape associated with polynesian research in Chile. The defining record highlighted in this atlas is moai quarry and sculpting. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	moai quarry and sculpting
Chronological frame	Polynesian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are moai quarry and sculpting linked to the cited phase of Polynesian?
- What natural and human processes altered deposits at Rapa Nui (Rano Raraku) after original use?
- Which nearby sites, especially Lelu Ruins, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Rapa Nui (Rano Raraku) with Lelu Ruins, Liang Bua and Watom Island. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Quarry remains allow investigators to examine carving stages and movement of moai; interpretations require the history and perspectives of Rapa Nui communities.

RESEARCH POINTER: https://whc.unesco.org/en/list/715/

Rapa Nui (Rano Raraku)

Chile • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Rapa Nui (Rano Raraku) begins with a bounded claim. For the material described here as moai quarry and sculpting, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Rapa Nui (Rano Raraku) using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Liang Bua and Watom Island using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ahu Tongariki

Oceania and Pacific • Evidence dossier 435 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Chile	Polynesian	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Ahu Tongariki is a site or landscape associated with polynesian research in Chile. The defining record highlighted in this atlas is ceremonial platform and moai. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	ceremonial platform and moai
Chronological frame	Polynesian; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are ceremonial platform and moai linked to the cited phase of Polynesian?
- What natural and human processes altered deposits at Ahu Tongariki after original use?
- Which nearby sites, especially Ha'amonga 'a Maui, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ahu Tongariki with Ha'amonga 'a Maui, Mata Menge and Teouma. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ahu Tongariki

Chile • Interpretive protocol

B | SCIENTIFIC TESTING AND FUTURE RESEARCH

05 RESEARCH DESIGN

A defensible study of Ahu Tongariki begins with a bounded claim. For the material described here as ceremonial platform and moai, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ahu Tongariki using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Mata Menge and Teouma using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Nan Madol

Oceania and Pacific • Evidence dossier 436 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Micronesia	medieval	Urban

01 WHAT SURVIVES AND WHY IT MATTERS

Nan Madol is a site or landscape associated with medieval research in Micronesia. The defining record highlighted in this atlas is basalt islet monumental complex. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	basalt islet monumental complex
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are basalt islet monumental complex linked to the cited phase of medieval?
- What natural and human processes altered deposits at Nan Madol after original use?
- Which nearby sites, especially Talietumu, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Nan Madol with Talietumu, Leang Bulu' Sipong 4 and Talepakemalai. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

05 SELECTED RESEARCH AND EXCAVATION HISTORY

Monumental basalt constructions on artificial islets testify to sophisticated island labor organization and maritime connections.

Nan Madol

Micronesia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Nan Madol begins with a bounded claim. For the material described here as basalt islet monumental complex, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Nan Madol using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Leang Bulu' Sipong 4 and Talepakemalai using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lelu Ruins

Oceania and Pacific • Evidence dossier 437 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Micronesia	medieval	Material

01 WHAT SURVIVES AND WHY IT MATTERS

Lelu Ruins is a site or landscape associated with medieval research in Micronesia. The defining record highlighted in this atlas is coral and basalt civic center. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	coral and basalt civic center
Chronological frame	medieval; exact ranges depend on excavated contexts
Analytical priority	establish artifact types, manufacturing traces, depositional contexts and environmental conditions
Important caveat	Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are coral and basalt civic center linked to the cited phase of medieval?
- What natural and human processes altered deposits at Lelu Ruins after original use?
- Which nearby sites, especially Lapaha, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lelu Ruins with Lapaha, Leang Tedongnge and Bourewa. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lelu Ruins

Micronesia • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Lelu Ruins begins with a bounded claim. For the material described here as coral and basalt civic center, the immediate objective is to establish artifact types, manufacturing traces, depositional contexts and environmental conditions. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic excavation and conservation

Record sample/context identifiers and validate the chain of custody.

02 Ceramic or material culture analysis

Choose independent controls, repeat measurements and quantify uncertainty.

03 Scientific dating appropriate to sampled material

Relate results to the local stratigraphic sequence and regional chronology.

04 GIS spatial and distribution analysis

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: Which observations distinguish local production, exchange, ritual deposition and later reuse? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Objects moved by reuse, flooding, digging or antiquarian collection may no longer mark their original behavior. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lelu Ruins using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Leang Tedongnge and Bourewa using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Ha'amonga 'a Maui

Oceania and Pacific • Evidence dossier 438 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tonga	**Polynesian**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Ha'amonga 'a Maui is a site or landscape associated with polynesian research in Tonga. The defining record highlighted in this atlas is monumental stone trilithon. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	monumental stone trilithon
Chronological frame	Polynesian; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are monumental stone trilithon linked to the cited phase of Polynesian?
- What natural and human processes altered deposits at Ha'amonga 'a Maui after original use?
- Which nearby sites, especially Niah Caves, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Ha'amonga 'a Maui with Niah Caves, Kuk Swamp and Sigatoka Sand Dunes. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Ha'amonga 'a Maui

Tonga • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Ha'amonga 'a Maui begins with a bounded claim. For the material described here as monumental stone trilithon, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Ha'amonga 'a Maui using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Kuk Swamp and Sigatoka Sand Dunes using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Talietumu

Oceania and Pacific • Evidence dossier 439 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Wallis and Futuna	**Polynesian**	**Urban**

01 WHAT SURVIVES AND WHY IT MATTERS

Talietumu is a site or landscape associated with polynesian research in Wallis and Futuna. The defining record highlighted in this atlas is fortified ceremonial complex. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	fortified ceremonial complex
Chronological frame	Polynesian; exact ranges depend on excavated contexts
Analytical priority	reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape
Important caveat	Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are fortified ceremonial complex linked to the cited phase of Polynesian?
- What natural and human processes altered deposits at Talietumu after original use?
- Which nearby sites, especially Madai Caves, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Talietumu with Madai Caves, Buang Merabak and Rapa Nui (Rano Raraku). The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Talietumu

Wallis and Futuna • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Talietumu begins with a bounded claim. For the material described here as fortified ceremonial complex, the immediate objective is to reconstruct building sequence, public infrastructure, domestic zones and connections to the wider landscape. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Stratigraphic architecture recording

Record sample/context identifiers and validate the chain of custody.

02 Aerial/LiDAR and geophysical survey

Choose independent controls, repeat measurements and quantify uncertainty.

03 Ceramic, coin or radiocarbon chronology

Relate results to the local stratigraphic sequence and regional chronology.

04 Household and environmental sampling

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: How did inhabitants use infrastructure, labor and domestic spaces over successive building phases? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Surviving monuments overrepresent durable public and elite structures compared with ordinary domestic life. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Talietumu using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Buang Merabak and Rapa Nui (Rano Raraku) using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.

Lapaha

Oceania and Pacific • Evidence dossier 440 of 440

A | SITE CONTEXT AND INTERPRETATION

COUNTRY / TERRITORY	BROAD PERIOD	RESEARCH FOCUS
Tonga	**Polynesian**	**Burial**

01 WHAT SURVIVES AND WHY IT MATTERS

Lapaha is a site or landscape associated with polynesian research in Tonga. The defining record highlighted in this atlas is royal tombs and stone architecture. Archaeological value rests not only in objects or monuments but in their exact provenience, associations, sequence of deposition, construction and later disturbance. The evidence must be interpreted at the scale documented by fieldwork, not enlarged into unsupported claims about an entire culture.

Its placement in oceania and pacific islands is comparative rather than exclusive: archaeological boundaries seldom match modern national borders. Read this entry as a research orientation to the material record, not as a replacement for excavation monographs, laboratory datasets, conservation inventories or the perspectives of affiliated communities.

02 EVIDENCE CLASSIFICATION

Primary material	royal tombs and stone architecture
Chronological frame	Polynesian; exact ranges depend on excavated contexts
Analytical priority	establish grave sequence, anatomical position, associated offerings and disturbance history
Important caveat	Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office.

03 QUESTIONS THE EXCAVATION RECORD MUST ANSWER

- How securely are royal tombs and stone architecture linked to the cited phase of Polynesian?
- What natural and human processes altered deposits at Lapaha after original use?
- Which nearby sites, especially Liang Bua, provide an independent comparative sequence?

04 REGIONAL COMPARISON AND SOURCE LIMITS

Compare Lapaha with Liang Bua, Watom Island and Ahu Tongariki. The goal is not to force a single historical narrative, but to test differences in environment, material preservation, site function and temporal resolution. A site-specific bibliography, excavation permit archive and primary report are required before making precise numerical, cultural or authorship claims.

Lapaha

Tonga • Interpretive protocol

05 RESEARCH DESIGN

A defensible study of Lapaha begins with a bounded claim. For the material described here as royal tombs and stone architecture, the immediate objective is to establish grave sequence, anatomical position, associated offerings and disturbance history. Investigators should define excavation units, deposits, sample IDs, analytical methods and uncertainty thresholds before drawing broader historical conclusions. Every inference should have a traceable link to a recorded observation.

06 DATA AND ANALYTIC METHODS

01 Burial stratigraphy and spatial mapping

Record sample/context identifiers and validate the chain of custody.

02 Osteology and taphonomy

Choose independent controls, repeat measurements and quantify uncertainty.

03 Radiocarbon or appropriate relative dating

Relate results to the local stratigraphic sequence and regional chronology.

04 Material sourcing and funerary comparison

Preserve raw observations, derived files, permissions and metadata for review.

07 INTERPRETATION TESTS AND ALTERNATIVES

Primary question: What variation in mortuary treatment reflects age, time, community practice or differential preservation? An initial interpretation should be compared against at least two alternatives: that the pattern reflects a different activity at the same time, or that it formed through reuse, erosion, mixing or collection practices at another time. Grave wealth does not provide an unambiguous measure of rank, ethnicity, gender or political office. Negative evidence is informative only if excavation coverage, recovery strategy and preservation make absence meaningful.

08 NEXT INVESTIGATIONS

- Reconstruct a phase-by-phase plan for Lapaha using published contexts, rather than a single composite image.
- Locate and assess the earliest primary excavation report, subsequent corrections and scientific dating publications.
- Compare the core signal with Watom Island and Ahu Tongariki using equivalent categories and time windows.
- Publish a source register that distinguishes direct observations from later hypotheses and popular retellings.

09 STEWARDSHIP AND EVIDENTIARY STATUS

Research should follow applicable laws and permissions, minimize unnecessary disturbance, consult descendant or local communities, and conserve archaeological contexts as finite evidence. This atlas provides a high-level, synthesized orientation, not an independently audited excavation catalogue. Claims about precise dates, artifact counts and interpretations must be checked against the primary record before citation as established fact.